

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
WASHINGTON, DC 20549

FORM 10-Q
QUARTERLY REPORT PURSUANT TO SECTION 13 or 15(d)
OF THE SECURITIES EXCHANGE ACT OF 1934
FOR THE QUARTER ENDED JUNE 30, 2023

1-2360
(Commission file number)

INTERNATIONAL BUSINESS MACHINES CORPORATION
(Exact name of registrant as specified in its charter)

New York
(State of incorporation)

One New Orchard Road
Armonk, New York
(Address of principal executive offices)

13-0871985
(IRS employer identification number)

10504
(Zip Code)

914-499-1900
(Registrant's telephone number)

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading symbol(s)	Name of each exchange on which registered
Capital stock, par value \$.20 per share	IBM	New York Stock Exchange NYSE Chicago
1.125% Notes due 2024	IBM 24A	New York Stock Exchange
2.875% Notes due 2025	IBM 25A	New York Stock Exchange
0.950% Notes due 2025	IBM 25B	New York Stock Exchange
0.875% Notes due 2025	IBM 25C	New York Stock Exchange
0.300% Notes due 2026	IBM 26B	New York Stock Exchange
1.250% Notes due 2027	IBM 27B	New York Stock Exchange
3.375% Notes due 2027	IBM 27F	New York Stock Exchange
0.300% Notes due 2028	IBM 28B	New York Stock Exchange
1.750% Notes due 2028	IBM 28A	New York Stock Exchange
1.500% Notes due 2029	IBM 29	New York Stock Exchange
0.875% Notes due 2030	IBM 30A	New York Stock Exchange
1.750% Notes due 2031	IBM 31	New York Stock Exchange
3.625% Notes due 2031	IBM 31B	New York Stock Exchange
0.650% Notes due 2032	IBM 32A	New York Stock Exchange
1.250% Notes due 2034	IBM 34	New York Stock Exchange
3.750% Notes due 2035	IBM 35	New York Stock Exchange
4.875% Notes due 2038	IBM 38	New York Stock Exchange
1.200% Notes due 2040	IBM 40	New York Stock Exchange
4.000% Notes due 2043	IBM 43	New York Stock Exchange
7.00% Debentures due 2025	IBM 25	New York Stock Exchange
6.22% Debentures due 2027	IBM 27	New York Stock Exchange
6.50% Debentures due 2028	IBM 28	New York Stock Exchange
5.875% Debentures due 2032	IBM 32D	New York Stock Exchange
7.00% Debentures due 2045	IBM 45	New York Stock Exchange
7.125% Debentures due 2096	IBM 96	New York Stock Exchange

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of “large accelerated filer,” “accelerated filer,” “smaller reporting company,” and “emerging growth company” in Rule 12b-2 of the Exchange Act.

Large accelerated filer ☒
Non-accelerated filer ☐

Accelerated filer ☐
Smaller reporting company ☐
Emerging growth company ☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

The registrant had 911,006,240 shares of common stock outstanding at June 30, 2023.

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Part I - Financial Information

Item 1. Consolidated Financial Statements:

**INTERNATIONAL BUSINESS MACHINES CORPORATION
AND SUBSIDIARY COMPANIES
CONSOLIDATED INCOME STATEMENT
(UNAUDITED)**

(Dollars in millions except per share amounts)	Three Months Ended June 30,		Six Months Ended June 30,	
	2023	2022	2023	2022
Revenue:				
Services	\$ 7,553	\$ 7,640	\$ 15,077	\$ 15,343
Sales	7,739	7,748	14,271	14,087
Financing	183	147	380	303
Total revenue	15,475	15,535	29,727	29,732
Cost:				
Services	5,294	5,399	10,604	10,747
Sales	1,587	1,750	2,910	3,165
Financing	93	96	203	194
Total cost	6,974	7,246	13,717	14,107
Gross profit	8,501	8,290	16,010	15,625
Expense and other (income):				
Selling, general and administrative	4,900	4,855	9,754	9,452
Research, development and engineering	1,687	1,673	3,342	3,352
Intellectual property and custom development income	(248)	(176)	(428)	(297)
Other (income) and expense	(261)	(81)	(506)	166
Interest expense	423	297	790	607
Total expense and other (income)	6,501	6,568	12,952	13,280
Income from continuing operations before income taxes	2,000	1,722	3,058	2,345
Provision for income taxes	419	257	543	218
Income from continuing operations	\$ 1,581	\$ 1,465	\$ 2,515	\$ 2,127
Income/(loss) from discontinued operations, net of tax	2	(73)	(4)	(2)
Net income	\$ 1,583	\$ 1,392	\$ 2,511	\$ 2,125
Earnings/(loss) per share of common stock:				
Assuming dilution:				
Continuing operations	\$ 1.72	\$ 1.61	\$ 2.74	\$ 2.34
Discontinued operations	0.00	(0.08)	0.00	0.00
Total	\$ 1.72	\$ 1.53	\$ 2.73	\$ 2.34
Basic:				
Continuing operations	\$ 1.74	\$ 1.62	\$ 2.77	\$ 2.36
Discontinued operations	0.00	(0.08)	0.00	0.00
Total	\$ 1.74	\$ 1.54	\$ 2.76	\$ 2.36
Weighted-average number of common shares outstanding: (millions)				
Assuming dilution	919.5	910.7	918.6	910.0
Basic	909.9	901.5	908.7	900.4

(Amounts may not add due to rounding.)
(The accompanying notes are an integral part of the financial statements.)

**INTERNATIONAL BUSINESS MACHINES CORPORATION
AND SUBSIDIARY COMPANIES
CONSOLIDATED STATEMENT OF COMPREHENSIVE INCOME
(UNAUDITED)**

(Dollars in millions)	Three Months Ended June 30,		Six Months Ended June 30,	
	2023	2022	2023	2022
Net income	\$ 1,583	\$ 1,392	\$ 2,511	\$ 2,125
Other comprehensive income/(loss), before tax:				
Foreign currency translation adjustments	116	213	29	655
Net changes related to available-for-sale securities:				
Unrealized gains/(losses) arising during the period	(17)	0	(1)	(1)
Reclassification of (gains)/losses to net income	—	—	—	—
Total net changes related to available-for-sale securities	(17)	0	(1)	(1)
Unrealized gains/(losses) on cash flow hedges:				
Unrealized gains/(losses) arising during the period	178	200	149	260
Reclassification of (gains)/losses to net income	(30)	16	(152)	16
Total unrealized gains/(losses) on cash flow hedges	148	217	(3)	276
Retirement-related benefit plans:				
Prior service costs/(credits)	—	—	—	(5)
Net (losses)/gains arising during the period	0	1	2	10
Curtailments and settlements	6	11	5	19
Amortization of prior service (credits)/costs	(2)	6	(4)	13
Amortization of net (gains)/losses	130	450	261	917
Total retirement-related benefit plans	134	468	263	954
Other comprehensive income/(loss), before tax	381	897	289	1,885
Income tax (expense)/benefit related to items of other comprehensive income	(101)	(534)	(48)	(819)
Other comprehensive income/(loss), net of tax	280	363	241	1,066
Total comprehensive income	\$ 1,863	\$ 1,755	\$ 2,751	\$ 3,191

(Amounts may not add due to rounding.)
(The accompanying notes are an integral part of the financial statements.)

**INTERNATIONAL BUSINESS MACHINES CORPORATION
AND SUBSIDIARY COMPANIES
CONSOLIDATED BALANCE SHEET
(UNAUDITED)**

ASSETS

(Dollars in millions)	At June 30, 2023	At December 31, 2022
Assets:		
Current assets:		
Cash and cash equivalents	\$ 9,394	\$ 7,886
Restricted cash	31	103
Marketable securities	6,904	852
Notes and accounts receivable — trade (net of allowances of \$217 in 2023 and \$233 in 2022)	5,673	6,541
Short-term financing receivables:		
Held for investment (net of allowances of \$143 in 2023 and \$145 in 2022)	5,564	6,851
Held for sale	865	939
Other accounts receivable (net of allowances of \$108 in 2023 and \$89 in 2022)	838	817
Inventory, at lower of average cost or net realizable value:		
Finished goods	174	158
Work in process and raw materials	1,327	1,394
Total inventory	1,501	1,552
Deferred costs	957	967
Prepaid expenses and other current assets	2,730	2,611
Total current assets	34,458	29,118
Property, plant and equipment	18,588	18,695
Less: Accumulated depreciation	13,145	13,361
Property, plant and equipment — net	5,443	5,334
Operating right-of-use assets — net	2,653	2,878
Long-term financing receivables (net of allowances of \$28 in 2023 and \$28 in 2022)	5,221	5,806
Prepaid pension assets	8,735	8,236
Deferred costs	897	866
Deferred taxes	6,340	6,256
Goodwill	56,385	55,949
Intangible assets — net	10,496	11,184
Investments and sundry assets	1,585	1,617
Total assets	\$ 132,213	\$ 127,243

(Amounts may not add due to rounding.)
(The accompanying notes are an integral part of the financial statements.)

**INTERNATIONAL BUSINESS MACHINES CORPORATION
AND SUBSIDIARY COMPANIES
CONSOLIDATED BALANCE SHEET – (CONTINUED)
(UNAUDITED)**

LIABILITIES AND EQUITY

(Dollars in millions except per share amounts)	At June 30, 2023	At December 31, 2022
Liabilities:		
Current liabilities:		
Taxes	\$ 1,606	\$ 2,196
Short-term debt	6,785	4,760
Accounts payable	3,732	4,051
Compensation and benefits	3,185	3,481
Deferred income	12,712	12,032
Operating lease liabilities	842	874
Other accrued expenses and liabilities	3,651	4,111
Total current liabilities	32,513	31,505
Long-term debt	50,691	46,189
Retirement and nonpension postretirement benefit obligations	9,385	9,596
Deferred income	3,264	3,499
Operating lease liabilities	1,986	2,190
Other liabilities	12,103	12,243
Total liabilities	109,942	105,222
Equity:		
IBM stockholders' equity:		
Common stock, par value \$0.20 per share, and additional paid-in capital	58,963	58,343
Shares authorized: 4,687,500,000		
Shares issued: 2023 - 2,262,697,191		
2022 - 2,257,116,920		
Retained earnings	149,318	149,825
Treasury stock - at cost	(169,581)	(169,484)
Shares: 2023 - 1,351,690,951		
2022 - 1,351,024,943		
Accumulated other comprehensive income/(loss)	(16,499)	(16,740)
Total IBM stockholders' equity	22,201	21,944
Noncontrolling interests	70	77
Total equity	22,271	22,021
Total liabilities and equity	\$ 132,213	\$ 127,243

(Amounts may not add due to rounding.)
(The accompanying notes are an integral part of the financial statements.)

**INTERNATIONAL BUSINESS MACHINES CORPORATION
AND SUBSIDIARY COMPANIES
CONSOLIDATED STATEMENT OF CASH FLOWS
(UNAUDITED)**

(Dollars in millions)	Six Months Ended June 30,	
	2023	2022*
Cash flows from operating activities:		
Net income	\$ 2,511	\$ 2,125
Adjustments to reconcile net income to cash provided by operating activities:		
Depreciation	1,047	1,251
Amortization of intangibles	1,104	1,251
Stock-based compensation	556	488
Net (gain)/loss on asset sales and other	(113)	(100)
Changes in operating assets and liabilities, net of acquisitions/divestitures	1,308	(446)
Net cash provided by operating activities	6,412	4,569
Cash flows from investing activities:		
Payments for property, plant and equipment	(664)	(620)
Proceeds from disposition of property, plant and equipment	25	90
Investment in software	(305)	(341)
Acquisition of businesses, net of cash acquired	(356)	(958)
Divestitures of businesses, net of cash transferred	6	1,268
Purchases of marketable securities and other investments	(9,260)	(2,336)
Proceeds from disposition of marketable securities and other investments	2,600	1,711
Net cash provided by/(used in) investing activities	(7,953)	(1,186)
Cash flows from financing activities:		
Proceeds from new debt	9,432	4,402
Payments to settle debt	(3,260)	(3,959)
Short-term borrowings/(repayments) less than 90 days — net	(3)	(9)
Common stock repurchases for tax withholdings	(240)	(315)
Financing — other	56	25
Cash dividends paid	(3,007)	(2,963)
Net cash provided by/(used in) financing activities	2,978	(2,819)
Effect of exchange rate changes on cash, cash equivalents and restricted cash	(1)	(267)
Net change in cash, cash equivalents and restricted cash	1,436	297
Cash, cash equivalents and restricted cash at January 1	7,988	6,957
Cash, cash equivalents and restricted cash at June 30	\$ 9,425	\$ 7,254

* Includes immaterial cash flows from discontinued operations.

(Amounts may not add due to rounding.)

(The accompanying notes are an integral part of the financial statements.)

**INTERNATIONAL BUSINESS MACHINES CORPORATION
AND SUBSIDIARY COMPANIES
CONSOLIDATED STATEMENT OF EQUITY
(UNAUDITED)**

(Dollars in millions except per share amounts)	Common Stock and Additional Paid-in Capital	Retained Earnings	Treasury Stock	Accumulated Other Comprehensive Income/(Loss)	Total IBM Stockholders' Equity	Non- Controlling Interests	Total Equity
Equity - April 1, 2023	\$ 58,675	\$ 149,253	\$ (169,544)	\$ (16,780)	\$ 21,604	\$ 68	\$ 21,672
Net income plus other comprehensive income/(loss):							
Net income		1,583			1,583		1,583
Other comprehensive income/(loss)				280	280		280
Total comprehensive income/(loss)					<u>\$ 1,863</u>		<u>\$ 1,863</u>
Cash dividends paid — common stock (\$1.66 per share)		(1,510)			(1,510)		(1,510)
Common stock issued under employee plans (3,199,344 shares)	288				288		288
Purchases (1,098,988 shares) and sales (860,470 shares) of treasury stock under employee plans — net		(8)	(37)		(45)		(45)
Changes in noncontrolling interests						2	2
Equity – June 30, 2023	<u>\$ 58,963</u>	<u>\$ 149,318</u>	<u>\$ (169,581)</u>	<u>\$ (16,499)</u>	<u>\$ 22,201</u>	<u>\$ 70</u>	<u>\$ 22,271</u>
(Dollars in millions except per share amounts)	Common Stock and Additional Paid-in Capital	Retained Earnings	Treasury Stock	Accumulated Other Comprehensive Income/(Loss)	Total IBM Stockholders' Equity	Non- Controlling Interests	Total Equity
Equity - April 1, 2022	\$ 57,603	\$ 153,401	\$ (169,422)	\$ (22,532)	\$ 19,050	\$ 62	\$ 19,112
Net income plus other comprehensive income/(loss):							
Net income		1,392			1,392		1,392
Other comprehensive income/(loss)				363	363		363
Total comprehensive income/(loss)					<u>\$ 1,755</u>		<u>\$ 1,755</u>
Cash dividends paid — common stock (\$1.65 per share)		(1,488)			(1,488)		(1,488)
Common stock issued under employee plans (4,398,589 shares)	199				199		199
Purchases (1,723,774 shares) and sales (1,070,214 shares) of treasury stock under employee plans — net		(7)	(100)		(107)		(107)
Changes in noncontrolling interests						6	6
Equity - June 30, 2022	<u>\$ 57,802</u>	<u>\$ 153,298</u>	<u>\$ (169,522)</u>	<u>\$ (22,169)</u>	<u>\$ 19,409</u>	<u>\$ 67</u>	<u>\$ 19,476</u>

(Amounts may not add due to rounding.)
(The accompanying notes are an integral part of the financial statements.)

**INTERNATIONAL BUSINESS MACHINES CORPORATION
AND SUBSIDIARY COMPANIES
CONSOLIDATED STATEMENT OF EQUITY – (CONTINUED)
(UNAUDITED)**

(Dollars in millions except per share amounts)	Common Stock and Additional Paid-in Capital	Retained Earnings	Treasury Stock	Accumulated Other Comprehensive Income/(Loss)	Total IBM Stockholders' Equity	Non- Controlling Interests	Total Equity
Equity - January 1, 2023	\$ 58,343	\$ 149,825	\$ (169,484)	\$ (16,740)	\$ 21,944	\$ 77	\$ 22,021
Net income plus other comprehensive income/(loss):							
Net income		2,511			2,511		2,511
Other comprehensive income/(loss)				241	241		241
Total comprehensive income/(loss)					\$ 2,751		\$ 2,751
Cash dividends paid — common stock (\$3.31 per share)		(3,007)			(3,007)		(3,007)
Common stock issued under employee plans (5,580,271 shares)	619				619		619
Purchases (1,810,313 shares) and sales (1,144,305 shares) of treasury stock under employee plans — net		(10)	(97)		(107)		(107)
Changes in noncontrolling interests						(7)	(7)
Equity - June 30, 2023	<u>\$ 58,963</u>	<u>\$ 149,318</u>	<u>\$ (169,581)</u>	<u>\$ (16,499)</u>	<u>\$ 22,201</u>	<u>\$ 70</u>	<u>\$ 22,271</u>
(Dollars in millions except per share amounts)	Common Stock and Additional Paid-in Capital	Retained Earnings	Treasury Stock	Accumulated Other Comprehensive Income/(Loss)	Total IBM Stockholders' Equity	Non- Controlling Interests	Total Equity
Equity - January 1, 2022	\$ 57,319	\$ 154,209	\$ (169,392)	\$ (23,234)	\$ 18,901	\$ 95	\$ 18,996
Net income plus other comprehensive income/(loss):							
Net income		2,125			2,125		2,125
Other comprehensive income/(loss)				1,066	1,066		1,066
Total comprehensive income/(loss)					\$ 3,191		\$ 3,191
Cash dividends paid — common stock (\$3.29 per share)		(2,963)			(2,963)		(2,963)
Common stock issued under employee plans (5,960,724 shares)	420				420		420
Purchases (2,319,484 shares) and sales (1,470,514 shares) of treasury stock under employee plans — net		(11)	(130)		(141)		(141)
Other equity	63	(63)			0		0
Changes in noncontrolling interests						(27)	(27)
Equity - June 30, 2022	<u>\$ 57,802</u>	<u>\$ 153,298</u>	<u>\$ (169,522)</u>	<u>\$ (22,169)</u>	<u>\$ 19,409</u>	<u>\$ 67</u>	<u>\$ 19,476</u>

(Amounts may not add due to rounding.)
(The accompanying notes are an integral part of the financial statements.)

Notes to Consolidated Financial Statements

1. Basis of Presentation:

The accompanying Consolidated Financial Statements and footnotes of the International Business Machines Corporation (IBM or the company) have been prepared in accordance with accounting principles generally accepted in the United States of America (GAAP). The financial statements and footnotes are unaudited. In the opinion of the company’s management, these statements include all adjustments, which are only of a normal recurring nature, necessary to present a fair statement of the company’s results of operations, financial position and cash flows.

The preparation of financial statements in conformity with GAAP requires management to make estimates and assumptions that affect the amount of assets, liabilities, revenue, costs, expenses and other comprehensive income/(loss) that are reported in the Consolidated Financial Statements and accompanying disclosures. These estimates are based on management’s best knowledge of current events, historical experience, actions that the company may undertake in the future and on various other assumptions that are believed to be reasonable under the circumstances. As a result, actual results may be different from these estimates.

In the fourth quarter of 2022, the company completed its annual assessment of the useful lives of its property, plant and equipment. Due to advances in technology, the company determined it should increase the estimated useful lives of its server and network equipment from five to six years for new assets and from three to four years for used assets. This change in accounting estimate was effective beginning January 1, 2023. Based on the carrying amount of server and network equipment included in property, plant and equipment-net in the company’s Consolidated Balance Sheet as of December 31, 2022, the effect of this change in estimate was an increase in income from continuing operations before income taxes of \$57 million, or \$0.05 per basic and diluted share for the three months ended June 30, 2023, and \$131 million, or \$0.12 and \$0.11 per basic and diluted share, respectively, for the six months ended June 30, 2023.

For the three and six months ended June 30, 2023, the company reported a provision for income taxes of \$419 million and \$543 million, respectively, and its effective tax rate was 21.0 percent and 17.8 percent, respectively. For the three and six months ended June 30, 2022, the company reported a provision for income taxes of \$257 million and \$218 million, respectively, and its effective tax rate was 14.9 percent and 9.3 percent, respectively. The rates are driven by many factors including the impacts of foreign tax credit regulations, geographical mix of income, incentives and changes in unrecognized tax benefits.

Noncontrolling interest amounts of \$3.9 million and \$5.5 million, net of tax, for the three months ended June 30, 2023 and 2022, respectively, and \$8.4 million and \$10.5 million, net of tax, for the six months ended June 30, 2023 and 2022, respectively, are included as a reduction within other (income) and expense in the Consolidated Income Statement.

The company has supplier finance programs with third-party financial institutions where the company agrees to pay the financial institutions the stated amounts of invoices from participating suppliers on the originally invoiced maturity date, which have an average term of 90 days. The financial institutions offer earlier payment of the invoices at the sole discretion of the supplier for a discounted amount. The company does not provide secured legal assets or other forms of guarantees under the arrangements. The company is not a party to the arrangements between its suppliers and the financial institutions. These obligations are recognized as accounts payable in the Consolidated Balance Sheet. The obligations outstanding under these programs at June 30, 2023 and December 31, 2022 were \$115 million and \$60 million, respectively.

Interim results are not necessarily indicative of financial results for a full year. The information included in this Form 10-Q should be read in conjunction with the company’s 2022 Annual Report.

Within the financial statements and tables presented, certain columns and rows may not add due to the use of rounded numbers for disclosure purposes. Percentages presented are calculated from the underlying whole-dollar amounts. Certain prior-period amounts have been reclassified to conform to the current-period presentation. This is annotated where applicable.

Notes to Consolidated Financial Statements — (continued)

2. Accounting Changes:

Standards Implemented

Disclosures of Supplier Finance Program Obligations

Standard/Description—Issuance date: September 2022. This guidance requires an entity to provide certain interim and annual disclosures about the use of supplier finance programs in connection with the purchase of goods or services.

Effective Date and Adoption Considerations—The guidance was effective January 1, 2023 with certain annual disclosures required beginning in 2024 and early adoption was permitted. The company adopted the guidance as of the effective date.

Effect on Financial Statements or Other Significant Matters—The guidance did not have a material impact in the consolidated financial results. Refer to Note 1, "Basis of Presentation," for additional information.

Troubled Debt Restructurings and Vintage Disclosures

Standard/Description—Issuance date: March 2022. This eliminates the accounting guidance for troubled debt restructurings and requires an entity to apply the general loan modification guidance to all loan modifications, including those made to customers experiencing financial difficulty, to determine whether the modification results in a new loan or a continuation of an existing loan. The guidance also requires presenting current-period gross write-offs by year of origination for financing receivables and net investment in leases.

Effective Date and Adoption Considerations—The amendment was effective January 1, 2023 and early adoption was permitted. The company adopted the guidance on a prospective basis as of the effective date.

Effect on Financial Statements or Other Significant Matters—The guidance did not have a material impact in the consolidated financial results. Refer to note 9, "Financing Receivables," for additional information.

3. Revenue Recognition:

Disaggregation of Revenue

The following tables provide details of revenue by major products/service offerings and revenue by geography.

Revenue by Major Products/Service Offerings

(Dollars in millions)	Three Months Ended June 30,		Six Months Ended June 30,	
	2023	2022	2023	2022
Hybrid Platform & Solutions	\$ 4,665	\$ 4,390	\$ 8,844	\$ 8,470
Transaction Processing	1,943	1,776	3,685	3,468
Total Software	\$ 6,608	\$ 6,166	\$ 12,529	\$ 11,938
Business Transformation	2,295	2,227	4,578	4,482
Application Operations	1,758	1,653	3,494	3,272
Technology Consulting	961	928	1,904	1,884
Total Consulting	\$ 5,013	\$ 4,809	\$ 9,975	\$ 9,637
Hybrid Infrastructure	2,260	2,760	3,969	4,461
Infrastructure Support	1,358	1,474	2,747	2,993
Total Infrastructure	\$ 3,618	\$ 4,235	\$ 6,716	\$ 7,453
Financing*	185	146	380	300
Other	51	180	126	404
Total revenue	\$ 15,475	\$ 15,535	\$ 29,727	\$ 29,732

* Contains lease and loan financing arrangements which are not subject to the guidance on revenue from contracts with customers.

Notes to Consolidated Financial Statements — (continued)

Revenue by Geography

(Dollars in millions)	Three Months Ended June 30,		Six Months Ended June 30,	
	2023	2022	2023	2022
Americas	\$ 8,046	\$ 8,142	\$ 15,124	\$ 15,198
Europe/Middle East/Africa	4,602	4,526	8,933	8,757
Asia Pacific	2,827	2,868	5,670	5,778
Total	\$ 15,475	\$ 15,535	\$ 29,727	\$ 29,732

Remaining Performance Obligations

The remaining performance obligation (RPO) disclosure provides the aggregate amount of the transaction price yet to be recognized as of the end of the reporting period and an explanation as to when the company expects to recognize these amounts in revenue. It is intended to be a statement of overall work under contract that has not yet been performed and does not include contracts in which the customer is not committed, such as certain as-a-Service, governmental, term software license and services offerings. The customer is not considered committed when they are able to terminate for convenience without payment of a substantive penalty. The disclosure includes estimates of variable consideration, except when the variable consideration is a sales-based or usage-based royalty promised in exchange for a license of intellectual property. Additionally, as a practical expedient, the company does not include contracts that have an original duration of one year or less. RPO estimates are subject to change and are affected by several factors, including terminations, changes in the scope of contracts, periodic revalidations, adjustment for revenue that has not materialized and adjustments for currency.

At June 30, 2023, the aggregate amount of the transaction price allocated to RPO related to customer contracts that are unsatisfied or partially unsatisfied was \$58 billion. Approximately 72 percent of the amount is expected to be recognized as revenue in the subsequent two years, approximately 26 percent in the subsequent three to five years and the balance thereafter.

Revenue Recognized for Performance Obligations Satisfied (or Partially Satisfied) in Prior Periods

For the three and six months ended June 30, 2023, revenue was reduced by \$14 million and \$29 million, respectively, for performance obligations satisfied (or partially satisfied) in previous periods mainly due to changes in estimates on contracts with cost-to-cost measures of progress.

Reconciliation of Contract Balances

The following table provides information about notes and accounts receivable—trade, contract assets and deferred income balances.

(Dollars in millions)	At June 30, 2023	At December 31, 2022
Notes and accounts receivable — trade (net of allowances of \$217 in 2023 and \$233 in 2022)	\$ 5,673	\$ 6,541
Contract assets*	\$ 447	\$ 464
Deferred income (current)	\$ 12,712	\$ 12,032
Deferred income (noncurrent)	\$ 3,264	\$ 3,499

* Included within prepaid expenses and other current assets in the Consolidated Balance Sheet.

Notes to Consolidated Financial Statements — (continued)

The amount of revenue recognized during the three and six months ended June 30, 2023 that was included within the deferred income balance at March 31, 2023 and December 31, 2022 was \$4.4 billion and \$6.8 billion, respectively, and was primarily related to services and software.

The following table provides roll forwards of the notes and accounts receivable–trade allowance for expected credit losses for the six months ended June 30, 2023 and the year ended December 31, 2022.

(Dollars in millions)					
January 1, 2023	Additions / (Releases)	Write-offs	Foreign currency and other	June 30, 2023	
\$ 233	\$ 31	\$ (47)	\$ 1	\$	217
January 1, 2022	Additions / (Releases)	Write-offs	Foreign currency and other	December 31, 2022	
\$ 218	\$ 59	\$ (31)	\$ (14)	\$	233

The contract assets allowance for expected credit losses was not material in any of the periods presented.

4. Segments:

The following tables reflect the results of continuing operations of the company’s segments consistent with the management and measurement system utilized within the company. Performance measurement is based on pre-tax income from continuing operations. These results are used by the chief operating decision maker, both in evaluating the performance of, and in allocating resources to, each of the segments.

Notes to Consolidated Financial Statements — (continued)
SEGMENT INFORMATION

(Dollars in millions)	Software	Consulting	Infrastructure	Financing	Total Segments
For the three months ended June 30, 2023:					
Revenue	\$ 6,608	\$ 5,013	\$ 3,618	\$ 185	\$ 15,424
Pre-tax income from continuing operations	\$ 1,504	\$ 446	\$ 633	\$ 64	\$ 2,647
Revenue year-to-year change	7.2 %	4.3 %	(14.6) %	26.2 %	0.4 %
Pre-tax income year-to-year change	9.4 %	30.1 %	(16.4) %	(36.8) %	2.7 %
Pre-tax income margin	22.8 %	8.9 %	17.5 %	34.9 %	17.2 %
For the three months ended June 30, 2022:					
Revenue	\$ 6,166	\$ 4,809	\$ 4,235	\$ 146	\$ 15,355
Pre-tax income from continuing operations	\$ 1,375	\$ 343	\$ 757	\$ 102	\$ 2,577
Pre-tax income margin	22.3 %	7.1 %	17.9 %	69.7 %	16.8 %

Reconciliations to IBM as Reported:

(Dollars in millions)	2023	2022
For the three months ended June 30:		
Revenue:		
Total reportable segments	\$ 15,424	\$ 15,355
Other—divested businesses	(1)	162
Other revenue	52	18
Total revenue from continuing operations	\$ 15,475	\$ 15,535
Pre-tax income from continuing operations:		
Total reportable segments	\$ 2,647	\$ 2,577
Amortization of acquired intangible assets	(389)	(458)
Acquisition-related (charges)/income	(7)	(2)
Non-operating retirement-related (costs)/income	(1)	(192)
Kyndryl-related impacts	—	(145) *
Workforce rebalancing charges**	(117)	—
Other—divested businesses	(1)	160
Unallocated corporate amounts and other	(133)	(219) +
Total pre-tax income/(loss) from continuing operations	\$ 2,000	\$ 1,722

* Unrealized loss on Kyndryl retained shares and related swap. Refer to note 5, "Acquisitions & Divestitures," and note 16, "Derivative Financial Instruments," for additional information.

** Beginning in the first quarter of 2023, the company updated its measure of segment pre-tax income, consistent with its management system, to no longer allocate workforce rebalancing charges to its segments. Workforce rebalancing charges in the second quarter of 2022 of \$3 million were included in the segments.

+ Recast to conform to 2023 presentation.

Notes to Consolidated Financial Statements — (continued)

SEGMENT INFORMATION					
(Dollars in millions)	Software	Consulting	Infrastructure	Financing	Total Segments
For the six months ended June 30, 2023:					
Revenue	\$ 12,529	\$ 9,975	\$ 6,716	\$ 380	\$ 29,601
Pre-tax income from continuing operations	\$ 2,668	\$ 828	\$ 849	\$ 165	\$ 4,510
Revenue year-to-year change	5.0 %	3.5 %	(9.9) %	26.8 %	0.9 %
Pre-tax income year-to-year change	6.3 %	19.8 %	(11.2) %	(11.3) %	3.8 %
Pre-tax income margin	21.3 %	8.3 %	12.6 %	43.3 %	15.2 %
For the six months ended June 30, 2022:					
Revenue	\$ 11,938	\$ 9,637	\$ 7,453	\$ 300	\$ 29,328
Pre-tax income from continuing operations	\$ 2,509	\$ 691	\$ 956	\$ 186	\$ 4,342
Pre-tax income margin	21.0 %	7.2 %	12.8 %	62.0 %	14.8 %

Reconciliations to IBM as Reported:

(Dollars in millions)		
For the six months ended June 30:	2023	2022
Revenue:		
Total reportable segments	\$ 29,601	\$ 29,328
Other—divested businesses	(1)	316
Other revenue	127	88
Total consolidated revenue	\$ 29,727	\$ 29,732
Pre-tax income from continuing operations:		
Total reportable segments	\$ 4,510	\$ 4,342
Amortization of acquired intangible assets	(781)	(919)
Acquisition-related charges	(10)	(9)
Non-operating retirement-related (costs)/income	4	(394)
Kyndryl-related impacts	—	(367) *
Workforce rebalancing charges**	(376)	—
Other—divested businesses	(5)	109
Unallocated corporate amounts	(284)	(418) +
Total pre-tax income/(loss) from continuing operations	\$ 3,058	\$ 2,345

* Unrealized loss on Kyndryl retained shares and related swap. Refer to note 5, "Acquisitions & Divestitures," and note 16, "Derivative Financial Instruments," for additional information.

** Beginning in the first quarter of 2023, the company updated its measure of segment pre-tax income, consistent with its management system, to no longer allocate workforce rebalancing charges to its segments. Workforce rebalancing charges in the first six months of 2022 of \$9 million were included in the segments.

+ Recast to conform to 2023 presentation.

Notes to Consolidated Financial Statements — (continued)

5. Acquisitions & Divestitures:

Acquisitions

Purchase price consideration for all acquisitions was paid primarily in cash. All acquisitions, unless otherwise stated, were for 100 percent of the acquired business and are reported in the Consolidated Statement of Cash Flows, net of acquired cash and cash equivalents.

During the six months ended June 30, 2023, the company completed six acquisitions at an aggregate cost of \$423 million. Each acquisition is expected to enhance the company’s portfolio of products and services capabilities and further advance IBM’s hybrid cloud and AI strategy.

Acquisition	Segment	Description of Acquired Business
<u>First Quarter</u>		
StepZen, Inc.	Software	Developer of GraphQL to help build application programming interfaces (APIs)
Asset Strategy Library (ASL) Portfolio of Uptake Technologies	Software	Library of industrial asset management data
NS1	Software	Leading provider of network automation SaaS solutions
<u>Second Quarter</u>		
Ahana Cloud, Inc.	Software	Expert in open-source-based solutions for data analytics
Polar Security	Software	Innovator in technology that helps companies discover, continuously monitor and secure cloud and SaaS application data
Agyla SAS	Consulting	Leading provider of cloud platform engineering services in France specializing in Cloud, DevOps and Security

At June 30, 2023, the remaining cash to be remitted by the company related to certain first-half 2023 acquisitions was \$42 million, most of which is expected to be paid in the first half of 2024. The unremitted cash associated with these acquisitions is primarily a non-cash financing activity for purposes of the company's Consolidated Statement of Cash Flows as of June 30, 2023.

The following table reflects the purchase price related to these acquisitions and the resulting purchase price allocations as of June 30, 2023.

(Dollars in millions)	Amortization Life (in years)	Total Acquisitions
Current assets		\$ 46
Property, plant and equipment/noncurrent assets		7
Intangible assets:		
Goodwill	N/A	301
Client relationships	7	37
Completed technology	5-7	66
Trademarks	2-5	2
Total assets acquired		\$ 458
Current liabilities		26
Noncurrent liabilities		10
Total liabilities assumed		\$ 36
Total purchase price		\$ 423

N/A – not applicable

Notes to Consolidated Financial Statements — (continued)

Goodwill of \$266 million, \$23 million and \$12 million was assigned to the Software, Consulting and Infrastructure segments, respectively, and is primarily attributable to the assembled workforce of the acquired businesses and the increased synergies expected to be achieved from the integration of the acquired businesses into the company’s various integrated solutions and services, neither of which qualifies as an amortizable intangible asset. It is expected that none of the goodwill will be deductible for tax purposes.

The overall weighted-average useful life of the identified amortizable intangible assets acquired was 6.8 years. The identified intangible assets will be amortized on a straight-line basis over their useful lives, which approximates the pattern that the assets economic benefits are expected to be consumed over time.

The valuation of the assets acquired and liabilities assumed is subject to revision. If additional information becomes available, the company may further revise the purchase price allocation as soon as practical, but no later than one year from the acquisition date; however, material changes are not expected.

Transactions Announced – The company signed a definitive agreement in June 2023 to acquire Apptio Inc. (Apptio), a leader in financial and operational IT management and optimization software, for estimated cash consideration of \$4.6 billion. The acquisition of Apptio will accelerate the advancement of IBM’s IT automation capabilities and enable enterprise leaders to deliver enhanced business value across technology investments. The transaction is expected to close in the second half of 2023, subject to customary closing conditions, including regulatory clearance. Upon closing, Apptio will be integrated into the Software segment.

Divestitures

Separation of Kyndryl — On November 3, 2021, the company completed the separation of its managed infrastructure services unit into a new public company with the distribution of 80.1 percent of the outstanding common stock of Kyndryl Holdings, Inc. (Kyndryl) to IBM stockholders on a pro rata basis. The company retained 19.9 percent of the shares of Kyndryl common stock immediately following the separation. During 2022, the company fully disposed of its retained interest in Kyndryl common stock pursuant to exchange agreements with a third-party financial institution, which were completed within twelve months of separation. As of November 2, 2022, the company no longer held an ownership interest in Kyndryl.

Income/(loss) from discontinued operations, net of tax for the three and six months ended June 30, 2023 of \$2 million and \$(4) million, respectively, reflects the net impact of changes in separation-related estimates and the settlement of assets and liabilities in accordance with the separation and distribution agreement. Income/(loss) from discontinued operations, net of tax for the three and six months ended June 30, 2022 of \$(73) million and \$(2) million, respectively, reflects the same drivers as above and also reflects a gain on sale of a joint venture historically managed by Kyndryl, which was sold to Kyndryl in the first quarter of 2022 upon receiving regulatory approval.

Other — The company completed two divestitures in the second quarter of 2023. The financial terms related to these transactions were not material.

Notes to Consolidated Financial Statements — (continued)

6. Other (Income) and Expense:

Components of other (income) and expense are as follows:

(Dollars in millions)	Three Months Ended June 30,		Six Months Ended June 30,	
	2023	2022	2023	2022
Other (income) and expense:				
Foreign currency transaction losses/(gains)*	\$ (166)	\$ (494)	\$ (78)	\$ (670)
(Gains)/losses on derivative instruments**	141	439	(1)	541
Interest income	(201)	(28)	(371)	(46)
Net (gains)/losses from securities and investment assets ⁺	3	54	8	273
Retirement-related costs/(income)	1	192	(4)	394
Other ⁺⁺	(39)	(243)	(61)	(327)
Total other (income) and expense	<u>\$ (261)</u>	<u>\$ (81)</u>	<u>\$ (506)</u>	<u>\$ 166</u>

* The company uses financial hedging instruments to limit specific currency risks related to foreign currency-based transactions. The hedging program does not hedge 100 percent of currency exposures and defers, versus eliminates, the impact of currency. Refer to note 16, "Derivative Financial Instruments," for additional information on foreign exchange risk.

** Prior year amounts include an unrealized loss on the cash-settled swap related to the Kyndryl retained shares of \$88 million recognized in the second quarter of 2022. Refer to note 16, "Derivative Financial Instruments," for additional information.

+ Prior year amounts include an unrealized loss on Kyndryl retained shares of \$56 million and \$278 million for the three and six months ended June 30, 2022, respectively. Refer to note 5, "Acquisitions & Divestitures," for additional information.

++ Other primarily consists of (gains)/losses from divestitures and dispositions of land/buildings. Prior year amounts include a pre-tax gain of \$232 million recognized in the second quarter of 2022 related to the divestiture of IBM's healthcare software assets.

Notes to Consolidated Financial Statements — (continued)

7. Earnings/(Loss) Per Share of Common Stock:

The following tables provide the computation of basic and diluted earnings per share of common stock for the three and six months ended June 30, 2023 and 2022.

(Dollars in millions except per share amounts)		
For the three months ended June 30:		
	2023	2022
Number of shares on which basic earnings per share is calculated:		
Weighted-average shares outstanding during period	909,855,943	901,470,793
Add — Incremental shares under stock-based compensation plans	7,584,035	7,518,749
Add — Incremental shares associated with contingently issuable shares	2,012,519	1,760,192
Number of shares on which diluted earnings per share is calculated	919,452,496	910,749,734
Income from continuing operations		
	\$ 1,581	\$ 1,465
Income/(loss) from discontinued operations, net of tax	2	(73)
Net income on which basic earnings per share is calculated	\$ 1,583	\$ 1,392
Income from continuing operations		
	\$ 1,581	\$ 1,465
Net income applicable to contingently issuable shares	—	—
Income from continuing operations on which diluted earnings per share is calculated	\$ 1,581	\$ 1,465
Income/(loss) from discontinued operations, net of tax, on which diluted earnings per share is calculated	2	(73)
Net income on which diluted earnings per share is calculated	\$ 1,583	\$ 1,392
Earnings/(loss) per share of common stock:		
Assuming dilution		
Continuing operations	\$ 1.72	\$ 1.61
Discontinued operations	0.00	(0.08)
Total	\$ 1.72	\$ 1.53
Basic		
Continuing operations	\$ 1.74	\$ 1.62
Discontinued operations	0.00	(0.08)
Total	\$ 1.74	\$ 1.54

Stock options to purchase 5,541,485 shares and 788,500 shares were outstanding as of June 30, 2023 and 2022, respectively, but were not included in the computation of diluted earnings per share because the exercise price of the options during the respective period was greater than the average market price of the common shares, and therefore, the effect would have been antidilutive.

Notes to Consolidated Financial Statements — (continued)

(Dollars in millions except per share amounts)

For the six months ended June 30:	2023	2022
Number of shares on which basic earnings per share is calculated:		
Weighted-average shares outstanding during period	908,691,415	900,393,410
Add — Incremental shares under stock-based compensation plans	8,096,637	7,946,998
Add — Incremental shares associated with contingently issuable shares	1,860,836	1,647,528
Number of shares on which diluted earnings per share is calculated	918,648,888	909,987,935
Income from continuing operations	\$ 2,515	\$ 2,127
Income/(loss) from discontinued operations, net of tax	(4)	(2)
Net income on which basic earnings per share is calculated	\$ 2,511	\$ 2,125
Income from continuing operations	\$ 2,515	\$ 2,127
Net income applicable to contingently issuable shares	—	—
Income from continuing operations on which diluted earnings per share is calculated	\$ 2,515	\$ 2,127
Income/(loss) from discontinued operations, net of tax, on which diluted earnings per share is calculated	(4)	(2)
Net income on which diluted earnings per share is calculated	\$ 2,511	\$ 2,125
Earnings/(loss) per share of common stock:		
Assuming dilution		
Continuing operations	\$ 2.74	\$ 2.34
Discontinued operations	0.00	0.00
Total	\$ 2.73	\$ 2.34
Basic		
Continuing operations	\$ 2.77	\$ 2.36
Discontinued operations	0.00	0.00
Total	\$ 2.76	\$ 2.36

Stock options to purchase 3,251,207 shares and 975,911 shares (average of first and second quarter share amounts) were outstanding as of June 30, 2023 and 2022, respectively, but were not included in the computation of diluted earnings per share because the exercise price of the options during the respective period was greater than the average market price of the common shares, and therefore, the effect would have been antidilutive.

8. Financial Assets & Liabilities:

Fair Value Measurements

Fair value is defined as the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date. The company classifies certain assets and liabilities based on the following fair value hierarchy:

- Level 1—Quoted prices (unadjusted) in active markets for identical assets or liabilities that can be accessed at the measurement date;
- Level 2—Inputs other than quoted prices included within Level 1 that are observable for the asset or liability, either directly or indirectly; and
- Level 3—Unobservable inputs for the asset or liability.

Notes to Consolidated Financial Statements — (continued)

When available, the company uses unadjusted quoted market prices in active markets to measure the fair value and classifies such items as Level 1. If quoted market prices are not available, fair value is based upon internally developed models that use current market-based or independently sourced market parameters such as interest rates and currency rates. Items valued using internally generated models are classified according to the lowest level input or value driver that is significant to the valuation.

The determination of fair value considers various factors including interest rate yield curves and time value underlying the financial instruments. For derivatives and debt securities, the company uses a discounted cash flow analysis using discount rates commensurate with the duration of the instrument.

In determining the fair value of financial instruments, the company considers certain market valuation adjustments to the “base valuations” calculated using the methodologies described below for several parameters that market participants would consider in determining fair value:

- Counterparty credit risk adjustments are applied to financial instruments, taking into account the actual credit risk of a counterparty as observed in the credit default swap market to determine the true fair value of such an instrument.
- Credit risk adjustments are applied to reflect the company’s own credit risk when valuing all liabilities measured at fair value. The methodology is consistent with that applied in developing counterparty credit risk adjustments, but incorporates the company’s own credit risk as observed in the credit default swap market.

The company holds investments primarily in time deposits, certificates of deposit, and U.S. government debt that are designated as available-for-sale. The primary objective of the company’s cash and debt investment portfolio is to protect principal by investing in very liquid investment securities with highly rated counterparties.

The company’s standard practice is to hold all of its debt security investments classified as available-for-sale until maturity. No impairments for credit losses and no material non-credit impairments were recorded for the three and six months ended June 30, 2023.

Certain non-financial assets such as property, plant and equipment, operating right-of-use assets, land, goodwill and intangible assets are also subject to nonrecurring fair value measurements if they are deemed to be impaired. The impairment models used for non-financial assets depend on the type of asset. There were no material impairments of non-financial assets for the three and six months ended June 30, 2023 and 2022, respectively.

Notes to Consolidated Financial Statements — (continued)

The following table presents the company’s financial assets and financial liabilities that are measured at fair value on a recurring basis at June 30, 2023 and December 31, 2022.

(Dollars in millions)	Fair Value Hierarchy Level	At June 30, 2023		At December 31, 2022	
		Assets ⁽⁵⁾	Liabilities ⁽⁶⁾	Assets ⁽⁵⁾	Liabilities ⁽⁶⁾
Cash equivalents: ⁽¹⁾					
Time deposits and certificates of deposit ⁽²⁾	2	\$ 4,383	N/A	\$ 3,712	N/A
Money market funds	1	399	N/A	306	N/A
Total cash equivalents		\$ 4,782	N/A	\$ 4,018	N/A
Debt securities-current ⁽²⁾⁽³⁾	2	6,904	N/A	852	N/A
Debt securities-noncurrent ⁽²⁾⁽⁴⁾	2,3	33	N/A	31	N/A
Derivatives designated as hedging instruments:					
Interest rate contracts	2	0	400	3	336
Foreign exchange contracts	2	342	374	184	674
Derivatives not designated as hedging instruments:					
Foreign exchange contracts	2	14	47	42	16
Equity contracts	2	21	4	49	8
Total		\$ 12,096	\$ 824	\$ 5,179	\$ 1,034

(1) Included within cash and cash equivalents in the Consolidated Balance Sheet.
(2) Available-for-sale debt securities with carrying values that approximate fair value.
(3) U.S. treasury bills and term deposits that are reported within marketable securities in the Consolidated Balance Sheet. The June 30, 2023 balance includes proceeds from the first quarter 2023 debt issuances. See note 12, "Borrowings," for additional information.
(4) Includes immaterial activity related to private company investments reported within investments and sundry assets in the Consolidated Balance Sheet.
(5) The gross balances of derivative assets contained within prepaid expenses and other current assets, and investments and sundry assets in the Consolidated Balance Sheet at June 30, 2023 were \$369 million and \$8 million, respectively, and at December 31, 2022 were \$271 million and \$7 million, respectively.
(6) The gross balances of derivative liabilities contained within other accrued expenses and liabilities, and other liabilities in the Consolidated Balance Sheet at June 30, 2023 were \$274 million and \$551 million, respectively, and at December 31, 2022 were \$546 million and \$488 million, respectively.

N/A – not applicable

Financial Assets and Liabilities Not Measured at Fair Value

Short-Term Receivables and Payables

Short-term receivables (excluding the current portion of long-term receivables) and other investments are financial assets with carrying values that approximate fair value. Accounts payable, other accrued expenses and short-term debt (excluding the current portion of long-term debt) are financial liabilities with carrying values that approximate fair value. If measured at fair value in the financial statements, these financial instruments would be classified as Level 3 in the fair value hierarchy, except for short-term debt which would be classified as Level 2.

Loans and Long-Term Receivables

Fair values are based on discounted future cash flows using current interest rates offered for similar loans to clients with similar credit ratings for the same remaining maturities. At June 30, 2023 and December 31, 2022, the difference between the carrying amount and estimated fair value for loans and long-term receivables was immaterial. If measured at fair value in the financial statements, these financial instruments would be classified as Level 3 in the fair value hierarchy.

Notes to Consolidated Financial Statements — (continued)

Long-Term Debt

Fair value of publicly traded long-term debt is based on quoted market prices for the identical liability when traded as an asset in an active market. For other long-term debt (including long-term finance lease liabilities) for which a quoted market price is not available, an expected present value technique that uses rates currently available to the company for debt with similar terms and remaining maturities is used to estimate fair value. The carrying amount of long-term debt was \$50,691 million and \$46,189 million, and the estimated fair value was \$47,250 million and \$42,514 million at June 30, 2023 and December 31, 2022, respectively. If measured at fair value in the financial statements, long-term debt (including the current portion) would be classified as Level 2 in the fair value hierarchy.

9. Financing Receivables:

Financing receivables primarily consist of client loan and installment payment receivables (loans), investment in sales-type and direct financing leases (collectively referred to as client financing receivables) and commercial financing receivables. Loans are provided primarily to clients to finance the purchase of hardware, software and services. Payment terms on these financing arrangements are for terms up to seven years. Investment in sales-type and direct financing leases relate principally to the company’s Infrastructure products and are for terms ranging generally from two to six years. Commercial financing receivables, which consist of both held-for-investment and held-for-sale receivables, relate primarily to working capital financing for dealers and remarketers of IBM products. Payment terms for working capital financing generally range from 30 to 90 days.

A summary of the components of the company’s financing receivables is presented as follows:

(Dollars in millions)	Client Financing Receivables		Commercial Financing Receivables		Total
	Client Loan and Installment Payment Receivables	Investment in Sales-Type and Direct Financing	Held for Investment	Held for Sale*	
	(Loans)	Leases			
At June 30, 2023					
Financing receivables, gross	\$ 7,262	\$ 3,853	\$ 251	\$ 865	\$ 12,230
Unearned income	(436)	(377)	—	—	(813)
Unguaranteed residual value	—	404	—	—	404
Amortized cost	\$ 6,826	\$ 3,880	\$ 251	\$ 865	\$ 11,821
Allowance for credit losses	(104)	(62)	(5)	—	(171)
Total financing receivables, net	\$ 6,722	\$ 3,818	\$ 245	\$ 865	\$ 11,650
Current portion	\$ 3,925	\$ 1,393	\$ 245	\$ 865	\$ 6,429
Noncurrent portion	\$ 2,796	\$ 2,425	\$ —	\$ —	\$ 5,221
(Dollars in millions)	Client Financing Receivables		Commercial Financing Receivables		Total
	Client Loan and Installment Payment Receivables	Investment in Sales-Type and Direct Financing	Held for Investment	Held for Sale*	
	(Loans)	Leases			
At December 31, 2022					
Financing receivables, gross	\$ 8,875	\$ 4,023	\$ 299	\$ 939	\$ 14,136
Unearned income	(439)	(351)	—	—	(790)
Unguaranteed residual value	—	422	—	—	422
Amortized cost	\$ 8,437	\$ 4,094	\$ 299	\$ 939	\$ 13,769
Allowance for credit losses	(108)	(60)	(5)	—	(173)
Total financing receivables, net	\$ 8,329	\$ 4,034	\$ 293	\$ 939	\$ 13,596
Current portion	\$ 5,073	\$ 1,485	\$ 293	\$ 939	\$ 7,790
Noncurrent portion	\$ 3,256	\$ 2,549	\$ —	\$ —	\$ 5,806

* The carrying value of the receivables classified as held for sale approximates fair value.

Notes to Consolidated Financial Statements — (continued)

The company has a long-standing practice of taking mitigation actions, in certain circumstances, to transfer credit risk to third parties. These actions may include credit insurance, financial guarantees, nonrecourse secured borrowings, transfers of receivables recorded as true sales in accordance with accounting guidance or sales of equipment under operating lease. Sale of receivables arrangements are also utilized in the normal course of business as part of the company’s cash and liquidity management.

Financing receivables pledged as collateral for secured borrowings were \$251 million and \$349 million at June 30, 2023 and December 31, 2022, respectively. These borrowings are included in note 12, “Borrowings.”

Transfer of Financial Assets

The company has an existing agreement with a third-party investor to sell IBM short-term commercial financing receivables on a revolving basis. In addition, the company enters into agreements with third-party financial institutions to sell certain of its client financing receivables, including both loan and lease receivables, for cash proceeds. There were no material client financing receivables transferred for the six months ended June 30, 2023 and 2022.

The following table presents the total amount of commercial financing receivables transferred.

(Dollars in millions)				
For the six months ended June 30:				
	2023		2022	
Commercial financing receivables:				
Receivables transferred during the period	\$	4,345	\$	3,914
Receivables uncollected at end of period*	\$	928	\$	815

* Of the total amount of commercial financing receivables sold and derecognized from the Consolidated Balance Sheet, the amounts presented remained uncollected from business partners as of June 30, 2023 and 2022.

The transfer of these receivables qualified as true sales and therefore reduced financing receivables. The cash proceeds from the sales are included in cash flows from operating activities. For the six months ended June 30, 2023 and 2022, the net loss, including fees, associated with the transfer of commercial financial receivables was \$45 million and \$22 million, respectively, and is included in other (income) and expense in the Consolidated Income Statement.

Notes to Consolidated Financial Statements — (continued)

Financing Receivables by Portfolio Segment

The following tables present the amortized cost basis for client financing receivables at June 30, 2023 and December 31, 2022, further segmented by three classes: Americas, Europe/Middle East/Africa (EMEA) and Asia Pacific. The commercial financing receivables portfolio segment is excluded from the tables in the sections below as the receivables are short term in nature and the current estimated risk of loss and resulting impact to the company’s financial results are not material.

(Dollars in millions)

At June 30, 2023:	Americas	EMEA	Asia Pacific	Total
Amortized cost	\$ 6,317	\$ 3,019	\$ 1,370	\$ 10,705
Allowance for credit losses:				
Beginning balance at January 1, 2023	\$ 88	\$ 60	\$ 20	\$ 168
Write-offs	\$ (3)	\$ 0	\$ 0	\$ (4)
Recoveries	—	0	0	1
Additions/(releases)	6	(12)	0	(6)
Other*	7	\$ 0	(1)	7
Ending balance at June 30, 2023	<u>\$ 98</u>	<u>\$ 48</u>	<u>\$ 19</u>	<u>\$ 165</u>

(Dollars in millions)

At December 31, 2022:	Americas	EMEA	Asia Pacific	Total
Amortized cost	\$ 7,281	\$ 3,546	\$ 1,704	\$ 12,531
Allowance for credit losses:				
Beginning balance at January 1, 2022	\$ 111	\$ 61	\$ 23	\$ 195
Write-offs	\$ (20)	\$ (3)	\$ (2)	\$ (25)
Recoveries	1	0	4	5
Additions/(releases)	(5)	6	(4)	(3)
Other*	2	(5)	(2)	(4)
Ending balance at December 31, 2022	<u>\$ 88</u>	<u>\$ 60</u>	<u>\$ 20</u>	<u>\$ 168</u>

* Primarily represents translation adjustments.

When determining the allowances, financing receivables are evaluated either on an individual or a collective basis. For the company’s policy on determining allowances for credit losses, refer to note A, “Significant Accounting Policies,” in the company’s 2022 Annual Report.

Notes to Consolidated Financial Statements — (continued)

Past Due Financing Receivables

The company summarizes information about the amortized cost basis for client financing receivables, including amortized cost aged over 90 days and still accruing, billed invoices aged over 90 days and still accruing, and amortized cost not accruing.

(Dollars in millions)	Total Amortized Cost	Amortized Cost > 90 Days*	Amortized Cost > 90 Days and Accruing*	Billed Invoices > 90 Days and Accruing	Amortized Cost Not Accruing**
At June 30, 2023:					
Americas	\$ 6,317	\$ 147	\$ 72	\$ 6	\$ 77
EMEA	3,019	35	4	1	31
Asia Pacific	1,370	17	2	1	16
Total client financing receivables	<u>\$ 10,705</u>	<u>\$ 200</u>	<u>\$ 78</u>	<u>\$ 8</u>	<u>\$ 124</u>

(Dollars in millions)	Total Amortized Cost	Amortized Cost > 90 Days*	Amortized Cost > 90 Days and Accruing*	Billed Invoices > 90 Days and Accruing	Amortized Cost Not Accruing**
At December 31, 2022:					
Americas	\$ 7,281	\$ 272	\$ 198	\$ 22	\$ 74
EMEA	3,546	52	8	1	46
Asia Pacific	1,704	20	3	1	17
Total client financing receivables	<u>\$ 12,531</u>	<u>\$ 344</u>	<u>\$ 208</u>	<u>\$ 23</u>	<u>\$ 137</u>

* At a contract level, which includes total billed and unbilled amounts for financing receivables aged greater than 90 days.
**Of the amortized cost not accruing, there was a related allowance of \$121 million and \$122 million at June 30, 2023 and December 31, 2022, respectively. Financing income recognized on these receivables was immaterial for the three and six months ended June 30, 2023, respectively.

Credit Quality Indicators

The company’s credit quality indicators, which are based on rating agency data, publicly available information and information provided by customers, are reviewed periodically based on the relative level of risk. The resulting indicators are a numerical rating system that maps to Moody’s Investors Service credit ratings as shown below. The company uses information provided by Moody’s, where available, as one of many inputs in its determination of customer credit ratings. The credit quality of the customer is evaluated based on these indicators and is assigned the same risk rating whether the receivable is a lease or a loan.

The following tables present the amortized cost basis for client financing receivables by credit quality indicator at June 30, 2023 and December 31, 2022, respectively. Receivables with a credit quality indicator ranging from Aaa to Baa3 are considered investment grade. All others are considered non-investment grade. The credit quality indicators reflect mitigating credit enhancement actions taken by customers which reduce the risk to IBM. Gross write-offs by vintage year at June 30, 2023 were not material.

(Dollars in millions)	Americas		EMEA		Asia Pacific	
At June 30, 2023:	Aaa – Baa3	Ba1 – D	Aaa – Baa3	Ba1 – D	Aaa – Baa3	Ba1 – D
Vintage year:						
2023	\$ 1,011	\$ 466	\$ 438	\$ 307	\$ 248	\$ 53
2022	2,363	507	895	498	476	60
2021	887	231	336	110	146	48
2020	337	161	145	99	140	30
2019	179	46	79	56	75	13
2018 and prior	63	65	18	37	57	24
Total	<u>\$ 4,841</u>	<u>\$ 1,476</u>	<u>\$ 1,911</u>	<u>\$ 1,108</u>	<u>\$ 1,142</u>	<u>\$ 227</u>

Notes to Consolidated Financial Statements — (continued)

(Dollars in millions) At December 31, 2022:	Americas		EMEA		Asia Pacific	
	Aaa – Baa3	Ba1 – D	Aaa – Baa3	Ba1 – D	Aaa – Baa3	Ba1 – D
Vintage year:						
2022	\$ 3,316	\$ 1,097	\$ 1,447	\$ 704	\$ 799	\$ 96
2021	1,197	323	451	159	203	65
2020	559	217	258	158	210	49
2019	251	91	161	99	127	22
2018	128	26	42	16	84	21
2017 and prior	32	45	14	38	12	17
Total	\$ 5,482	\$ 1,800	\$ 2,373	\$ 1,173	\$ 1,434	\$ 269

Modifications and Troubled Debt Restructurings

The company did not have any significant modifications due to financial difficulty during the six months ended June 30, 2023. The company did not have any significant troubled debt restructurings during the year ended December 31, 2022.

10. Leases:

Accounting for Leases as a Lessor

The following table presents amounts included in the Consolidated Income Statement related to lessor activity.

(Dollars in millions)	Three Months Ended June 30,		Six Months Ended June 30,	
	2023	2022	2023	2022
Lease income — sales-type and direct financing leases:				
Sales-type lease selling price	\$ 248	\$ 735	\$ 338	\$ 789
Less: Carrying value of underlying assets*	(61)	(120)	(91)	(139)
Gross profit	\$ 187	\$ 615	\$ 247	\$ 651
Interest income on lease receivables	59	45	118	90
Total sales-type and direct financing lease income	\$ 246	\$ 660	\$ 365	\$ 741
Lease income — operating leases	25	27	51	56
Variable lease income	14	28	35	56
Total lease income	\$ 284	\$ 715	\$ 451	\$ 853

* Excludes unguaranteed residual value.

Sales-type lease revenue was \$284 million and \$451 million for the three and six months ended June 30, 2023, respectively, compared to \$715 million and \$853 million for the three and six months ended June 30, 2022, respectively. The decreases in both the three and six month periods were predominantly due to the zSystems product cycle dynamics.

Notes to Consolidated Financial Statements — (continued)

11. Intangible Assets Including Goodwill:

Intangible Assets

The following tables present the company's intangible asset balances by major asset class.

(Dollars in millions)	At June 30, 2023		
	Gross Carrying Amount	Accumulated Amortization	Net Carrying Amount*
Intangible asset class:			
Capitalized software	\$ 1,622	\$ (707)	\$ 915
Client relationships	8,274	(3,056)	5,217
Completed technology	5,131	(2,174)	2,957
Patents/trademarks	1,781	(378)	1,403
Other**	19	(16)	3
Total	\$ 16,827	\$ (6,331)	\$ 10,496

(Dollars in millions)	At December 31, 2022		
	Gross Carrying Amount	Accumulated Amortization	Net Carrying Amount*
Intangible asset class:			
Capitalized software	\$ 1,650	\$ (705)	\$ 945
Client relationships	8,559	(2,951)	5,608
Completed technology	5,220	(2,045)	3,175
Patents/trademarks	2,140	(688)	1,452
Other**	19	(15)	4
Total	\$ 17,588	\$ (6,404)	\$ 11,184

* Amounts as of June 30, 2023 and December 31, 2022 include an increase in net intangible asset balances of \$24 million and a decrease in net intangible asset balances of \$198 million, respectively, due to foreign currency translation.

** Other intangibles are primarily acquired proprietary and non-proprietary business processes, methodologies and systems.

The net carrying amount of intangible assets decreased \$688 million during the first six months of 2023, primarily due to intangible asset amortization, partially offset by additions of capitalized software and acquired intangibles. The aggregate intangible asset amortization expense was \$556 million and \$1,104 million for the second quarter and first six months of 2023, respectively, compared to \$625 million and \$1,251 million for the second quarter and first six months of 2022, respectively. In the first six months of 2023, the company retired \$1,189 million of fully amortized intangible assets, impacting both the gross carrying amount and accumulated amortization by this amount.

The future amortization expense relating to intangible assets currently recorded in the Consolidated Balance Sheet was estimated to be the following at June 30, 2023:

(Dollars in millions)	Capitalized Software	Acquired Intangibles	Total
Remainder of 2023	\$ 303	\$ 786	\$ 1,089
2024	410	1,557	1,967
2025	175	1,538	1,713
2026	27	1,515	1,542
2027	—	1,496	1,496
Thereafter	—	2,688	2,688

Notes to Consolidated Financial Statements — (continued)

Goodwill

The changes in the goodwill balances by segment for the six months ended June 30, 2023 and for the year ended December 31, 2022 were as follows:

(Dollars in millions) Segment	Balance 1/1/2023	Goodwill Additions	Purchase Price Adjustments	Divestitures	Foreign Currency Translation and Other Adjustments*	Balance 6/30/2023
Software	\$ 43,657	\$ 267	\$ (7)	\$ —	\$ 88	\$ 44,005
Consulting	7,928	23	6	—	43	8,000
Infrastructure	4,363	12	—	—	4	4,380
Other	—	—	—	—	—	—
Total	<u>\$ 55,949</u>	<u>\$ 302</u>	<u>\$ (1)</u>	<u>\$ —</u>	<u>\$ 136</u>	<u>\$ 56,385</u>

(Dollars in millions) Segment	Balance 1/1/2022	Goodwill Additions	Purchase Price Adjustments	Divestitures	Foreign Currency Translation and Other Adjustments*	Balance 12/31/2022
Software	\$ 43,966	\$ 568	\$ (118)	\$ —	\$ (760)	\$ 43,657
Consulting	6,797	1,366	(42)	—	(192)	7,928
Infrastructure	4,396	—	—	(1)	(32)	4,363
Other**	484	—	—	(484)	—	—
Total	<u>\$ 55,643</u>	<u>\$ 1,934</u>	<u>\$ (159)</u>	<u>\$ (485)</u>	<u>\$ (984)</u>	<u>\$ 55,949</u>

* Primarily driven by foreign currency translation.

** The company derecognized goodwill related to the divestiture of its healthcare software assets in the second quarter of 2022.

There were no goodwill impairment losses recorded during the first six months of 2023 or full-year 2022 and the company has no accumulated impairment losses. Purchase price adjustments recorded in the first six months of 2023 and full-year 2022 were related to acquisitions that were still subject to the measurement period that ends at the earlier of 12 months from the acquisition date or when information becomes available. Net purchase price adjustments recorded in the first six months of 2023 were not material. Net purchase price adjustments recorded in 2022 primarily related to deferred tax assets and liabilities associated with the Turbonomic acquisition.

12. Borrowings:

Short-Term Debt

(Dollars in millions)	At June 30, 2023	At December 31, 2022
Short-term loans	\$ 5	\$ 8
Long-term debt — current maturities	6,780	4,751
Total	<u>\$ 6,785</u>	<u>\$ 4,760</u>

The weighted-average interest rate for short-term loans was 8.6 percent and 7.6 percent at June 30, 2023 and December 31, 2022, respectively.

Notes to Consolidated Financial Statements — (continued)

Long-Term Debt

Pre-Swap Borrowing

(Dollars in millions)	Maturities	Balance 6/30/2023	Balance 12/31/2022
U.S. dollar debt (weighted-average interest rate at June 30, 2023):*			
3.4%	2023	\$ 1,508	\$ 1,529
3.3%	2024	5,005	5,009
5.1%	2025	1,602	1,603
3.5%	2026	5,201	4,351
3.1%	2027	3,620	3,620
5.0%	2028	1,313	313
3.5%	2029	3,250	3,250
2.0%	2030	1,350	1,350
4.4%	2032	1,850	1,850
4.8%	2033	750	—
8.0%	2038	83	83
4.5%	2039	2,745	2,745
2.9%	2040	650	650
4.0%	2042	1,107	1,107
7.0%	2045	27	27
4.7%	2046	650	650
4.3%	2049	3,000	3,000
3.0%	2050	750	750
4.2%	2052	1,400	1,400
5.1%	2053	650	—
7.1%	2096	316	316
		\$ 36,828	\$ 33,605
Other currencies (weighted-average interest rate at June 30, 2023, in parentheses):*			
Euro (1.8%)	2024–2043	\$ 19,095	\$ 17,087
Pound sterling (4.9%)	2038	954	—
Japanese yen (0.5%)	2024–2028	1,221	694
Other (16.1%)	2023–2026	272	361
		\$ 58,369	\$ 51,747
Finance lease obligations (3.9%)	2023–2030	268	239
		\$ 58,637	\$ 51,986
Less: net unamortized discount		859	835
Less: net unamortized debt issuance costs		164	138
Add: fair value adjustment**		(144)	(73)
		\$ 57,471	\$ 50,940
Less: current maturities		6,780	4,751
Total		\$ 50,691	\$ 46,189

* Includes notes, debentures, bank loans and secured borrowings.

** The portion of the company’s fixed-rate debt obligations that is hedged is reflected in the Consolidated Balance Sheet as an amount equal to the sum of the debt’s carrying value and a fair value adjustment representing changes in the fair value of the hedged debt obligations attributable to movements in benchmark interest rates.

Notes to Consolidated Financial Statements — (continued)

The company’s indenture governing its debt securities and its various credit facilities each contain significant covenants which obligate the company to promptly pay principal and interest, limit the aggregate amount of secured indebtedness and sale and leaseback transactions to 10 percent of the company’s consolidated net tangible assets, and restrict the company’s ability to merge or consolidate unless certain conditions are met. The credit facilities also include a covenant on the company’s consolidated net interest expense ratio, which cannot be less than 2.20 to 1.0, as well as a cross default provision with respect to other defaulted indebtedness of at least \$500 million.

The company is in compliance with its debt covenants and provides periodic certifications to its lenders. The failure to comply with its debt covenants could constitute an event of default with respect to the debt to which such provisions apply. If certain events of default were to occur, the principal and interest on the debt to which such event of default applied would become immediately due and payable.

In the first quarter of 2023, the company issued \$0.7 billion of Japanese yen floating-rate syndicated bank loans with a maturity of 5 years; \$4.6 billion of Euro fixed-rate notes in tranches with maturities ranging from 4 to 20 years and coupons ranging from 3.375 percent to 4 percent; \$0.9 billion of Pound sterling fixed-rate notes with a maturity of 15 years and a coupon of 4.875 percent; and \$3.25 billion of U.S. dollar fixed-rate notes in tranches with maturities ranging from 3 to 30 years and coupons ranging from 4.5 to 5.1 percent.

Pre-swap annual contractual obligations of long-term debt outstanding at June 30, 2023, were as follows:

(Dollars in millions)	Total	
Remainder of 2023	\$	1,668
2024		6,364
2025		4,962
2026		5,534
2027		5,829
Thereafter		34,280
Total	\$	58,637

Interest on Debt

(Dollars in millions)		
For the six months ended June 30:	2023	2022
Cost of financing	\$ 173	\$ 165
Interest expense	790	607
Interest capitalized	6	2
Total interest paid and accrued	\$ 969	\$ 775

Lines of Credit

On June 15, 2023, the company amended its existing \$2.5 billion Three-Year Credit Agreement and \$7.5 billion Five-Year Credit Agreement (the Credit Agreements) to extend the maturity dates to June 20, 2026 and June 22, 2028, respectively. The Credit Agreements permit the company and its subsidiary borrowers to borrow up to \$10 billion on a revolving basis. At June 30, 2023, there were no borrowings by the company, or its subsidiaries, under these credit facilities.

Notes to Consolidated Financial Statements — (continued)

13. Commitments:

The company’s extended lines of credit to third-party entities include unused amounts of \$1.2 billion and \$1.6 billion at June 30, 2023 and December 31, 2022, respectively. A portion of these amounts was available to the company’s business partners to support their working capital needs. In addition, the company has committed to provide future financing to its clients in connection with client purchase agreements for \$1.7 billion and \$2.1 billion at June 30, 2023 and December 31, 2022, respectively. The reduction in the future financing commitments is primarily due to lower services financing in the current year. The company collectively evaluates the allowance for these arrangements using a provision methodology consistent with the portfolio of the commitments. Refer to note A, “Significant Accounting Policies,” in the company’s 2022 Annual Report for additional information. The allowance for these commitments is recorded in other liabilities in the Consolidated Balance Sheet and was not material at June 30, 2023.

The company has applied the guidance requiring a guarantor to disclose certain types of guarantees, even if the likelihood of requiring the guarantor’s performance is remote. The following is a description of arrangements in which the company is the guarantor.

The company is a party to a variety of agreements pursuant to which it may be obligated to indemnify the other party with respect to certain matters. Typically, these obligations arise in the context of contracts entered into by the company, under which the company customarily agrees to hold the party harmless against losses arising from a breach of representations and covenants related to such matters as title to the assets sold, certain intellectual property rights, specified environmental matters, third-party performance of nonfinancial contractual obligations and certain income taxes. In each of these circumstances, payment by the company is conditioned on the other party making a claim pursuant to the procedures specified in the particular contract, the procedures of which typically allow the company to challenge the other party’s claims. While indemnification provisions typically do not include a contractual maximum on the company’s payment, the company’s obligations under these agreements may be limited in terms of time and/or nature of claim, and in some instances, the company may have recourse against third parties for certain payments made by the company.

It is not possible to predict the maximum potential amount of future payments under these or similar agreements due to the conditional nature of the company’s obligations and the unique facts and circumstances involved in each particular agreement. Historically, payments made by the company under these agreements have not had a material effect on the company’s business, financial condition or results of operations.

In addition, the company guarantees certain loans and financial commitments. The maximum potential future payment under these financial guarantees and the fair value of these guarantees recognized in the Consolidated Balance Sheet at June 30, 2023 and December 31, 2022 was not material.

Changes in the company’s warranty liability for standard warranties, which are included in other accrued expenses and liabilities and other liabilities in the Consolidated Balance Sheet, and for extended warranty contracts, which are included in deferred income in the Consolidated Balance Sheet, are presented in the following tables.

Standard Warranty Liability

(Dollars in millions)	2023	2022
Balance at January 1	\$ 79	\$ 77
Current-period accruals	35	39
Accrual adjustments to reflect actual experience	(14)	(1)
Charges incurred	(43)	(41)
Balance at June 30	<u>\$ 57</u>	<u>\$ 74</u>

Notes to Consolidated Financial Statements — (continued)

Extended Warranty Liability

(Dollars in millions)	2023		2022	
Balance at January 1	\$	272	\$	350
Revenue deferred for new extended warranty contracts		20		84
Amortization of deferred revenue		(73)		(83)
Other*		(1)		(12)
Balance at June 30	\$	218	\$	339
Current portion	\$	119	\$	172
Noncurrent portion	\$	99	\$	167

* Other primarily consists of foreign currency translation adjustments.

14. Contingencies:

As a company with a substantial employee population and with clients in more than 175 countries, IBM is involved, either as plaintiff or defendant, in a variety of ongoing claims, demands, suits, investigations, tax matters and proceedings that arise from time to time in the ordinary course of its business. The company is a leader in the information technology industry and, as such, has been and will continue to be subject to claims challenging its IP rights and associated products and offerings, including claims of copyright and patent infringement and violations of trade secrets and other IP rights. In addition, the company enforces its own IP against infringement, through license negotiations, lawsuits or otherwise. Further, given the rapidly evolving external landscape of cybersecurity, privacy and data protection laws, regulations and threat actors, the company and its clients have been and will continue to be subject to actions or proceedings in various jurisdictions. Also, as is typical for companies of IBM’s scope and scale, the company is party to actions and proceedings in various jurisdictions involving a wide range of labor and employment issues (including matters related to contested employment decisions, country-specific labor and employment laws, and the company’s pension, retirement and other benefit plans), as well as actions with respect to contracts, product liability, securities, foreign operations, competition law and environmental matters. These actions may be commenced by a number of different parties, including competitors, clients, current or former employees, government and regulatory agencies, stockholders and representatives of the locations in which the company does business. Some of the actions to which the company is party may involve particularly complex technical issues, and some actions may raise novel questions under the laws of the various jurisdictions in which these matters arise.

The company records a provision with respect to a claim, suit, investigation or proceeding when it is probable that a liability has been incurred and the amount of the loss can be reasonably estimated. Any recorded liabilities, including any changes to such liabilities for the quarter ended June 30, 2023 were not material to the Consolidated Financial Statements.

In accordance with the relevant accounting guidance, the company provides disclosures of matters for which the likelihood of material loss is at least reasonably possible. In addition, the company also discloses matters based on its consideration of other matters and qualitative factors, including the experience of other companies in the industry, and investor, customer and employee relations considerations.

With respect to certain of the claims, suits, investigations and proceedings discussed herein, the company believes at this time that the likelihood of any material loss is remote, given, for example, the procedural status, court rulings, and/or the strength of the company’s defenses in those matters. With respect to the remaining claims, suits, investigations and proceedings discussed in this note, except as specifically discussed herein, the company is unable to provide estimates of reasonably possible losses or range of losses, including losses in excess of amounts accrued, if any, for the following reasons. Claims, suits, investigations and proceedings are inherently uncertain, and it is not possible to predict the ultimate outcome of these matters. It is the company’s experience that damage amounts claimed in litigation against it are unreliable and unrelated to possible outcomes, and as such are not meaningful indicators of the company’s potential liability. Further, the company is unable to provide such an estimate due to a number of other factors with respect to these claims, suits, investigations and proceedings, including considerations of the procedural status of the matter in question, the presence of complex or novel legal theories, and/or the ongoing discovery and development of information important to the matters. The company reviews claims, suits, investigations and proceedings at least quarterly, and decisions are made with respect to recording or adjusting provisions and disclosing reasonably possible losses or range of losses (individually or in the

Notes to Consolidated Financial Statements — (continued)

aggregate), to reflect the impact and status of settlement discussions, discovery, procedural and substantive rulings, reviews by counsel and other information pertinent to a particular matter.

Whether any losses, damages or remedies finally determined in any claim, suit, investigation or proceeding could reasonably have a material effect on the company’s business, financial condition, results of operations or cash flows will depend on a number of variables, including: the timing and amount of such losses or damages; the structure and type of any such remedies; the significance of the impact any such losses, damages or remedies may have in the Consolidated Financial Statements; and the unique facts and circumstances of the particular matter that may give rise to additional factors. While the company will continue to defend itself vigorously, it is possible that the company’s business, financial condition, results of operations or cash flows could be affected in any particular period by the resolution of one or more of these matters.

The following is a summary of the more significant legal matters involving the company.

On June 8, 2021, IBM sued GlobalFoundries U.S. Inc. (GF) in New York State Supreme Court for claims including fraud and breach of contract relating to a long-term strategic relationship between IBM and GF for researching, developing, and manufacturing advanced semiconductor chips for IBM. GF walked away from its obligations and IBM is now suing to recover amounts paid to GF, and other compensatory and punitive damages, totaling more than \$1.5 billion. On September 14, 2021, the court ruled on GF’s motion to dismiss. On April 7, 2022, the Appellate Division unanimously reversed the lower court’s dismissal of IBM’s fraud claim. IBM’s claims for breaches of contract, promissory estoppel, and fraud are proceeding.

On April 5, 2022, a putative securities law class action was commenced in the United States District Court for the Southern District of New York alleging that during the period from April 4, 2017 through October 20, 2021, certain strategic imperatives revenues were misclassified. The company, two current IBM senior executives, and two former IBM senior executives are named as defendants. On June 23, 2022, the court entered an order appointing Iron Workers Local 580 Joint Funds as lead plaintiff. On September 21, 2022, the plaintiff voluntarily dismissed the case, without prejudice. On January 13, 2023, a putative securities law class action making allegations substantially similar to those in the dismissed case was filed in the same court. On April 4, 2023, the court entered an order appointing June E. Adams Irrevocable Trust Dated 7/21/14 FBO Edward Adams, the same entity that filed the since dismissed April 5, 2022 class action, as lead plaintiff. On June 5, 2023, the plaintiff voluntarily dismissed the case, with prejudice.

On June 2, 2022, a putative class action lawsuit was filed in the United States District Court for the Southern District of New York alleging that the IBM Pension Plan miscalculated certain joint and survivor annuity pension benefits by using outdated actuarial tables in violation of the Employee Retirement Income Security Act of 1974. IBM, the Plan Administrator Committee, and the IBM Pension Plan are named as defendants.

As disclosed in the Kyndryl Form 10 and subsequent Kyndryl public filings, in 2017 BMC Software, Inc. (BMC) filed suit against IBM in the United States District Court for the Southern District of Texas in a dispute involving IBM’s former managed infrastructure services business. On May 30, 2022, the trial court awarded BMC \$718 million in direct damages and \$718 million in punitive damages, plus interest and fees. IBM filed a notice of appeal. IBM does not believe it has any material exposure relating to this litigation. No material liability or related indemnification asset has been recorded by IBM.

The company is party to, or otherwise involved in, proceedings brought by U.S. federal or state environmental agencies under the Comprehensive Environmental Response, Compensation and Liability Act (CERCLA), known as “Superfund,” or laws similar to CERCLA. Such statutes require potentially responsible parties to participate in remediation activities regardless of fault or ownership of sites. The company is also conducting environmental investigations, assessments or remediations at or in the vicinity of several current or former operating sites globally pursuant to permits, administrative orders or agreements with country, state or local environmental agencies, and is involved in lawsuits and claims concerning certain current or former operating sites.

The company is also subject to ongoing tax examinations and governmental assessments in various jurisdictions. Along with many other U.S. companies doing business in Brazil, the company is involved in various challenges with Brazilian tax authorities regarding non-income tax assessments and non-income tax litigation matters. The total potential amount related to all these matters for all applicable years is approximately \$450 million. The company believes it will prevail on these matters and that this amount is not a meaningful indicator of liability.

Notes to Consolidated Financial Statements — (continued)

15. Equity Activity:

Reclassifications and Taxes Related to Items of Other Comprehensive Income

(Dollars in millions)				
For the three months ended June 30, 2023:		Before Tax Amount	Tax (Expense)/ Benefit	Net of Tax Amount
Other comprehensive income/(loss):				
Foreign currency translation adjustments	\$	116	\$ (34)	\$ 82
Net changes related to available-for-sale securities:				
Unrealized gains/(losses) arising during the period	\$	(17)	\$ 4	\$ (12)
Reclassification of (gains)/losses to other (income) and expense		—	—	—
Total net changes related to available-for-sale securities	\$	(17)	\$ 4	\$ (12)
Unrealized gains/(losses) on cash flow hedges:				
Unrealized gains/(losses) arising during the period	\$	178	\$ (48)	\$ 130
Reclassification of (gains)/losses to:				
Cost of services		3	(1)	2
Cost of sales		(2)	1	(1)
Cost of financing		4	(1)	3
SG&A expense		(1)	0	0
Other (income) and expense		(55)	14	(41)
Interest expense		22	(5)	16
Total unrealized gains/(losses) on cash flow hedges	\$	148	\$ (40)	\$ 109
Retirement-related benefit plans:*				
Prior service costs/(credits)	\$	—	\$ 0	\$ 0
Net (losses)/gains arising during the period		0	7	7
Curtailments and settlements		6	(1)	5
Amortization of prior service (credits)/costs		(2)	1	(2)
Amortization of net (gains)/losses		130	(38)	92
Total retirement-related benefit plans	\$	134	\$ (31)	\$ 102
Other comprehensive income/(loss)	\$	381	\$ (101)	\$ 280

* These accumulated other comprehensive income (AOCI) components are included in the computation of net periodic pension cost. Refer to note 18, “Retirement-Related Benefits,” for additional information.

Notes to Consolidated Financial Statements — (continued)

Reclassifications and Taxes Related to Items of Other Comprehensive Income

(Dollars in millions)				
For the three months ended June 30, 2022:	Before Tax Amount	Tax (Expense)/ Benefit	Net of Tax Amount	
Other comprehensive income/(loss):				
Foreign currency translation adjustments	\$ 213	\$ (347)	\$ (134)	
Net changes related to available-for-sale securities:				
Unrealized gains/(losses) arising during the period	\$ 0	\$ 0	\$ 0	
Reclassification of (gains)/losses to other (income) and expense	—	—	—	
Total net changes related to available-for-sale securities	\$ 0	\$ 0	\$ 0	
Unrealized gains/(losses) on cash flow hedges:				
Unrealized gains/(losses) arising during the period	\$ 200	\$ (53)	\$ 147	
Reclassification of (gains)/losses to:				
Cost of services	(13)	3	(10)	
Cost of sales	(23)	7	(17)	
Cost of financing	6	(2)	5	
SG&A expense	(14)	4	(10)	
Other (income) and expense	38	(10)	29	
Interest expense	22	(6)	16	
Total unrealized gains/(losses) on cash flow hedges	\$ 217	\$ (56)	\$ 161	
Retirement-related benefit plans:*				
Prior service costs/(credits)	\$ —	\$ 0	\$ 0	
Net (losses)/gains arising during the period	1	(3)	(2)	
Curtailments and settlements	11	(3)	8	
Amortization of prior service (credits)/costs	6	(2)	5	
Amortization of net (gains)/losses	450	(125)	325	
Total retirement-related benefit plans	\$ 468	\$ (132)	\$ 336	
Other comprehensive income/(loss)	\$ 897	\$ (534)	\$ 363	

* These AOCI components are included in the computation of net periodic pension cost. Refer to note 18, “Retirement-Related Benefits,” for additional information.

Notes to Consolidated Financial Statements — (continued)

Reclassifications and Taxes Related to Items of Other Comprehensive Income

(Dollars in millions)				
For the six months ended June 30, 2023:	Before Tax Amount	Tax (Expense)/ Benefit	Net of Tax Amount	
Other comprehensive income/(loss):				
Foreign currency translation adjustments	\$ 29	\$ 22	\$ 52	
Net changes related to available-for-sale securities:				
Unrealized gains/(losses) arising during the period	\$ (1)	\$ 0	\$ (1)	
Reclassification of (gains)/losses to other (income) and expense	—	—	—	
Total net changes related to available-for-sale securities	\$ (1)	\$ 0	\$ (1)	
Unrealized gains/(losses) on cash flow hedges:				
Unrealized gains/(losses) arising during the period	\$ 149	\$ (41)	\$ 107	
Reclassification of (gains)/losses to:				
Cost of services	5	(1)	4	
Cost of sales	(17)	5	(12)	
Cost of financing	9	(2)	7	
SG&A expense	(10)	3	(7)	
Other (income) and expense	(181)	45	(135)	
Interest expense	42	(11)	32	
Total unrealized gains/(losses) on cash flow hedges	\$ (3)	\$ (1)	\$ (4)	
Retirement-related benefit plans:*				
Prior service costs/(credits)	\$ —	\$ 1	\$ 1	
Net (losses)/gains arising during the period	2	7	8	
Curtailments and settlements	5	(1)	4	
Amortization of prior service (credits)/costs	(4)	1	(3)	
Amortization of net (gains)/losses	261	(76)	185	
Total retirement-related benefit plans	\$ 263	\$ (69)	\$ 194	
Other comprehensive income/(loss)	\$ 289	\$ (48)	\$ 241	

* These AOCI components are included in the computation of net periodic pension cost. Refer to note 18, “Retirement-Related Benefits,” for additional information.

Notes to Consolidated Financial Statements — (continued)

Reclassifications and Taxes Related to Items of Other Comprehensive Income

(Dollars in millions)				
For the six months ended June 30, 2022:	Before Tax Amount	Tax (Expense)/ Benefit	Net of Tax Amount	
Other comprehensive income/(loss):				
Foreign currency translation adjustments	\$ 655	\$ (483)	\$ 172	
Net changes related to available-for-sale securities:				
Unrealized gains/(losses) arising during the period	\$ (1)	\$ 0	\$ 0	
Reclassification of (gains)/losses to other (income) and expense	—	—	—	
Total net changes related to available-for-sale securities	\$ (1)	\$ 0	\$ 0	
Unrealized gains/(losses) on cash flow hedges:				
Unrealized gains/(losses) arising during the period	\$ 260	\$ (69)	\$ 191	
Reclassification of (gains)/losses to:				
Cost of services	(28)	7	(21)	
Cost of sales	(35)	10	(25)	
Cost of financing	12	(3)	9	
SG&A expense	(20)	5	(14)	
Other (income) and expense	45	(11)	34	
Interest expense	43	(11)	32	
Total unrealized gains/(losses) on cash flow hedges	\$ 276	\$ (71)	\$ 205	
Retirement-related benefit plans:*				
Prior service costs/(credits)	\$ (5)	\$ 5	\$ 0	
Net (losses)/gains arising during the period	10	(7)	3	
Curtailments and settlements	19	(5)	14	
Amortization of prior service (credits)/costs	13	(3)	10	
Amortization of net (gains)/losses	917	(256)	662	
Total retirement-related benefit plans	\$ 954	\$ (266)	\$ 689	
Other comprehensive income/(loss)	\$ 1,885	\$ (819)	\$ 1,066	

* These AOCI components are included in the computation of net periodic pension cost. Refer to note 18, “Retirement-Related Benefits,” for additional information.

Notes to Consolidated Financial Statements — (continued)
Accumulated Other Comprehensive Income/(Loss) (net of tax)

(Dollars in millions)	Net Unrealized Gains/(Losses) on Cash Flow Hedges	Foreign Currency Translation Adjustments*	Net Change Retirement-Related Benefit Plans	Net Unrealized Gains/(Losses) on Available-For-Sale Securities	Accumulated Other Comprehensive Income/ (Loss)
January 1, 2023	\$ (135)	\$ (3,591)	\$ (13,013)	\$ (1)	\$ (16,740)
Other comprehensive income before reclassifications	107	52	9	(1)	167
Amount reclassified from accumulated other comprehensive income	(112)	—	185	—	74
Total change for the period	\$ (4)	\$ 52	\$ 194	\$ (1)	\$ 241
June 30, 2023	<u>\$ (139)</u>	<u>\$ (3,539)</u>	<u>\$ (12,819)</u>	<u>\$ (2)</u>	<u>\$ (16,499)</u>

(Dollars in millions)	Net Unrealized Gains/(Losses) on Cash Flow Hedges	Foreign Currency Translation Adjustments*	Net Change Retirement-Related Benefit Plans	Net Unrealized Gains/(Losses) on Available-For-Sale Securities	Accumulated Other Comprehensive Income/ (Loss)
January 1, 2022	\$ (18)	\$ (3,362)	\$ (19,854)	\$ (1)	\$ (23,234)
Other comprehensive income before reclassifications	191	172	3	0	366
Amount reclassified from accumulated other comprehensive income	14	—	686	—	699
Total change for the period	\$ 205	\$ 172	\$ 689	\$ 0	\$ 1,066
June 30, 2022	<u>\$ 187</u>	<u>\$ (3,189)</u>	<u>\$ (19,165)</u>	<u>\$ (1)</u>	<u>\$ (22,169)</u>

* Foreign currency translation adjustments are presented gross except for any associated hedges which are presented net of tax.

16. Derivative Financial Instruments:

The company operates in multiple functional currencies and is a significant lender and borrower in the global markets. In the normal course of business, the company is exposed to the impact of interest rate changes and foreign currency fluctuations, and to a lesser extent equity and commodity price changes and client credit risk. The company limits these risks by following established risk management policies and procedures, including the use of derivatives, and, where cost effective, financing with debt in the currencies in which assets are denominated. For interest rate exposures, derivatives are used to better align rate movements between the interest rates associated with the company's lease and other financial assets and the interest rates associated with its financing debt. Derivatives are also used to manage the related cost of debt. For foreign currency exposures, derivatives are used to better manage the cash flow volatility arising from foreign exchange rate fluctuations.

In the Consolidated Balance Sheet, the company does not offset derivative assets against liabilities in master netting arrangements nor does it offset receivables or payables recognized upon payment or receipt of cash collateral against the fair values of the related derivative instruments. At June 30, 2023 and December 31, 2022, the amount recognized in other accounts receivables for the right to reclaim cash collateral was \$46 million and \$140 million, respectively. At June 30, 2023, there was no amount recognized in accounts payable for the obligation to return cash collateral. At December 31, 2022, the amount recognized in accounts payable for such obligation was \$8 million. The company restricts the use of cash collateral received to rehypothecation, and therefore reports it in restricted cash in the Consolidated Balance Sheet. At June 30, 2023, there was no cash collateral rehypothecated. At December 31, 2022, the amount rehypothecated was \$8 million. Additionally, if derivative exposures covered by a qualifying master netting agreement had been netted in the Consolidated Balance Sheet at June 30, 2023 and December 31, 2022, the total derivative asset and liability positions each would have been reduced by \$267 million and \$220 million, respectively.

Notes to Consolidated Financial Statements — (continued)

On May 19, 2022, in connection with the disposition of 22.3 million shares of Kyndryl common stock, the company entered into a cash-settled swap that maintained IBM’s continued economic exposure in those shares. The notional value of the swap was \$311 million. For the three and six months ended June 30, 2022, an unrealized loss of \$88 million was recorded in other (income) and expense in the Consolidated Income Statement. The company settled the swap on November 2, 2022.

In its hedging programs, the company may use forward contracts, futures contracts, interest-rate swaps, cross-currency swaps, equity swaps, and options depending upon the underlying exposure. The company is not a party to leveraged derivative instruments.

A brief description of the major hedging programs, categorized by underlying risk, follows.

Interest Rate Risk

Fixed and Variable Rate Borrowings

The company issues debt in the global capital markets to fund its operations and financing business. Access to cost-effective financing can result in interest rate mismatches with the underlying assets. To manage these mismatches and to reduce overall interest cost, the company may use interest-rate swaps to convert specific fixed-rate debt issuances into variable-rate debt (i.e., fair value hedges) and to convert specific variable-rate debt issuances into fixed-rate debt (i.e., cash flow hedges). At June 30, 2023 and December 31, 2022, the total notional amount of the company’s interest-rate swaps was \$6.9 billion and \$6.5 billion, respectively. The weighted-average remaining maturity of these instruments at June 30, 2023 and December 31, 2022 was approximately 5.8 years and 6.0 years, respectively. These interest-rate contracts were accounted for as fair value hedges. The company did not have any cash flow hedges relating to this program outstanding at June 30, 2023 and December 31, 2022.

Forecasted Debt Issuance

The company is exposed to interest rate volatility on future debt issuances. To manage this risk, the company may use instruments such as forward starting interest-rate swaps to lock in the rate on the interest payments related to the forecasted debt issuances. There were no instruments outstanding at June 30, 2023 and December 31, 2022.

In connection with cash flow hedges of forecasted interest payments related to the company's borrowings, the company recorded net losses (before taxes) of \$130 million and \$139 million at June 30, 2023 and December 31, 2022, respectively, in AOCI. The company estimates that \$17 million of the deferred net losses (before taxes) on derivatives in AOCI at June 30, 2023 will be reclassified to net income within the next 12 months, providing an offsetting economic impact against the underlying interest payments.

Foreign Exchange Risk

Long-Term Investments in Foreign Subsidiaries (Net Investment)

A large portion of the company’s foreign currency denominated debt portfolio is designated as a hedge of net investment in foreign subsidiaries to reduce the volatility in stockholders’ equity caused by changes in foreign currency exchange rates in the functional currency of major foreign subsidiaries with respect to the U.S. dollar. At June 30, 2023 and December 31, 2022, the carrying value of debt designated as hedging instruments was \$15.7 billion and \$13.4 billion, respectively. The company also uses cross-currency swaps and foreign exchange forward contracts for this risk management purpose. At June 30, 2023 and December 31, 2022, the total notional amount of derivative instruments designated as net investment hedges was \$4.9 billion and \$4.7 billion, respectively. At June 30, 2023 and December 31, 2022, the weighted-average remaining maturity of these instruments was approximately 0.2 years and 0.1 years, respectively.

Anticipated Royalties and Cost Transactions

The company’s operations generate significant nonfunctional currency, third-party vendor payments and intercompany payments for royalties and goods and services among the company’s non-U.S. subsidiaries and with the company. In anticipation of these foreign currency cash flows and in view of the volatility of the currency markets, the company selectively employs foreign exchange forward contracts to manage its currency risk. These forward contracts are accounted

Notes to Consolidated Financial Statements — (continued)

for as cash flow hedges. At June 30, 2023, the maximum remaining length of time over which the company hedged its exposure is approximately two years. At June 30, 2023 and December 31, 2022, the total notional amount of forward contracts designated as cash flow hedges of forecasted royalty and cost transactions was \$8.6 billion and \$8.1 billion, respectively. At both June 30, 2023 and December 31, 2022, the weighted-average remaining maturity of these instruments was approximately 0.6 years.

At June 30, 2023 and December 31, 2022, in connection with cash flow hedges of anticipated royalties and cost transactions, the company recorded net gains (before taxes) of \$63 million and \$66 million, respectively, in AOCI. The company estimates that \$3 million of deferred net losses (before taxes) on derivatives in AOCI at June 30, 2023 will be reclassified to net income within the next 12 months, providing an offsetting economic impact against the underlying anticipated transactions.

Foreign Currency Denominated Borrowings

The company is exposed to exchange rate volatility on foreign currency denominated debt. To manage this risk, the company employs cross-currency swaps to convert fixed-rate foreign currency denominated debt to fixed-rate debt denominated in the functional currency of the borrowing entity. These swaps are accounted for as cash flow hedges. At June 30, 2023, the maximum length of time remaining over which the company hedged its exposure is approximately eight years. At June 30, 2023 and December 31, 2022, the total notional amount of cross-currency swaps designated as cash flow hedges of foreign currency denominated debt was \$3.9 billion and \$3.1 billion, respectively.

At June 30, 2023 and December 31, 2022, in connection with cash flow hedges of foreign currency denominated borrowings, the company recorded net losses (before taxes) of \$109 million and \$101 million, respectively, in AOCI. The company estimates that \$48 million of deferred net gains (before taxes) on derivatives in AOCI at June 30, 2023 will be reclassified to net income within the next 12 months, providing an offsetting economic impact against the underlying exposure.

Subsidiary Cash and Foreign Currency Asset/Liability Management

The company uses its Global Treasury Centers to manage the cash of its subsidiaries. These centers principally use currency swaps to convert cash flows in a cost-effective manner. In addition, the company uses foreign exchange forward contracts to economically hedge, on a net basis, the foreign currency exposure of a portion of the company’s nonfunctional currency assets and liabilities. The terms of these forward and swap contracts are generally less than one year. The changes in the fair values of these contracts and of the underlying hedged exposures are generally offsetting and are recorded in other (income) and expense in the Consolidated Income Statement. At June 30, 2023 and December 31, 2022, the total notional amount of derivative instruments in economic hedges of foreign currency exposure was \$7.5 billion and \$5.9 billion, respectively.

Equity Risk Management

The company is exposed to market price changes in certain broad market indices and in the company’s own stock primarily related to certain obligations to employees. Changes in the overall value of these employee compensation obligations are recorded in SG&A expense in the Consolidated Income Statement. Although not designated as accounting hedges, the company utilizes derivatives, including equity swaps and futures, to economically hedge the exposures related to its employee compensation obligations. The derivatives are linked to the total return on certain broad market indices or the total return on the company’s common stock, and are recorded at fair value with gains or losses also reported in SG&A expense in the Consolidated Income Statement. At June 30, 2023 and December 31, 2022, the total notional amount of derivative instruments in economic hedges of these compensation obligations was \$1.2 billion and \$1.1 billion, respectively.

Notes to Consolidated Financial Statements — (continued)

Cumulative Basis Adjustments for Fair Value Hedges

At June 30, 2023 and December 31, 2022, the following amounts were recorded in the Consolidated Balance Sheet related to cumulative basis adjustments for fair value hedges:

(Dollars in millions)	June 30, 2023	December 31, 2022
Short-term debt:		
Carrying amount of the hedged item	\$ (203)	\$ (199)
Cumulative hedging adjustments included in the carrying amount — assets/(liabilities)*	\$ (3)	\$ 1
Long-term debt:		
Carrying amount of the hedged item	\$ (6,540)	\$ (6,216)
Cumulative hedging adjustments included in the carrying amount — assets/(liabilities)*	\$ 147	\$ 72

* Includes (\$225) million and (\$250) million of hedging adjustments on discontinued hedging relationships at June 30, 2023 and December 31, 2022, respectively.

The Effect of Derivative Instruments in the Consolidated Income Statement

The total amounts of income and expense line items presented in the Consolidated Income Statement in which the effects of fair value hedges, cash flow hedges, net investment hedges and derivatives not designated as hedging instruments are recorded and the total effect of hedge activity on these income and expense line items are as follows:

(Dollars in millions)	Total		Gains/(Losses) of Total Hedge Activity	
	2023	2022	2023	2022
For the three months ended June 30:				
Cost of services	\$ 5,294	\$ 5,399	\$ (3)	\$ 13
Cost of sales	\$ 1,587	\$ 1,750	\$ 2	\$ 23
Cost of financing	\$ 93	\$ 96	\$ (3)	\$ 1
SG&A expense	\$ 4,900	\$ 4,855	\$ 43	\$ (152)
Other (income) and expense	\$ (261)	\$ (81)	\$ (141)	\$ (439)
Interest expense	\$ 423	\$ 297	\$ (18)	\$ 3

Notes to Consolidated Financial Statements — (continued)

	Gain (Loss) Recognized in Consolidated Income Statement				
(Dollars in millions)	Consolidated Income Statement Line Item	Recognized on Derivatives		Attributable to Risk Being Hedged ⁽²⁾	
For the three months ended June 30:		2023	2022	2023	2022
Derivative instruments in fair value hedges ⁽¹⁾ :					
Interest rate contracts	Cost of financing	\$ (30)	\$ (17)	\$ 25	\$ 23
	Interest expense	(153)	(61)	130	81
Derivative instruments not designated as hedging instruments:					
Foreign exchange contracts	Other (income) and expense	(196)	(313)	N/A	N/A
Equity contracts	SG&A expense	42	(166)	N/A	N/A
	Other (income) and expense	—	(88)	N/A	N/A
Total		\$ (337)	\$ (645)	\$ 155	\$ 104

	Gain (Loss) Recognized in Consolidated Income Statement and Other Comprehensive Income												
(Dollars in millions)	Recognized in OCI		Consolidated Income Statement Line Item	Reclassified from AOCI		Amounts Excluded from Effectiveness Testing ⁽³⁾							
For the three months ended June 30:	2023	2022		2023	2022	2023	2022						
Derivative instruments in cash flow hedges:													
Interest rate contracts	\$	—	\$	—	Cost of financing	\$	(1)	\$	(1)	\$	—	\$	—
					Interest expense		(4)		(3)		—		—
Foreign exchange contracts		178		200	Cost of services		(3)		13		—		—
					Cost of sales		2		23		—		—
					Cost of financing		(3)		(5)		—		—
					SG&A expense		1		14		—		—
					Other (income) and expense		55		(38)		—		—
					Interest expense		(18)		(19)		—		—
Instruments in net investment hedges ⁽⁴⁾ :													
Foreign exchange contracts		136		1,379	Cost of financing		—		—		5		1
					Interest expense		—		—		27		5
Total	\$	313	\$	1,579		\$	30	\$	(16)	\$	32	\$	6

(1) The amount includes changes in clean fair values of the derivative instruments in fair value hedging relationships and the periodic accrual for coupon payments required under these derivative contracts.

(2) The amount includes basis adjustments to the carrying value of the hedged item recorded during the period and amortization of basis adjustments recorded on de-designated hedging relationships during the period.

(3) The company's policy is to recognize all fair value changes in amounts excluded from effectiveness testing in net income each period.

(4) Instruments in net investment hedges include derivative and non-derivative instruments with the amounts recognized in OCI providing an offset to the translation of foreign subsidiaries.

N/A - not applicable

(Dollars in millions) For the six months ended June 30:	Total		Gains/(Losses) of Total Hedge Activity	
	2023	2022	2023	2022
Cost of services	\$ 10,604	\$ 10,747	\$ (5)	\$ 28
Cost of sales	\$ 2,910	\$ 3,165	\$ 17	\$ 35
Cost of financing	\$ 203	\$ 194	\$ (7)	\$ (1)
SG&A expense	\$ 9,754	\$ 9,452	\$ 102	\$ (223)
Other (income) and expense	\$ (506)	\$ 166	\$ 1	\$ (541)
Interest expense	\$ 790	\$ 607	\$ (31)	\$ (3)

Notes to Consolidated Financial Statements — (continued)

	Gain (Loss) Recognized in Consolidated Income Statement				
(Dollars in millions)	Consolidated Income Statement Line Item	Recognized on Derivatives		Attributable to Risk Being Hedged ⁽²⁾	
For the six months ended June 30:		2023	2022	2023	2022
Derivative instruments in fair value hedges ⁽¹⁾ :					
Interest rate contracts	Cost of financing	\$ (21)	\$ (18)	\$ 13	\$ 26
	Interest expense	(96)	(65)	58	97
Derivative instruments not designated as hedging instruments:					
Foreign exchange contracts	Other (income) and expense	(180)	(409)	N/A	N/A
Equity contracts	SG&A expense	91	(243)	N/A	N/A
	Other (income) and expense	—	(88)	N/A	N/A
Total		\$ (206)	\$ (821)	\$ 71	\$ 123

	Gain (Loss) Recognized in Consolidated Income Statement and Other Comprehensive Income												
(Dollars in millions)	Recognized in OCI		Consolidated Income Statement Line Item	Reclassified from AOCI		Amounts Excluded from Effectiveness Testing ⁽³⁾							
For six months ended June 30:	2023	2022		2023	2022	2023	2022						
Derivative instruments in cash flow hedges:													
Interest rate contracts	\$	—	\$	—	Cost of financing	\$	(2)	\$	(2)	\$	—	\$	—
					Interest expense		(7)		(7)		—		—
Foreign exchange contracts		149		260	Cost of services		(5)		28		—		—
					Cost of sales		17		35		—		—
					Cost of financing		(8)		(10)		—		—
					SG&A expense		10		20		—		—
					Other (income) and expense		181		(45)		—		—
					Interest expense		(35)		(36)		—		—
Instruments in net investment hedges ⁽⁴⁾ :													
Foreign exchange contracts		(88)		1,920	Cost of financing		—		—		11		2
					Interest expense		—		—		49		7
Total	\$	61	\$	2,180		\$	152	\$	(16)	\$	60	\$	9

(1) The amount includes changes in clean fair values of the derivative instruments in fair value hedging relationships and the periodic accrual for coupon payments required under these derivative contracts.

(2) The amount includes basis adjustments to the carrying value of the hedged item recorded during the period and amortization of basis adjustments recorded on de-designated hedging relationships during the period.

(3) The company's policy is to recognize all fair value changes in amounts excluded from effectiveness testing in net income each period.

(4) Instruments in net investment hedges include derivative and non-derivative instruments with the amounts recognized in OCI providing an offset to the translation of foreign subsidiaries.

N/A - not applicable

For the three and six months ended June 30, 2023 and 2022, there were no material gains or losses excluded from the assessment of hedge effectiveness (for fair value or cash flow hedges), or associated with an underlying exposure that did not or was not expected to occur (for cash flow hedges); nor are there any anticipated in the normal course of business.

Notes to Consolidated Financial Statements — (continued)

17. Stock-Based Compensation:

Stock-based compensation cost for stock awards and stock options is measured at grant date, based on the fair value of the award, and is recognized over the employee requisite service period. The following table presents total stock-based compensation cost included in income from continuing operations.

(Dollars in millions)	Three Months Ended June 30,		Six Months Ended June 30,	
	2023	2022	2023	2022
Cost	\$ 47	\$ 43	\$ 93	\$ 84
Selling, general and administrative	168	153	317	289
Research, development and engineering	73	58	146	115
Pre-tax stock-based compensation cost	\$ 288	\$ 254	\$ 556	\$ 488
Income tax benefits	(75)	(82)	(142)	(139)
Total net stock-based compensation cost	\$ 213	\$ 172	\$ 414	\$ 348

Pre-tax stock-based compensation cost for the three months ended June 30, 2023 increased \$34 million compared to the corresponding period in the prior year, including increases in restricted stock units (\$15 million), performance share units (\$11 million) and stock options (\$7 million). The increases are driven by stock-based compensation awards granted by the company as part of its annual cycles for executives and other employees.

Pre-tax stock-based compensation cost for the six months ended June 30, 2023 increased \$69 million compared to the corresponding period in the prior year, including increases in stock options (\$21 million), Employees Stock Purchase Plan (ESPP) (\$20 million), restricted stock units (\$17 million) and performance share units (\$11 million). The increases are driven by stock-based compensation awards granted by the company as part of its annual cycles for executives and other employees and the ESPP being considered compensatory effective April 1, 2022.

Total unrecognized compensation cost related to non-vested awards at June 30, 2023 was \$1.6 billion and is expected to be recognized over a weighted-average period of approximately 2.5 years.

18. Retirement-Related Benefits:

The company offers defined benefit (DB) pension plans, defined contribution pension plans, as well as nonpension postretirement plans primarily consisting of retiree medical benefits. The following tables provide the pre-tax cost for all retirement-related plans.

(Dollars in millions)			Yr. to Yr. Percent Change
For the three months ended June 30:	2023	2022	
Retirement-related plans — cost:			
Defined benefit and contribution pension plans — cost	\$ 259	\$ 456	(43.2)%
Nonpension postretirement plans — cost	33	33	(2.2)
Total	\$ 292	\$ 489	(40.4)%

(Dollars in millions)			Yr. to Yr. Percent Change
For the six months ended June 30:	2023	2022	
Retirement-related plans — cost:			
Defined benefit and contribution pension plans — cost	\$ 541	\$ 934	(42.1)%
Nonpension postretirement plans — cost	65	66	(1.9)
Total	\$ 606	\$ 1,000	(39.4)%

Notes to Consolidated Financial Statements — (continued)

Cost/(Income) of Pension Plans

The following tables provide the components of the cost/(income) for the company’s pension plans.

(Dollars in millions) For the three months ended June 30:	U.S. Plans		Non-U.S. Plans	
	2023	2022	2023	2022
Service cost	\$ —	\$ —	\$ 44	\$ 59
Interest cost*	272	302	292	131
Expected return on plan assets*	(382)	(475)	(362)	(259)
Amortization of prior service costs/(credits)*	0	2	5	3
Recognized actuarial losses*	27	179	101	260
Curtailments and settlements*	—	—	6	11
Multi-employer plans	—	—	3	3
Other costs/(credits)*	—	—	10	6
Total net periodic pension (income)/cost of defined benefit plans	\$ (82)	\$ 8	\$ 99	\$ 215
Cost of defined contribution plans	151	141	91	92
Total defined benefit and contribution pension plans cost recognized in the Consolidated Income Statement	\$ 69	\$ 149	\$ 190	\$ 307

(Dollars in millions) For the six months ended June 30:	U.S. Plans		Non-U.S. Plans	
	2023	2022	2023	2022
Service cost	\$ —	\$ —	\$ 88	\$ 123
Interest cost*	545	603	580	270
Expected return on plan assets*	(764)	(950)	(718)	(532)
Amortization of prior service costs/(credits)*	0	4	10	7
Recognized actuarial losses*	55	359	203	537
Curtailments and settlements*	—	—	5	19
Multi-employer plans	—	—	6	7
Other costs/(credits)*	—	—	19	15
Total net periodic pension (income)/cost of defined benefit plans	\$ (165)	\$ 15	\$ 194	\$ 446
Cost of defined contribution plans	323	282	188	190
Total defined benefit and contribution pension plans cost recognized in the Consolidated Income Statement	\$ 158	\$ 298	\$ 382	\$ 636

* These components of net periodic pension cost are included in other (income) and expense in the Consolidated Income Statement.

Notes to Consolidated Financial Statements — (continued)

Cost of Nonpension Postretirement Plans

The following tables provide the components of the cost for the company's nonpension postretirement plans.

(Dollars in millions) For the three months ended June 30:	U.S. Plan		Non-U.S. Plans	
	2023	2022	2023	2022
Service cost	\$ 1	\$ 1	\$ 1	\$ 1
Interest cost*	29	18	10	9
Expected return on plan assets*	—	—	(1)	(1)
Amortization of prior service costs/(credits)*	(7)	1	0	0
Recognized actuarial losses*	—	2	0	1
Curtailments and settlements*	—	—	—	—
Total nonpension postretirement plans cost recognized in the Consolidated Income Statement	<u>\$ 23</u>	<u>\$ 23</u>	<u>\$ 10</u>	<u>\$ 10</u>

(Dollars in millions) For the six months ended June 30:	U.S. Plan		Non-U.S. Plans	
	2023	2022	2023	2022
Service cost	\$ 2	\$ 2	\$ 1	\$ 2
Interest cost*	58	37	19	18
Expected return on plan assets*	—	—	(1)	(1)
Amortization of prior service costs/(credits)*	(15)	2	0	0
Recognized actuarial losses*	—	4	(1)	2
Curtailments and settlements*	—	—	—	—
Total nonpension postretirement plans cost recognized in the Consolidated Income Statement	<u>\$ 46</u>	<u>\$ 46</u>	<u>\$ 19</u>	<u>\$ 20</u>

* These components of net periodic pension cost are included in other (income) and expense in the Consolidated Income Statement.

The company does not anticipate any significant changes to the expected plan contributions in 2023 from the amounts disclosed in the 2022 Annual Report.

The table below includes contributions to the following plans:

(Dollars in millions) For the six months ended June 30:	Plan Contributions	
	2023	2022
U.S. nonpension postretirement benefit plans	\$ 134	\$ 202
Non-U.S. DB and multi-employer plans*	29	43
Total plan contributions	<u>\$ 163</u>	<u>\$ 245</u>

* Amounts reported net of refunds.

During the six months ended June 30, 2023 and 2022, the company contributed \$134 million and \$182 million of U.S. Treasury Securities, respectively, to the U.S. nonpension postretirement benefit plan. Additionally, during the six months ended June 30, 2023 and 2022, the company contributed \$347 million and \$261 million of U.S. Treasury securities, respectively, to the Active Medical Trust. Contributions made with U.S. Treasury securities are considered a non-cash transaction.

19. Subsequent Events:

On July 24, 2023, the company announced that the Board of Directors approved a quarterly dividend of \$1.66 per common share. The dividend is payable September 9, 2023 to shareholders of record on August 10, 2023.

Item 2.

**MANAGEMENT’S DISCUSSION AND ANALYSIS
OF RESULTS OF OPERATIONS AND FINANCIAL CONDITION
FOR THE THREE AND SIX MONTHS ENDED JUNE 30, 2023**

Snapshot

Organization of Information:

In the fourth quarter of 2022, we completed our annual assessment of the useful lives of our property, plant and equipment. Due to advances in technology, we determined we should increase the estimated useful lives of our server and network equipment from five to six years for new assets and from three to four years for used assets. This change in accounting estimate was effective beginning January 1, 2023. Based on the carrying amount of server and network equipment included in property, plant and equipment-net in our Consolidated Balance Sheet as of December 31, 2022, the effect of this change in estimate was an increase in income from continuing operations before income taxes of \$57 million or \$0.05 per basic and diluted share for the three months ended June 30, 2023, and \$131 million, or \$0.12 and \$0.11 per basic and diluted share, respectively, for the six months ended June 30, 2023.

In the first half of 2023, we initiated workforce rebalancing actions to address remaining stranded costs from portfolio actions over the last couple of years resulting in a charge to pre-tax income from continuing operations of \$117 million and \$376 million for the three and six months ended June 30, 2023. In addition, beginning in the first quarter of 2023, we updated our measure of segment pre-tax income to no longer allocate workforce rebalancing actions to our segments, consistent with our management system. Workforce rebalancing charges in the second quarter and first half of 2022 of \$3 million and \$9 million, respectively, were included in the segments.

Within the tables presented, certain columns and rows may not add due to the use of rounded numbers for disclosure purposes. Percentages presented are calculated from the underlying whole-dollar amounts. Certain prior-period amounts have been reclassified to conform to the current period presentation. This is annotated where applicable.

Currency:

The references to “adjusted for currency” or “at constant currency” in the Management Discussion do not include operational impacts that could result from fluctuations in foreign currency rates. When we refer to growth rates at constant currency or adjust such growth rates for currency, it is done so that certain financial results can be viewed without the impact of fluctuations in foreign currency exchange rates, thereby facilitating period-to-period comparisons of its business performance. Financial results adjusted for currency are calculated by translating current period activity in local currency using the comparable prior-year period’s currency conversion rate. This approach is used for countries where the functional currency is the local currency. Generally, when the dollar either strengthens or weakens against other currencies, the growth at constant currency rates or adjusting for currency will be higher or lower than growth reported at actual exchange rates. Refer to “Currency Rate Fluctuations” for additional information.

Operating (non-GAAP) Earnings:

In an effort to provide better transparency into the operational results of the business, supplementally, management separates business results into operating and non-operating categories. Operating earnings from continuing operations is a non-GAAP measure that excludes the effects of certain acquisition-related charges, intangible asset amortization, expense resulting from basis differences on equity method investments, retirement-related costs, certain impacts from the Kyndryl separation and their related tax impacts. Due to the unique, non-recurring nature of the enactment of the U.S. Tax Cuts and Jobs Act (U.S. tax reform), management characterizes the one-time provisional charge recorded in the fourth quarter of 2017 and adjustments to that charge as non-operating. Adjustments primarily include true-ups, accounting elections and any changes to regulations, laws, audit adjustments that affect the recorded one-time charge. Management characterizes direct and incremental charges incurred related to the Kyndryl separation as non-operating given their unique and non-recurring nature. These charges primarily relate to any net gains or losses on the Kyndryl common stock and the related cash-settled swap with a third-party financial institution, which were recorded in other (income) and expense in the Consolidated Income Statement. As of November 2, 2022, the company no longer held an ownership interest in Kyndryl. For acquisitions, operating (non-GAAP) earnings exclude the amortization of purchased intangible assets and acquisition-related charges such as in-process research and development, transaction costs, applicable retention, restructuring and

Management Discussion – (continued)

related expenses, tax charges related to acquisition integration and pre-closing charges, such as financing costs. These charges are excluded as they may be inconsistent in amount and timing from period to period and are significantly impacted by the size, type and frequency of the company’s acquisitions. All other spending for acquired companies is included in both earnings from continuing operations and in operating (non-GAAP) earnings. For retirement-related costs, management characterizes certain items as operating and others as non-operating, consistent with GAAP. We include defined benefit plan and nonpension postretirement benefit plan service costs, multi-employer plan costs and the cost of defined contribution plans in operating earnings. Non-operating retirement-related costs include defined benefit plan and nonpension postretirement benefit plan amortization of prior service costs, interest cost, expected return on plan assets, amortized actuarial gains/losses, the impacts of any plan curtailments/settlements and pension insolvency costs and other costs. Non-operating retirement-related costs are primarily related to changes in pension plan assets and liabilities which are tied to financial market performance, and the company considers these costs to be outside of the operational performance of the business.

Overall, management believes that supplementally providing investors with a view of operating earnings as described above provides increased transparency and clarity into both the operational results of the business and the performance of the company’s pension plans; improves visibility to management decisions and their impacts on operational performance; enables better comparison to peer companies; and allows the company to provide a long-term strategic view of the business going forward. In addition, these non-GAAP measures provide a perspective consistent with areas of interest we routinely receive from investors and analysts. Our reportable segment financial results reflect pre-tax operating earnings from continuing operations, consistent with our management and measurement system.

Financial Results Summary — Three Months Ended June 30

(Dollars and shares in millions except per share amounts)			
For the three months ended June 30:	2023	2022	Yr. to Yr. Percent/ Margin Change
Revenue	\$ 15,475	\$ 15,535	(0.4) %*
Gross profit margin	54.9 %	53.4 %	1.6 pts.
Total expense and other (income)	\$ 6,501	\$ 6,568	(1.0) %
Income from continuing operations before income taxes	\$ 2,000	\$ 1,722	16.2 %
Provision for income taxes from continuing operations	\$ 419	\$ 257	63.4 %
Income from continuing operations	\$ 1,581	\$ 1,465	7.9 %
Income from continuing operations margin	10.2 %	9.4 %	0.8 pts.
Income/(loss) from discontinued operations, net of tax	\$ 2	\$ (73)	nm
Net income	\$ 1,583	\$ 1,392	13.7 %
Earnings per share from continuing operations - assuming dilution	\$ 1.72	\$ 1.61	6.8 %
Consolidated earnings per share - assuming dilution	\$ 1.72	\$ 1.53	12.4 %
Weighted-average shares outstanding - assuming dilution	919.5	910.7	1.0 %

* 0.4% percent adjusted for currency.
nm - not meaningful

Management Discussion – (continued)

The following table provides the company’s operating (non-GAAP) earnings for the second quarter of 2023 and 2022.

(Dollars in millions except per share amounts)				Yr. to Yr. Percent Change
For the three months ended June 30:		2023	2022	
Net income as reported	\$	1,583	\$ 1,392	13.7 %
Income/(loss) from discontinued operations, net of tax		2	(73)	nm
Income from continuing operations	\$	1,581	\$ 1,465	7.9 %
Non-operating adjustments (net of tax):				
Acquisition-related charges	\$	308	\$ 345	(10.8) %
Non-operating retirement-related costs/(income)		5	146	(96.9)
U.S. tax reform impacts		110	4	nm
Kyndryl-related impacts		—	145	(100)
Operating (non-GAAP) earnings*	\$	2,003	\$ 2,105	(4.8) %
Diluted operating (non-GAAP) earnings per share*	\$	2.18	\$ 2.31	(5.6) %

* Refer to page 81 for a more detailed reconciliation of net income to operating earnings.

nm - not meaningful

Macroeconomic Environment:

Our business profile positions us well in challenging times. Our diversification across geographies, industries, clients and business mix and our recurring revenue base provides some stability in revenue, profit and cash generation.

In the first half of 2023, we saw progress from the actions we have taken to mitigate the impacts of escalating labor and component costs and a strong U.S. dollar. Consulting gross profit and pre-tax margin increased in the second quarter of 2023 reflecting the pricing and productivity actions we have taken. We expect these actions to continue to contribute to margin improvement throughout 2023. In the current environment, clients and partners continue to view technology as a source of competitive advantage, and are prioritizing larger digital transformation projects that focus on cost savings and increased productivity. In Consulting, while demand for our offerings that support these priorities remains solid, we continue to experience some delays for other projects considered to be more discretionary, especially in the U.S. The strength of the U.S. dollar has continued to impact our reported year-to-year revenue and pre-tax profit. We execute hedging programs which defer but do not eliminate the impact of currency. The (gains)/losses from these hedging programs are reflected primarily in other income and expense. See “Currency Rate Fluctuations,” for additional information.

In March 2023, the bank failures of Silicon Valley Bank and Signature Bank created significant market disruption and uncertainty within the U.S. banking sector, in particular with respect to regional banks. We hold minimal cash balances with regional banks in the U.S. We have a robust and disciplined cash management process to protect our cash and maintain financial stability. Further, while we serve many clients in the Financial Services Sector (FSS), less than one percent of our total revenue comes from regional banks. We have not seen any notable changes in the buying behaviors of our FSS clients in the U.S. In addition, we have not experienced a significant impact to our results of operations or financial position and continue to monitor the impacts of this situation.

Financial Performance Summary — Three Months Ended June 30:

In the second quarter of 2023, we reported \$15.5 billion in revenue, income from continuing operations of \$1.6 billion and operating (non-GAAP) earnings of \$2.0 billion. Diluted earnings per share from continuing operations was \$1.72 as reported and \$2.18 on an operating (non-GAAP) basis. We generated \$2.6 billion in cash from operations and \$2.1 billion in free cash flow, and delivered shareholder returns of \$1.5 billion in dividends. Our second-quarter results reflect the continued execution of our hybrid cloud and AI strategy. We had continued strength in our growth areas of Software and Consulting, and solid cash generation. We continued to invest in innovation while transforming our business processes and driving productivity.

Total revenue decreased 0.4 percent as reported and was up modestly adjusted for currency compared to the prior-year period. Software delivered revenue growth of 7.2 percent as reported and 8 percent adjusted for currency, as clients

Management Discussion – (continued)

leverage our hybrid cloud and AI platform capabilities. Hybrid Platform & Solutions revenue was up 6.3 percent as reported and 7 percent adjusted for currency, led by growth in Red Hat, Data & AI and Automation. Transaction Processing grew 9.4 percent as reported and 10 percent adjusted for currency, as this software remains core to supporting clients' mission-critical workloads. Consulting revenue increased 4.3 percent as reported and 6 percent adjusted for currency, with growth across all lines of business, and reflecting the continued demand for technology-driven transformation as clients prioritize projects that drive cost savings and increase productivity. Infrastructure revenue decreased 14.6 percent year to year as reported and 14 percent adjusted for currency, reflecting product cycle dynamics which impacted both Hybrid Infrastructure and Infrastructure Support.

From a geographic perspective, Americas revenue decreased 1.2 percent year to year as reported and was flat adjusted for currency, including the impact from declines in Infrastructure and the divestiture of our healthcare software assets in the second quarter of 2022. Europe/Middle East/Africa (EMEA) increased 1.7 percent and was flat adjusted for currency. Asia Pacific decreased 1.4 percent, but grew 3 percent adjusted for currency.

Gross margin of 54.9 percent increased 1.6 points year to year with continued margin expansion across all reportable segments driven by our improving portfolio mix and productivity actions. Operating (non-GAAP) gross margin of 55.9 percent increased 1.4 points compared to the prior-year period due to the same dynamics.

Total expense and other (income) decreased 1.0 percent in the second quarter of 2023 versus the prior-year period primarily driven by lower non-operating retirement-related costs, higher interest income, prior-year impacts related to the Kyndryl retained shares and swap, and benefits from productivity and transformation of our business processes. This was partially offset by lower gains from divestitures, higher interest expense, higher workforce rebalancing charges, the effects of currency and higher net spending to drive our hybrid cloud and AI strategy. Total operating (non-GAAP) expense and other (income) increased 5.1 percent year to year, primarily driven by lower gains from divestitures, higher interest expense, the effects of currency, higher workforce rebalancing charges and higher net spending to drive our strategy; partially offset by higher interest income and benefits from productivity and transformation initiatives.

Pre-tax income from continuing operations of \$2.0 billion increased 16.2 percent and pre-tax margin was 12.9 percent, an increase of 1.8 points versus the second quarter of 2022. Performance this quarter benefited from the expense dynamics described above, and improvements in business mix and ongoing productivity initiatives. In the second quarter of 2022, we recorded a pre-tax gain of approximately \$230 million from the sale of our healthcare software assets which impacted the pre-tax income from continuing operations year-to-year performance by approximately 17 points and the pre-tax margin by 1.4 points. In addition, in the second-quarter 2023 workforce rebalancing charges of \$117 million impacted the pre-tax income from continuing operations year-to-year performance by approximately 5 points and the pre-tax margin by 0.6 points. The continuing operations provision for income taxes in the second quarter of 2023 was \$419 million compared to \$257 million in the second quarter of 2022. The increase was primarily driven by the impact of foreign tax credit regulations. Net income from continuing operations of \$1.6 billion increased 7.9 percent and the net income from continuing operations margin was 10.2 percent, up 0.8 points year to year. The gains from the healthcare software assets divestiture impacted net income from continuing operations year-to-year performance by approximately 15 points and net income margin by 1.2 points.

Operating (non-GAAP) pre-tax income from continuing operations of \$2.4 billion decreased 4.8 percent compared to the prior-year period and the operating (non-GAAP) pre-tax margin from continuing operations decreased 0.7 points to 15.5 percent. The year-to-year gains from the healthcare software assets divestiture impacted these results by approximately 9 points and 1.4 points, respectively. The higher workforce rebalancing charges in the second quarter 2023 impacted these results by approximately 4 points and 0.6 points, respectively. The operating (non-GAAP) income tax provision for the second quarter of 2023 was \$393 million, compared to \$413 million in the second quarter of 2022. Operating (non-GAAP) net income from continuing operations of \$2.0 billion decreased 4.8 percent and the operating (non-GAAP) net income margin from continuing operations of 12.9 percent was down 0.6 points year to year. The gains from the healthcare software assets divestiture impacted these results by approximately 9 points and 1.2 points, respectively.

Diluted earnings per share from continuing operations of \$1.72 in the second quarter of 2023 increased 6.8 percent and operating (non-GAAP) diluted earnings per share of \$2.18 decreased 5.6 percent versus the prior-year period. The gains from the divestiture of our healthcare software assets resulted in an impact of 15 points and 9 points to diluted earnings per share from continuing operations and diluted operating (non-GAAP) earnings per share, respectively.

Management Discussion – (continued)

Cash provided by operating activities was \$2.6 billion in the second quarter of 2023, an increase of \$1.3 billion compared to the second quarter of 2022. Net cash provided by investing activities was immaterial, a decline of \$0.2 billion and net cash used in financing activities of \$2.7 billion decreased \$1.5 billion compared to the second quarter of 2022.

Financial Results Summary —Six Months Ended June 30:

(Dollars and shares in millions except per share amounts)			Yr. to Yr. Percent/ Margin Change
For the six months ended June 30:	2023	2022	
Revenue	\$ 29,727	\$ 29,732	0.0 %*
Gross profit margin	53.9 %	52.6 %	1.3 pts.
Total expense and other (income)	\$ 12,952	\$ 13,280	(2.5) %
Income from continuing operations before income taxes	\$ 3,058	\$ 2,345	30.4 %
Provision for income taxes from continuing operations	\$ 543	\$ 218	149.7 %
Income from continuing operations	\$ 2,515	\$ 2,127	18.2 %
Income from continuing operations margin	8.5 %	7.2 %	1.3 pts.
Loss from discontinued operations, net of tax	\$ (4)	\$ (2)	128.1 %
Net income	\$ 2,511	\$ 2,125	18.1 %
Earnings per share from continuing operations - assuming dilution	\$ 2.74	\$ 2.34	17.1 %
Consolidated earnings per share - assuming dilution	\$ 2.73	\$ 2.34	16.7 %
Weighted-average shares outstanding - assuming dilution	918.6	910.0	1.0 %
	At 6/30/2023	At 12/31/2022	
Assets	\$ 132,213	\$ 127,243	3.9 %
Liabilities	\$ 109,942	\$ 105,222	4.5 %
Equity	\$ 22,271	\$ 22,021	1.1 %

* 2.3% percent adjusted for currency.

The following table provides the company's operating (non-GAAP) earnings for the first six months of 2023 and 2022.

(Dollars in millions except per share amounts)			Yr. to Yr. Percent Change
For the six months ended June 30:	2023	2022	
Net income as reported	\$ 2,511	\$ 2,125	18.1 %
Loss from discontinued operations, net of tax	(4)	(2)	128.1
Income from continuing operations	\$ 2,515	\$ 2,127	18.2 %
Non-operating adjustments (net of tax):			
Acquisition-related charges	\$ 613	\$ 704	(13.0) %
Non-operating retirement-related costs/(income)	10	290	(96.7)
U.S. tax reform impacts	115	(112)	nm
Kyndryl-related impacts	—	367	(100.0)
Operating (non-GAAP) earnings	\$ 3,252	\$ 3,376	(3.7) %
Diluted operating (non-GAAP) earnings per share	\$ 3.54	\$ 3.71	(4.6) %

nm - not meaningful

Management Discussion – (continued)

Financial Performance Summary —Six Months Ended June 30:

In the first six months of 2023, we reported \$29.7 billion in revenue, income from continuing operations of \$2.5 billion and operating (non-GAAP) earnings of \$3.3 billion. Diluted earnings per share from continuing operations was \$2.74 as reported and \$3.54 on an operating (non-GAAP) basis. We generated \$6.4 billion in cash from operations and \$3.4 billion in free cash flow, and delivered shareholder returns of \$3.0 billion in dividends. Our first-half performance reflects the momentum in our growth areas of Software and Consulting, and a solid recurring revenue base driven by our high-value software.

Total revenue was flat as reported and grew 2 percent adjusted for currency compared to the prior-year period. Software delivered revenue growth of 5.0 percent as reported and 7 percent adjusted for currency, with growth in both Hybrid Platform & Solutions and Transaction Processing. Consulting revenue increased 3.5 percent as reported and 7 percent adjusted for currency, with growth across all lines of business. Infrastructure revenue decreased 9.9 percent as reported and 8 percent adjusted for currency, reflecting product cycle dynamics which impacted both Hybrid Infrastructure and Infrastructure Support.

From a geographic perspective, Americas revenue decreased 0.5 percent year to year as reported and was flat adjusted for currency. EMEA increased 2.0 percent (4 percent adjusted for currency). Asia Pacific decreased 1.9 percent but grew 5 percent adjusted for currency.

Gross margin of 53.9 percent increased 1.3 points year to year with continued gross profit expansion across all reportable segments driven by our improving portfolio mix and productivity initiatives. Operating (non-GAAP) gross margin of 54.9 percent increased 1.1 points compared to the prior-year period due to the same dynamics.

Total expense and other (income) decreased 2.5 percent in the first six months of 2023 versus the prior-year period primarily driven by lower non-operating retirement-related costs, prior-year impacts related to the Kyndryl retained shares and swap, higher interest income and benefits from productivity and transformation of our business processes. This was partially offset by higher workforce rebalancing charges, lower gains from divestitures, higher interest expense and higher net spending to drive our hybrid cloud and AI strategy. Total operating (non-GAAP) expense and other (income) increased 4.3 percent year to year, driven primarily by higher workforce rebalancing charges, lower gains from divestitures, higher interest expense and higher net spending to drive our strategy; partially offset by higher interest income and benefits from productivity and transformation initiatives.

Pre-tax income from continuing operations of \$3.1 billion increased 30.4 percent and pre-tax margin was 10.3 percent, an increase of 2.4 points versus the first half of 2022. Performance in the first six months of 2023 benefited from the expense dynamics described above, improvements in business mix and ongoing productivity initiatives. The continuing operations provision for income taxes in the first six months of 2023 was \$543 million, compared to \$218 million in the first six months of 2022. The increase was primarily driven by the impact of foreign tax credit regulations. Net income from continuing operations of \$2.5 billion increased 18.2 percent and the net income from continuing operations margin was 8.5 percent, up 1.3 points year to year.

Operating (non-GAAP) pre-tax income from continuing operations of \$3.8 billion declined 4.7 percent compared to the prior-year period and the operating (non-GAAP) pre-tax margin from continuing operations decreased 0.6 points to 12.9 percent. The operating (non-GAAP) provision for income taxes was \$593 million in the first six months of 2023, compared to \$657 million in the first six months of 2022. Operating (non-GAAP) income from continuing operations of \$3.3 billion decreased 3.7 percent and the operating (non-GAAP) income margin from continuing operations of 10.9 percent decreased 0.4 points year to year.

Diluted earnings per share from continuing operations of \$2.74 in the first six months of 2023 increased 17.1 percent and operating (non-GAAP) diluted earnings per share of \$3.54 decreased 4.6 percent versus the prior-year period.

At June 30, 2023, the balance sheet remained strong with the flexibility to support and invest in the business. Cash and cash equivalents, restricted cash and marketable securities at June 30, 2023 of \$16.3 billion increased \$7.5 billion from December 31, 2022 and debt of \$57.5 billion at June 30, 2023 increased \$6.5 billion.

Total assets increased \$5.0 billion (\$4.5 billion adjusted for currency) from December 31, 2022 primarily driven by an increase in cash and cash equivalents and marketable securities; partially offset by a decrease in receivables. Total liabilities increased \$4.7 billion (\$4.3 billion adjusted for currency) from December 31, 2022 primarily driven by the

Management Discussion – (continued)

increase in debt and an increase in deferred income; partially offset by decreases in tax liabilities. Total equity of \$22.3 billion increased \$0.3 billion from December 31, 2022 primarily driven by first-half 2023 net income and common stock issuances; partially offset by dividends paid.

Cash provided by operating activities was \$6.4 billion in the first six months of 2023, an increase of \$1.8 billion. Net cash used in investing activities of \$8.0 billion increased \$6.8 billion compared to the prior-year period. Cash from financing activities was a net source of cash of \$3.0 billion in the first six months of 2023, compared to a net use of cash of \$2.8 billion in the prior period.

Second Quarter and First Six Months in Review

Results of Continuing Operations

Segment Details

The following tables present each reportable segment’s revenue and gross margin results, followed by an analysis of the second quarter and first six months of 2023 versus the second quarter and first six months of 2022 reportable segments results.

(Dollars in millions)				Yr. to Yr. Percent/Margin Change	Yr. to Yr. Percent Change Adjusted For Currency
For the three months ended June 30:	2023	2022			
Revenue:					
Software	\$ 6,608	\$ 6,166		7.2 %	7.5 %
Gross margin	79.3 %	79.2 %		0.1 pts.	
Consulting	5,013	4,809		4.3 %	5.9 %
Gross margin	25.9 %	24.2 %		1.8 pts.	
Infrastructure	3,618	4,235		(14.6) %	(13.8) %
Gross margin	55.8 %	53.8 %		2.0 pts.	
Financing	185	146		26.2 %	27.1 %
Gross margin	49.2 %	35.3 %		13.9 pts.	
Other	51	180		(71.9) %	(72.9) %
Gross margin	(295.7) %	(49.3) %		(246.5) pts.	
Total revenue	\$ 15,475	\$ 15,535		(0.4) %	0.4 %
Total gross profit	\$ 8,501	\$ 8,290		2.5 %	
Total gross margin	54.9 %	53.4 %		1.6 pts.	
Non-operating adjustments:					
Amortization of acquired intangible assets	150	180		(17.1) %	
Operating (non-GAAP) gross profit	\$ 8,650	\$ 8,470		2.1 %	
Operating (non-GAAP) gross margin	55.9 %	54.5 %		1.4 pts.	

Management Discussion – (continued)

(Dollars in millions)			Yr. to Yr. Percent/Margin Change	Yr. to Yr. Percent Change Adjusted For Currency
For the six months ended June 30:	2023	2022		
Revenue:				
Software	\$ 12,529	\$ 11,938	5.0 %	6.6 %
Gross margin	79.4 %	79.0 %	0.3 pts.	
Consulting	9,975	9,637	3.5 %	7.0 %
Gross margin	25.6 %	24.3 %	1.3 pts.	
Infrastructure	6,716	7,453	(9.9) %	(7.8) %
Gross margin	53.9 %	52.4 %	1.6 pts.	
Financing	380	300	26.8 %	29.1 %
Gross margin	46.5 %	36.5 %	10.0 pts.	
Other	126	404	(68.9) %	(68.6) %
Gross margin	(228.2) %	(40.2) %	(188.0) pts.	
Total revenue	\$ 29,727	\$ 29,732	0.0 %	2.3 %
Total gross profit	\$ 16,010	\$ 15,625	2.5 %	
Total gross margin	53.9 %	52.6 %	1.3 pts.	
Non-operating adjustments:				
Amortization of acquired intangible assets	298	361	(17.5) %	
Operating (non-GAAP) gross profit	\$ 16,308	\$ 15,986	2.0 %	
Operating (non-GAAP) gross margin	54.9 %	53.8 %	1.1 pts.	

Software

(Dollars in millions)			Yr. to Yr. Percent Change	Yr. to Yr. Percent Change Adjusted For Currency
For the three months ended June 30:	2023	2022		
Software revenue:	\$ 6,608	\$ 6,166	7.2 %	7.5 %
Hybrid Platform & Solutions	\$ 4,665	\$ 4,390	6.3 %	6.6 %
Red Hat			10.7	10.8
Automation			1.2	1.6
Data & AI			10.0	10.6
Security			(1.5)	(1.0)
Transaction Processing	1,943	1,776	9.4	9.7

(Dollars in millions)			Yr. to Yr. Percent Change	Yr. to Yr. Percent Change Adjusted For Currency
For the six months ended June 30:	2023	2022		
Software revenue:	\$ 12,529	\$ 11,938	5.0 %	6.6 %
Hybrid Platform & Solutions	\$ 8,844	\$ 8,470	4.4 %	5.9 %
Red Hat			9.5	10.8
Automation			0.2	1.8
Data & AI			5.7	7.2
Security			(1.4)	0.4
Transaction Processing	3,685	3,468	6.3	8.1

Management Discussion – (continued)

Software revenue of \$6,608 million increased 7.2 percent as reported (8 percent adjusted for currency) in the second quarter of 2023 compared to the prior-year period, driven by revenue growth in both Hybrid Platform & Solutions and Transaction Processing as clients leverage our hybrid cloud and AI platform capabilities. This revenue performance reflects continued growth across our recurring revenue base, which is approximately 80 percent of our annual software revenue, as well as transactional revenue.

Hybrid Platform & Solutions revenue of \$4,665 million increased 6.3 percent as reported (7 percent adjusted for currency) in the second quarter of 2023 compared to the prior-year period, led by growth in Red Hat, Data & AI, and Automation. Red Hat revenue increased 10.7 percent as reported (11 percent adjusted for currency). OpenShift, our leading hybrid cloud platform, grew more than 30 percent year to year in the second-quarter 2023 and had \$1.1 billion in annual recurring revenue. Ansible also had double-digit revenue growth in the second quarter of 2023 compared to 2022 and gained market share. Automation revenue increased 1.2 percent as reported (2 percent adjusted for currency), reflecting growth across Integration, Application Servers and Business Automation, as clients drive enhanced business value through productivity and performance optimization. Data & AI revenue increased 10.0 percent as reported (11 percent adjusted for currency) with broad-based growth in areas such as Data Management and Business Analytics given enterprise needs for data visualization, organization, analysis, and insights as the underpinnings for AI workloads. Security revenue decreased 1.5 percent as reported (1 percent adjusted for currency) driven by declines in security services, partially offset by growth in security software led by Data Security with Guardium Insights.

Across Hybrid Platform & Solutions, our annual recurring revenue (ARR) was \$13.6 billion. ARR is a key performance metric management uses to assess the health and growth trajectory of our Hybrid Platform & Solutions business within the Software segment. ARR is calculated by estimating the current quarter’s recurring, committed value for certain types of active contracts as of the period-end date and then multiplying that value by four. This value is based on each arrangement’s contract value and start date, mitigating fluctuations during the contract term, and includes the following consumption models: (1) software subscription agreements, including committed term licenses, (2) as-a-service arrangements such as SaaS and PaaS, (3) maintenance and support contracts, and (4) security managed services contracts. ARR should be viewed independently of revenue as this performance metric and its inputs may not represent the amount of revenue recognized in the period and therefore is not intended to represent current period revenue or revenue that will be recognized in future periods. ARR is calculated at estimated constant currency.

Transaction Processing revenue of \$1,943 million increased 9.4 percent as reported (10 percent adjusted for currency) in the second quarter of 2023 compared to the prior-year period. The increase in zSystems installed capacity over the last couple of product cycles, strong renewal rates and higher price increases contributed to growth in recurring and transactional revenue.

For the first six months of 2023, Software revenue of \$12,529 million increased 5.0 percent as reported (7 percent adjusted for currency) compared to the same period in 2022, driven by solid growth in Hybrid Platform & Solutions, led by Red Hat and Data & AI, and in Transaction Processing.

(Dollars in millions)					Yr. to Yr. Percent/ Margin Change
For the three months ended June 30:	2023		2022		
Software:					
Gross profit	\$	5,239	\$	4,884	7.3 %
Gross profit margin		79.3 %		79.2 %	0.1 pts.
Pre-tax income	\$	1,504	\$	1,375	9.4 %
Pre-tax margin		22.8 %		22.3 %	0.5 pts.

Management Discussion – (continued)

(Dollars in millions)				Yr. to Yr. Percent/ Margin Change
For the six months ended June 30:		2023	2022	
Software:				
Gross profit	\$	9,944	\$ 9,434	5.4 %
Gross profit margin		79.4 %	79.0 %	0.3 pts.
Pre-tax income	\$	2,668	\$ 2,509	6.3 %
Pre-tax margin		21.3 %	21.0 %	0.3 pts.

Software gross profit margin increased 0.1 points to 79.3 percent in the second quarter of 2023 compared to the prior-year period, driven primarily by portfolio mix, partially offset by a margin decline in software products. For the first six months of 2023, gross profit margin increased 0.3 points to 79.4 percent, driven primarily by portfolio mix.

In the second quarter, pre-tax income of \$1,504 million increased 9.4 percent and pre-tax margin of 22.8 percent increased 0.5 points compared to the prior year. The margin expansion reflects operating leverage from higher revenue and product mix while absorbing more than a point of impact from currency. For the first six months of 2023, pre-tax income of \$2,668 million increased 6.3 percent and pre-tax margin of 21.3 percent increased 0.3 points compared to the prior-year period, which included approximately a point of impact from currency.

Consulting

(Dollars in millions)				Yr. to Yr. Percent Change	Yr. to Yr. Percent Change Adjusted For Currency
For the three months ended June 30:		2023	2022		
Consulting revenue:	\$	5,013	\$ 4,809	4.3 %	5.9 %
Business Transformation	\$	2,295	\$ 2,227	3.0 %	4.5 %
Technology Consulting		961	928	3.5	5.1
Application Operations		1,758	1,653	6.3	8.1

(Dollars in millions)				Yr. to Yr. Percent Change	Yr. to Yr. Percent Change Adjusted For Currency
For the six months ended June 30:		2023	2022		
Consulting revenue:	\$	9,975	\$ 9,637	3.5 %	7.0 %
Business Transformation	\$	4,578	\$ 4,482	2.1 %	5.5 %
Technology Consulting		1,904	1,884	1.1	4.6
Application Operations		3,494	3,272	6.8	10.5

Consulting revenue of \$5,013 million increased 4.3 percent as reported (6 percent adjusted for currency) in the second quarter of 2023 compared to the prior-year period. Revenue grew in all lines of business in Consulting, driven by broad-based growth across our service offerings. This growth reflects sustained demand for larger transformations that delivered meaningful returns on investment, while some clients, predominantly in the U.S., continued to delay projects considered to be more discretionary. Our Red Hat consulting practice had double-digit growth in signings and revenue, and our strategic partnerships grew signings and revenue in the second quarter at a double-digit rate on a year-to-year basis.

In the second quarter of 2023, Business Transformation revenue of \$2,295 million increased 3.0 percent as reported (5 percent adjusted for currency) compared to the prior-year period, driven by data and technology transformations including AI and analytics focused projects. Digital transformations continue to be underpinned by clients' embracing a hybrid cloud strategy.

Management Discussion – (continued)

Technology Consulting revenue of \$961 million increased 3.5 percent as reported (5 percent adjusted for currency) in the second quarter of 2023 compared to the prior-year period, led by our cloud application development and cloud modernization offerings.

Application Operations revenue of \$1,758 million increased 6.3 percent as reported (8 percent adjusted for currency) compared to the second quarter of 2022, driven by growth in cloud application management services as we helped clients optimize their operations and reduce cost by managing clients’ applications in hybrid and multi-cloud environments.

For the first six months of 2023, Consulting revenue of \$9,975 million increased 3.5 percent as reported (7 percent adjusted for currency) reflecting year-to-year growth across all three lines of business. Business Transformation revenue grew year to year led by growth in data and technology and customer experience transformation projects. In our Technology Consulting business, we led client engagements around cloud application development and modernization. Through our Application Operations offerings, we continued to provide application and cloud platform services required to operationalize and run cloud platforms.

(Dollars in millions)					Yr. to Yr. Percent/ Margin Change
For the three months ended June 30:	2023		2022		
Consulting:					
Gross profit	\$	1,300	\$	1,163	11.8 %
Gross profit margin		25.9 %		24.2 %	1.8 pts.
Pre-tax income	\$	446	\$	343	30.1 %
Pre-tax margin		8.9 %		7.1 %	1.8 pts.

(Dollars in millions)					Yr. to Yr. Percent/ Margin Change
For the six months ended June 30:	2023		2022		
Consulting:					
Gross profit	\$	2,553	\$	2,339	9.1 %
Gross profit margin		25.6 %		24.3 %	1.3 pts.
Pre-tax income	\$	828	\$	691	19.8 %
Pre-tax margin		8.3 %		7.2 %	1.1 pts.

In the second quarter of 2023, Consulting gross profit margin of 25.9 percent increased 1.8 points on a year-to-year basis. Pre-tax income of \$446 million increased 30.1 percent and pre-tax margin of 8.9 percent increased 1.8 points in second-quarter 2023 compared to the same prior-year period. Our gross profit margin expansion and pre-tax margin performance are a reflection of the pricing and productivity actions we have taken during the past year, which are more than offsetting the increased labor costs and investments.

For the first six months of 2023, Consulting gross profit margin of 25.6 percent increased 1.3 points compared to the prior-year period. Pre-tax income of \$828 million increased 19.8 percent and pre-tax margin of 8.3 percent increased 1.1 points in the first six months of 2023 compared to the prior-year period. The six-month margin performance was driven by the same factors as described above for the second quarter.

Management Discussion – (continued)

Consulting Signings and Book-to-Bill

(Dollars in millions)			Yr. to Yr. Percent Change	Yr. to Yr. Percent Change Adjusted For Currency
For the three months ended June 30:	2023	2022		
Total Consulting signings	\$ 5,667	\$ 4,654	21.8 %	24.5 %

(Dollars in millions)			Yr. to Yr. Percent Change	Yr. to Yr. Percent Change Adjusted For Currency
For the six months ended June 30:	2023	2022		
Total Consulting signings	\$ 10,860	\$ 9,791	10.9 %	15.1 %

In the second quarter of 2023, Consulting signings grew 22 percent as reported and 24 percent adjusted for currency, with solid growth in both large and small engagements, and our book-to-bill ratio was 1.1 over the last twelve months. We addressed continued demand for technology-driven transformation as clients prioritize projects that drive cost savings and increase productivity.

Book-to-bill represents the ratio of IBM Consulting signings to its revenue over the same period. The metric is a useful indicator of the demand of our business over time. Signings are management’s initial estimate of the value of a client’s commitment under a services contract within IBM Consulting. There are no third-party standards or requirements governing the calculation of signings. The calculation used by management involves estimates and judgments to gauge the extent of a client’s commitment, including the type and duration of the agreement, and the presence of termination charges or wind-down costs.

Contract extensions and increases in scope are treated as signings only to the extent of the incremental new value. Total signings can vary over time due to a variety of factors including, but not limited to, the timing of signing a small number of larger contracts. Signings associated with an acquisition will be recognized on a prospective basis.

Management believes the estimated values of signings disclosed provide an indication of our forward-looking revenue. Signings are used to monitor the performance of the business and viewed as useful information for management and shareholders. The conversion of signings into revenue may vary based on the types of services and solutions, contract duration, customer decisions, and other factors, which may include, but are not limited to, the macroeconomic environment.

Management Discussion – (continued)

Infrastructure

(Dollars in millions)				Yr. to Yr. Percent Change	Yr. to Yr. Percent Change Adjusted For Currency
For the three months ended June 30:		2023	2022		
Infrastructure revenue:	\$	3,618	\$ 4,235	(14.6) %	(13.8) %
Hybrid Infrastructure	\$	2,260	\$ 2,760	(18.1) %	(17.8) %
zSystems				(30.1)	(30.0)
Distributed Infrastructure				(6.5)	(5.9)
Infrastructure Support		1,358	1,474	(7.9)	(6.3)

(Dollars in millions)				Yr. to Yr. Percent Change	Yr. to Yr. Percent Change Adjusted For Currency
For the six months ended June 30:		2023	2022		
Infrastructure revenue:	\$	6,716	\$ 7,453	(9.9) %	(7.8) %
Hybrid Infrastructure	\$	3,969	\$ 4,461	(11.0) %	(9.6) %
zSystems				(19.0)	(17.8)
Distributed Infrastructure				(4.9)	(3.2)
Infrastructure Support		2,747	2,993	(8.2)	(5.1)

Infrastructure revenue of \$3,618 million decreased 14.6 percent as reported (14 percent adjusted for currency) in the second quarter of 2023 compared to the prior-year period, reflecting product cycle dynamics which impacted both Hybrid Infrastructure and Infrastructure Support.

Hybrid Infrastructure revenue of \$2,260 million decreased 18.1 percent as reported (18 percent adjusted for currency) in the second quarter of 2023 compared to the prior-year period. Within Hybrid Infrastructure, zSystems revenue decreased 30.1 percent as reported (30 percent adjusted for currency), driven by strong revenue performance in the prior year as the z16 launched in a seasonally strong quarter. Through the first five quarters of availability, z16 program revenue exceeded the prior cycles, as it brings the power of embedded AI at scale, cyber-resilient security and cloud-native development for hybrid cloud to our clients. Distributed Infrastructure revenue decreased 6.5 percent as reported (6 percent adjusted for currency), driven by strong revenue growth in the prior year led by Storage and Power10 high-end systems.

Infrastructure Support revenue of \$1,358 million decreased 7.9 percent as reported (6 percent adjusted for currency) in the second quarter of 2023 compared to the prior-year period, reflecting product cycle dynamics.

For the first six months of 2023, Infrastructure revenue of \$6,716 million decreased 9.9 percent as reported (8 percent adjusted for currency) compared to the prior-year period, driven by declines in Hybrid Infrastructure and Infrastructure Support. Within Hybrid Infrastructure, the revenue decline was primarily driven by zSystems. The revenue decline in Infrastructure Support for the first six months of 2023 reflects product cycle dynamics.

(Dollars in millions)				Yr. to Yr. Percent/ Margin Change
For the three months ended June 30:		2023	2022	
Infrastructure:				
Gross profit	\$	2,021	\$ 2,280	(11.4) %
Gross profit margin		55.8 %	53.8 %	2.0 pts.
Pre-tax income	\$	633	\$ 757	(16.4) %
Pre-tax margin		17.5 %	17.9 %	(0.4) pts.

Management Discussion – (continued)

(Dollars in millions)					Yr. to Yr. Percent/ Margin Change
For the six months ended June 30:		2023	2022		
Infrastructure:					
Gross profit	\$	3,623	\$	3,905	(7.2) %
Gross profit margin		53.9 %		52.4 %	1.6 pts.
Pre-tax income	\$	849	\$	956	(11.2) %
Pre-tax margin		12.6 %		12.8 %	(0.2) pts.

Infrastructure gross profit margin of 55.8 percent increased 2.0 points in the second quarter of 2023 compared to the prior-year period, driven by margin expansion in Hybrid Infrastructure, partially offset by a margin decline in Infrastructure Support reflecting product cycle dynamics. For the first six months of 2023, gross profit margin of 53.9 percent increased 1.6 points compared to the prior-year period, driven by the same factors as described above for the second quarter.

In the second quarter of 2023, Infrastructure pre-tax income of \$633 million decreased 16.4 percent and pre-tax margin of 17.5 percent decreased 0.4 points compared to the prior-year period. This performance reflects declines in gross profit contributions from Hybrid Infrastructure and Infrastructure Support primarily driven by product cycle dynamics. The declines are partially offset by a year-to-year increase in IP and custom development income and the benefit of the changes in the useful life of servers and network equipment effective January 1, 2023. The decline in pre-tax margin in the second quarter also included approximately a point of impact from currency. For the first six months of 2023, Infrastructure pre-tax income of \$849 million decreased 11.2 percent and pre-tax margin of 12.6 percent decreased 0.2 points compared to the prior-year period, driven primarily by the same factors described above for the second quarter.

Financing

See pages 78 through 80 for a discussion of Financing’s segment results.

Geographic Revenue

In addition to the revenue presentation by reportable segment, we also measure revenue performance on a geographic basis.

(Dollars in millions)				Yr. to Yr. Percent Change	Yr. to Yr. Percent Change Adjusted For Currency
For the three months ended June 30:		2023	2022		
Total Revenue	\$	15,475	\$	15,535	(0.4) %
Americas	\$	8,046	\$	8,142	(1.2) %
Europe/Middle East/Africa (EMEA)		4,602		4,526	1.7
Asia Pacific		2,827		2,868	(1.4)

(Dollars in millions)				Yr. to Yr. Percent Change	Yr. to Yr. Percent Change Adjusted For Currency
For the six months ended June 30:		2023	2022		
Total Revenue	\$	29,727	\$	29,732	0.0 %
Americas	\$	15,124	\$	15,198	(0.5) %
Europe/Middle East/Africa (EMEA)		8,933		8,757	2.0
Asia Pacific		5,670		5,778	(1.9)

Management Discussion – (continued)

Total revenue of \$15,475 million decreased 0.4 percent as reported and was up modestly adjusted for currency in the second quarter of 2023 compared to the prior-year period.

Americas revenue of \$8,046 million decreased 1.2 percent as reported and was flat adjusted for currency. The U.S. decreased 2.2 percent compared to the prior year. Canada decreased 13.6 percent as reported and 9 percent adjusted for currency. Latin America increased 23.2 percent as reported and 26 percent adjusted for currency, with Brazil increasing 15.4 percent as reported and 16 percent adjusted for currency.

In EMEA, total revenue of \$4,602 million increased 1.7 percent as reported and was flat adjusted for currency. Italy and the UK increased 5.8 percent and 1.3 percent, respectively, as reported, and 4 percent and 1 percent, respectively, adjusted for currency. France was flat as reported and decreased 3 percent adjusted for currency. Germany decreased 11.3 percent as reported and 13 percent adjusted for currency.

Asia Pacific revenue of \$2,827 million decreased 1.4 percent as reported, but increased 3 percent adjusted for currency. Japan increased 2.6 percent as reported and 9 percent adjusted for currency. India increased 4.3 percent as reported and 11 percent adjusted for currency. China and Australia decreased 19.0 percent and 13.0 percent, respectively, as reported, and 15 percent and 7 percent, respectively, adjusted for currency.

For the first six months of 2023, total revenue of \$29,727 million was flat as reported and increased 2.3 percent adjusted for currency compared to the prior-year period.

Americas revenue of \$15,124 million decreased 0.5 percent as reported and was flat adjusted for currency. The U.S. decreased 1.4 percent compared to the prior-year period. Canada decreased 6.2 percent as reported and 1 percent adjusted for currency. Latin America increased 16.3 percent as reported and 19 percent adjusted for currency, with Brazil increasing 14.7 percent as reported and 15 percent adjusted for currency.

In EMEA, total revenue of \$8,933 million increased 2.0 percent as reported and 4 percent adjusted for currency. Italy and France increased 4.2 percent and 1.3 percent, respectively, as reported, and 5 percent and 2 percent, respectively, adjusted for currency. The UK decreased 5.0 percent as reported and was flat adjusted for currency. Germany decreased 8.5 percent as reported and 8 percent adjusted for currency.

Asia Pacific revenue of \$5,670 million decreased 1.9 percent as reported, but grew 5 percent adjusted for currency. Japan was flat as reported and increased 10 percent adjusted for currency. India increased 6.3 percent as reported and 15 percent adjusted for currency. China and Australia decreased 21.7 percent and 14.5 percent, respectively, as reported, and 18 percent and 9 percent, respectively, adjusted for currency.

Management Discussion – (continued)

Expense

Total Expense and Other (Income)

(Dollars in millions)			
For the three months ended June 30:	2023	2022	Yr. to Yr. Percent Change
Total expense and other (income)	\$ 6,501	\$ 6,568	(1.0) %
Non-operating adjustments:			
Amortization of acquired intangible assets	\$ (239)	\$ (278)	(14.0) %
Acquisition-related charges	(7)	(2)	nm
Non-operating retirement-related (costs)/income	(1)	(192)	(99.4)
Kyndryl-related impacts	—	(145)	(100.0)
Operating (non-GAAP) expense and other (income)	\$ 6,254	\$ 5,952	5.1 %
Total expense-to-revenue ratio	42.0 %	42.3 %	(0.3) pts.
Operating (non-GAAP) expense-to-revenue ratio	40.4 %	38.3 %	2.1 pts.

(Dollars in millions)			
For the six months ended June 30:	2023	2022	Yr. to Yr. Percent Change
Total expense and other (income)	\$ 12,952	\$ 13,280	(2.5) %
Non-operating adjustments:			
Amortization of acquired intangible assets	\$ (483)	\$ (558)	(13.4) %
Acquisition-related charges	(10)	(9)	17.6
Non-operating retirement-related (costs)/income	4	(394)	nm
Kyndryl-related impacts	—	(367)	(100.0)
Operating (non-GAAP) expense and other (income)	\$ 12,463	\$ 11,953	4.3 %
Total expense-to-revenue ratio	43.6 %	44.7 %	(1.1) pts.
Operating (non-GAAP) expense-to-revenue ratio	41.9 %	40.2 %	1.7 pts.

nm - not meaningful

For additional information regarding total expense and other (income) for both expense presentations, see the following analyses by category.

Management Discussion – (continued)

Selling, General and Administrative Expense

(Dollars in millions)			
For the three months ended June 30:	2023	2022	Yr. to Yr. Percent Change
Selling, general and administrative expense:			
Selling, general and administrative — other	\$ 3,991	\$ 3,996	(0.1) %
Advertising and promotional expense	372	395	(5.8)
Workforce rebalancing charges	117	28	nm
Amortization of acquired intangible assets	239	277	(13.9)
Stock-based compensation	168	153	10.3
Provision for/(benefit from) expected credit loss expense	14	6	129.1
Total selling, general and administrative expense	<u>\$ 4,900</u>	<u>\$ 4,855</u>	<u>0.9 %</u>
Non-operating adjustments:			
Amortization of acquired intangible assets	\$ (239)	\$ (277)	(13.9) %
Acquisition-related charges	(7)	(2)	nm
Kyndryl-related impacts	—	0	nm
Operating (non-GAAP) selling, general and administrative expense	<u>\$ 4,655</u>	<u>\$ 4,576</u>	<u>1.7 %</u>

(Dollars in millions)			
For the six months ended June 30:	2023	2022	Yr. to Yr. Percent Change
Selling, general and administrative expense:			
Selling, general and administrative — other	\$ 7,877	\$ 7,820	0.7 %
Advertising and promotional expense	687	732	(6.1)
Workforce rebalancing charges	376	33	nm
Amortization of acquired intangible assets	482	557	(13.4)
Stock-based compensation	317	289	9.9
Provision for/(benefit from) expected credit loss expense	16	22	(29.5)
Total selling, general and administrative expense	<u>\$ 9,754</u>	<u>\$ 9,452</u>	<u>3.2 %</u>
Non-operating adjustments:			
Amortization of acquired intangible assets	\$ (482)	\$ (557)	(13.4) %
Acquisition-related charges	(9)	(9)	6.4
Kyndryl-related impacts	—	0	nm
Operating (non-GAAP) selling, general and administrative expense	<u>\$ 9,263</u>	<u>\$ 8,887</u>	<u>4.2 %</u>

nm - not meaningful

Total selling, general and administrative (SG&A) expense increased 0.9 percent in the second quarter of 2023 versus the prior-year period driven primarily by the following factors:

- Higher workforce rebalancing charges (2 points) to address remaining stranded cost from portfolio actions; partially offset by
- Benefits from productivity actions, partially offset by higher spending to drive our hybrid cloud and AI strategy.

Operating (non-GAAP) expense increased 1.7 percent year to year, driven primarily by the same factors.

Management Discussion – (continued)

SG&A expense increased 3.2 percent in the first six months of 2023 versus the prior-year period driven primarily by the following factors:

- Higher workforce rebalancing charges (4 points); and
- Higher net spending driven by the same factors above; partially offset by
- The effects of currency (1 point).

Operating (non-GAAP) expense increased 4.2 percent year to year, driven primarily by the same factors.

Provisions for expected credit loss expense in the first six months of 2023 decreased \$6 million compared to the prior-year period, driven primarily by lower specific reserve requirements in the current year. The receivables provision coverage was 2.8 percent at June 30, 2023, excluding receivables classified as held for sale, an increase of 40 basis points from December 31, 2022, primarily due to seasonality declines in the total receivables balance, and an increase in coverage of 60 basis points from June 30, 2022.

Research, Development and Engineering

(Dollars in millions)				
For the three months ended June 30:		2023	2022	Yr. to Yr. Percent Change
Research, development and engineering expense	\$	1,687	\$ 1,673	0.8 %

(Dollars in millions)				
For the six months ended June 30:		2023	2022	Yr. to Yr. Percent Change
Research, development and engineering expense	\$	3,342	\$ 3,352	(0.3) %

Research, development and engineering (RD&E) expense in the second quarter of 2023 increased 0.8 percent year to year primarily due to higher spending in the current-year period (2 points); partially offset by the effects of currency (1 point). RD&E expense in the first six months of 2023 was approximately flat compared to 2022 driven primarily by the effects of currency (1 point); partially offset by higher spending (1 point). We continue to invest to deliver innovation in AI, hybrid cloud and emerging areas such as quantum.

Intellectual Property and Custom Development Income

(Dollars in millions)				
For the three months ended June 30:		2023	2022	Yr. to Yr. Percent Change
Intellectual property and custom development income:				
Licensing of intellectual property including royalty-based fees	\$	127	\$ 113	12.5 %
Custom development income		119	57	111.2
Sales/other transfers of intellectual property		2	6	(73.4)
Total	\$	248	\$ 176	41.3 %

(Dollars in millions)				
For the six months ended June 30:		2023	2022	Yr. to Yr. Percent Change
Intellectual property and custom development income:				
Licensing of intellectual property including royalty-based fees	\$	188	\$ 184	2.2 %
Custom development income		235	105	124.0
Sales/other transfers of intellectual property		5	8	(32.5)
Total	\$	428	\$ 297	44.3 %

Management Discussion – (continued)

Total intellectual property and custom development income increased 41.3 percent year to year in the second quarter, and 44.3 percent in the first six months of 2023 compared to the prior-year period. The increase was primarily driven by a three-year joint development and licensing agreement signed in the fourth quarter of 2022 with a Japanese consortium to leverage our intellectual property and expertise on advanced semiconductors.

The timing and amount of licensing, sales or other transfers of IP may vary significantly from period to period depending upon the timing of licensing agreements, economic conditions, industry consolidation and the timing of new patents and know-how development.

Other (Income) and Expense
(Dollars in millions)
For the three months ended June 30:

	2023	2022	Yr. to Yr. Percent Change
Other (income) and expense:			
Foreign currency transaction losses/(gains)	\$ (166)	\$ (494)	(66.4) %
(Gains)/losses on derivative instruments	141	439	(67.8)
Interest income	(201)	(28)	nm
Net (gains)/losses from securities and investment assets	3	54	(94.1)
Retirement-related costs/(income)	1	192	(99.4)
Other	(39)	(243)	(84.0)
Total other (income) and expense	<u>\$ (261)</u>	<u>\$ (81)</u>	<u>223.6 %</u>
Non-operating adjustments:			
Amortization of acquired intangible assets	\$ 0	\$ (1)	(66.7) %
Non-operating retirement-related (costs)/income	(1)	(192)	(99.4)
Kyndryl-related impacts	—	(145)	(100.0)
Operating (non-GAAP) other (income) and expense	<u>\$ (262)</u>	<u>\$ (418)</u>	<u>(37.2) %</u>

(Dollars in millions)
For the six months ended June 30:

	2023	2022	Yr. to Yr. Percent Change
Other (income) and expense:			
Foreign currency transaction losses/(gains)	\$ (78)	\$ (670)	(88.3) %
(Gains)/losses on derivative instruments	(1)	541	nm
Interest income	(371)	(46)	nm
Net (gains)/losses from securities and investment assets	8	273	(96.9)
Retirement-related costs/(income)	(4)	394	nm
Other	(61)	(327)	(81.4)
Total other (income) and expense	<u>\$ (506)</u>	<u>\$ 166</u>	<u>nm</u>
Non-operating adjustments:			
Amortization of acquired intangible assets	\$ (1)	\$ (1)	(33.3) %
Acquisition-related charges	(1)	—	nm
Non-operating retirement-related (costs)/income	4	(394)	nm
Kyndryl-related impacts	—	(367)	(100.0)
Operating (non-GAAP) other (income) and expense	<u>\$ (504)</u>	<u>\$ (596)</u>	<u>(15.5) %</u>

nm - not meaningful

Management Discussion – (continued)

Total other (income) and expense was income of \$261 million in the second quarter of 2023 compared to \$81 million in the prior-year period. The year-to-year change was primarily driven by:

- Lower non-operating retirement-related cost (\$191 million). Refer to “Retirement-Related Plans” for additional information; and
- Higher interest income (\$173 million) driven by higher average interest rates and a higher average cash balance in the current year; and
- Kyndryl-related impacts in the prior period (\$145 million), including unrealized losses on Kyndryl retained shares (\$56 million) and related cash-settled swap (\$88 million); partially offset by
- Lower gains on divestitures (\$222 million) primarily driven by the divestiture of our healthcare software assets in the second quarter 2022 (included in “Other”); and
- The effects of currency.

Operating (non-GAAP) other (income) and expense was income of \$262 million in the second quarter of 2023 and decreased \$155 million compared to the prior-year period. The year-to-year change was primarily driven by the factors described above, excluding the lower non-operating retirement-related costs and benefits related to the prior year Kyndryl retained shares and swap.

Total other (income) and expense was income of \$506 million in the first six months of 2023 compared to expense of \$166 million in the prior-year period. The year-to-year change was primarily driven by:

- Lower non-operating retirement-related cost (\$398 million). Refer to “Retirement-Related Plans” for additional information; and
- Kyndryl-related impacts in the prior period (\$367 million), including unrealized losses on Kyndryl retained shares (\$278 million) and related cash-settled swap (\$88 million); and
- Higher interest income (\$325 million) driven by higher average interest rates and a higher average cash balance in the current year; partially offset by
- Lower gains on divestitures (\$240 million) primarily driven by the divestiture of our healthcare software assets in the second quarter 2022 (included in “Other”); and
- The effects of currency.

Operating (non-GAAP) other (income) and expense was income of \$504 million in the first six months of 2023 and decreased \$92 million compared to the prior-year period. The year-to-year change was primarily driven by the factors described above, excluding the lower non-operating retirement-related costs and benefits related to the prior year Kyndryl retained shares and swap.

Interest Expense

(Dollars in millions)			Yr. to Yr. Percent Change
For the three months ended June 30:	2023	2022	
Interest expense	\$ 423	\$ 297	42.5 %

(Dollars in millions)			Yr. to Yr. Percent Change
For the six months ended June 30:	2023	2022	
Interest expense	\$ 790	\$ 607	30.1 %

Management Discussion – (continued)

Interest expense increased \$126 million and \$183 million year to year in the second quarter and first six months of 2023, respectively. Interest expense is presented in cost of financing in the Consolidated Income Statement if the related external borrowings are to support the Financing external business. Overall interest expense (excluding capitalized interest) for the second quarter and first six months of 2023 was \$505 million and \$963 million, respectively, an increase of \$126 million and \$191 million, respectively, compared to the prior-year periods. The year-to-year dynamics for both the second quarter and first six months of 2023 were primarily driven by higher average interest rates and a higher average debt balance in the current year.

Retirement-Related Plans

The following tables provide the total pre-tax cost for all retirement-related plans. The operating cost amounts are included in the Consolidated Income Statement within the caption (e.g., Cost, SG&A, RD&E) relating to the job function of the plan participants. The non-operating cost amounts are included in other (income) and expense.

(Dollars in millions)			Yr. to Yr. Percent Change
For the three months ended June 30:			
Retirement-related plans — cost:			
Service cost	\$ 46	\$ 61	(25.9) %
Multi-employer plans	3	3	(17.6)
Cost of defined contribution plans	242	233	4.1
Total operating costs	\$ 291	\$ 298	(2.3) %
Interest cost	\$ 604	\$ 460	31.3 %
Expected return on plan assets	(745)	(734)	1.5
Recognized actuarial losses	128	442	(71.0)
Amortization of prior service costs/(credits)	(2)	6	nm
Curtailments/settlements	6	11	(47.0)
Other costs	10	6	67.7
Total non-operating costs/(income)	\$ 1	\$ 192	(99.4) %
Total retirement-related plans — cost	\$ 292	\$ 489	(40.4) %

(Dollars in millions)			Yr. to Yr. Percent Change
For the six months ended June 30:			
Retirement-related plans — cost:			
Service cost	\$ 92	\$ 127	(27.8) %
Multi-employer plans	6	7	(11.5)
Cost of defined contribution plans	511	472	8.4
Total operating costs	\$ 610	\$ 606	0.6 %
Interest cost	\$ 1,203	\$ 927	29.7 %
Expected return on plan assets	(1,484)	(1,483)	0.0
Recognized actuarial losses	257	902	(71.5)
Amortization of prior service costs/(credits)	(4)	13	nm
Curtailments/settlements	5	19	(74.0)
Other costs	19	15	22.6
Total non-operating costs/(income)	\$ (4)	\$ 394	nm
Total retirement-related plans — cost	\$ 606	\$ 1,000	(39.4) %

nm - not meaningful

Management Discussion – (continued)

Total pre-tax retirement-related plan cost decreased by \$198 million compared to the second quarter of 2022 primarily driven by a decrease in recognized actuarial losses (\$314 million), partially offset by higher interest costs (\$144 million). Total cost for the first six months of 2023 decreased \$394 million compared to the first six months of 2022, primarily driven by a decrease in recognized actuarial losses (\$645 million), partially offset by higher interest costs (\$276 million).

As described in the “Operating (non-GAAP) Earnings” section, management characterizes certain retirement-related costs as operating and others as non-operating. Utilizing this characterization, operating retirement-related costs in the second quarter of 2023 were \$291 million, a decrease of \$7 million compared to the second quarter of 2022, primarily driven by lower service cost (\$16 million), partially offset by higher cost of defined contribution plans (\$10 million). For the first six months of 2023, operating retirement-related costs were \$610 million, an increase of \$3 million compared to the prior-year period, primarily driven by higher cost of defined contribution plans (\$40 million), partially offset by lower service cost (\$35 million). Non-operating costs/(income) of \$1 million in the second quarter of 2023 decreased \$191 million year to year and for the first six months of 2023 was \$4 million of income compared to cost of \$394 million in the prior-year period. The year-to-year changes were primarily driven by a decrease in recognized actuarial losses, partially offset by higher interest costs.

The year-to-year decrease in recognized actuarial losses was primarily driven by the December 2022 remeasurement of our retirement and postretirement plans which resulted in a significant reduction to our pension plan benefit obligations and an improvement in our overall funded status primarily due to higher discount rates. In addition, we transferred \$16 billion of our U.S. Qualified PPP obligations and related plan assets to third-party insurers in 2022 to further reduce the risk profile of our plans. The transfer resulted in a pre-tax pension settlement charge of \$5.9 billion in the third quarter of 2022 primarily related to the accelerated recognition of actuarial losses.

Taxes

The continuing operations provision for income taxes for the second quarter of 2023 was \$419 million, compared to \$257 million in the second quarter of 2022. The operating (non-GAAP) income tax provision for the second quarter of 2023 was \$393 million, compared to \$413 million in the second quarter of 2022.

The continuing operations provision for income taxes for the first six months of 2023 was \$543 million, compared to \$218 million for the first six months of 2022. The operating (non-GAAP) provision for income taxes for the first six months of 2023 was \$593 million, compared to \$657 million for the first six months of 2022.

The increase in the continuing operations provision for income taxes for the second quarter and first six months of 2023 compared to prior year was primarily driven by the impact of foreign tax credit regulations.

IBM’s tax provision and effective tax rate are impacted by recurring factors including the geographical mix of income before taxes, incentives, changes in unrecognized tax benefits and discrete tax events, such as the settlement of income tax audits and changes in or new interpretations of tax laws. The GAAP tax provision and effective tax rate could also be affected by adjustments to the previously recorded charges for U.S. tax reform attributable to any changes in law, new regulations and guidance, and audit adjustments, among others.

During the fourth quarter of 2020, the U.S. Internal Revenue Service (IRS) concluded its examination of the company’s U.S. income tax returns for 2013 and 2014, which had a specific focus on certain cross-border transactions that occurred in 2013, and issued a final Revenue Agent’s Report (RAR). The IRS’ proposed adjustments relative to these cross-border transactions, if sustained, would result in additional taxable income of approximately \$4.5 billion. The company strongly disagrees with the IRS on these specific matters and filed its IRS Appeals protest in the first quarter of 2021. In the third quarter of 2018, the IRS commenced its audit of the company’s U.S. tax returns for 2015 and 2016. The company anticipates that this audit will be completed in 2023. In the fourth quarter of 2021, the IRS commenced its audit of the company’s U.S. tax returns for 2017 and 2018. With respect to major U.S. state and foreign taxing jurisdictions, the company is generally no longer subject to tax examinations for years prior to 2015. The company is no longer subject to income tax examination of its U.S. federal tax return for years prior to 2013. The open years contain matters that could be subject to differing interpretations of applicable tax laws and regulations as they relate to the amount and/or timing of income, deductions, and tax credits. Although the outcome of tax audits is always uncertain, the company believes that adequate amounts of tax, interest and penalties have been provided for any adjustments that are expected to result for these years.

Management Discussion – (continued)

The company is involved in a number of income tax-related matters in India challenging tax assessments issued by the India Tax Authorities. As of June 30, 2023, the company had recorded \$639 million as prepaid income taxes in India. A significant portion of this balance represents cash tax deposits paid over time to protect the company's right to appeal various income tax assessments made by the India Tax Authorities. Although the outcome of tax audits are always uncertain, the company believes that adequate amounts of tax, interest and penalties have been provided for any adjustments that are expected to result for these years.

The amount of unrecognized tax benefits at June 30, 2023 is \$8,796 million which can be reduced by \$548 million associated with timing adjustments, potential transfer pricing adjustments, and state income taxes. The net amount of \$8,248 million, if recognized, would favorably affect the company's effective tax rate.

Earnings Per Share

Basic earnings per share is computed on the basis of the weighted-average number of shares of common stock outstanding during the period. Diluted earnings per share is computed on the basis of the weighted-average number of shares of common stock outstanding plus the effect of dilutive potential common shares outstanding during the period using the treasury stock method. Dilutive potential common shares include outstanding stock options and stock awards.

For the three months ended June 30:	2023	2022	Yr. to Yr. Percent Change
Earnings per share of common stock from continuing operations:			
Assuming dilution	\$ 1.72	\$ 1.61	6.8 %
Basic	\$ 1.74	\$ 1.62	7.4 %
Diluted operating (non-GAAP)	\$ 2.18	\$ 2.31	(5.6) %
Weighted-average shares outstanding: (in millions)			
Assuming dilution	919.5	910.7	1.0 %
Basic	909.9	901.5	0.9 %
For the six months ended June 30:	2023	2022	Yr. to Yr. Percent Change
Earnings per share of common stock from continuing operations:			
Assuming dilution	\$ 2.74	\$ 2.34	17.1 %
Basic	\$ 2.77	\$ 2.36	17.4 %
Diluted operating (non-GAAP)	\$ 3.54	\$ 3.71	(4.6) %
Weighted-average shares outstanding: (in millions)			
Assuming dilution	918.6	910.0	1.0 %
Basic	908.7	900.4	0.9 %

Actual shares outstanding at June 30, 2023 were 911.0 million. The weighted-average number of common shares outstanding assuming dilution during the second quarter and first six months of 2023 were 8.7 million shares (1.0 percent) higher than the same periods of 2022.

Financial Position

Dynamics

Our balance sheet at June 30, 2023 continues to provide us with flexibility to support and invest in the business.

Cash and cash equivalents, restricted cash and marketable securities at June 30, 2023 were \$16,329 million, an increase of \$7,489 million compared to December 31, 2022. Total debt of \$57,476 million at June 30, 2023 increased \$6,527 million from December 31, 2022 primarily due to new debt issuances. We were opportunistic in accessing the debt market and issued \$9,463 million of debt in the first quarter of 2023 to prudently plan for our debt maturity obligations in

Management Discussion – (continued)

2023 and 2024 as well as capital allocation priorities. We continue to manage our debt levels while being acquisitive and without sacrificing investments in our business or our secure and modestly growing dividend policy.

In the first six months of 2023, we generated \$6,412 million in cash from operating activities, an increase of \$1,843 million compared to the first six months of 2022. Our free cash flow for the six months ended June 30, 2023 was \$3,441 million, an increase of \$110 million versus the prior year. See pages 76 through 77 for additional information on free cash flow. Our solid cash generation supports investment and deployment of capital to areas with the most attractive long-term opportunities. We completed six acquisitions and returned \$3,007 million to shareholders through dividends in the first half of 2023.

Our pension plans were well funded at the end of 2022, with worldwide qualified plans funded at 114 percent. Overall pension funded status as of the end of June 2023 was fairly consistent with year-end 2022, and we currently have no change to expected plan contributions in 2023.

IBM Working Capital

(Dollars in millions)	At June 30, 2023	At December 31, 2022
Current assets	\$ 34,458	\$ 29,118
Current liabilities	32,513	31,505
Working capital	\$ 1,945	\$ (2,387)
Current ratio	1.06:1	0.92:1

Working capital increased \$4,332 million from the year-end 2022 position. Current assets increased \$5,340 million (\$5,279 million adjusted for currency) primarily in cash and cash equivalents and marketable securities mainly driven by new debt issuances; partially offset by a decrease in receivables mainly from collections of seasonally higher year-end balances. Current liabilities increased \$1,008 million (\$948 million adjusted for currency) primarily in short-term debt driven by reclassifications from long-term debt net of maturities; partially offset by a decrease in taxes payable and accounts payable.

Receivables and Allowances

Roll Forward of Total IBM Receivables Allowance for Credit Losses

(Dollars in millions)					
January 1, 2023	Additions / (Releases) *	Write-offs **	Foreign currency and other+	June 30, 2023	
\$ 495	\$ 14	\$ (51)	\$ 38	\$ 496	

* Additions/(Releases) for allowance for credit losses are recorded in expense.
** Refer to note A, “Significant Accounting Policies,” in our 2022 Annual Report for additional information regarding allowance for credit loss write-offs.
+ Other includes additions/(releases) related to discontinued operations.

Excluding receivables classified as held for sale, the total IBM receivables provision coverage was 2.8 percent at June 30, 2023, an increase of 40 basis points compared to December 31, 2022. The increase in coverage was primarily driven by the overall decrease in total receivables. The majority of the write-offs during the six months ended June 30, 2023 related to receivables which had been previously reserved.

Management Discussion – (continued)

Financing Segment Receivables and Allowances

The following table presents external Financing segment receivables excluding receivables classified as held for sale, and immaterial miscellaneous receivables.

(Dollars in millions)	At June 30, 2023	At December 31, 2022
Amortized cost *	\$ 10,969	\$ 12,843
Specific allowance for credit losses	126	127
Unallocated allowance for credit losses	45	46
Total allowance for credit losses	171	173
Net financing receivables	\$ 10,798	\$ 12,670
Allowance for credit losses coverage	1.6 %	1.3 %

* Includes deferred initial direct costs which are expensed in IBM’s consolidated financial results.

The percentage of Financing segment receivables reserved increased from 1.3 percent at December 31, 2022 to 1.6 percent at June 30, 2023, primarily driven by the decline in amortized cost and higher unallocated reserve requirements.

Roll Forward of Financing Segment Receivables Allowance for Credit Losses (included in Total IBM)

(Dollars in millions)						
January 1, 2023		Additions / (Releases)*		Write-offs **		Foreign currency and other
\$ 173	\$	(6)	\$	(4)	\$	7
						June 30, 2023
						171

* Additions/(Releases) for allowance for credit losses are recorded in expense.

** Refer to note A, “Significant Accounting Policies,” in our 2022 Annual Report for additional information regarding allowance for credit loss write-offs.

Financing’s expected credit loss expense (including reserves for off-balance sheet commitments which are recorded in other liabilities) was a net addition of \$10 million for the three months ended June 30, 2023, compared to a net release of \$2 million for the three months ended June 30, 2022. The year-to-year increase in expected credit loss expense was due to higher unallocated reserve requirements in the current year and higher specific reserve releases in the prior year.

Expected credit loss expense was a net release of \$5 million for the six months ended June 30, 2023, compared to a net release of \$12 million for the six months ended June 30, 2022. The lower year-to-year net release was due to lower specific and unallocated reserve releases in the current year.

Noncurrent Assets and Liabilities

(Dollars in millions)	At June 30, 2023	At December 31, 2022
Noncurrent assets	\$ 97,755	\$ 98,125
Long-term debt	\$ 50,691	\$ 46,189
Noncurrent liabilities (excluding debt)	\$ 26,738	\$ 27,528

The decrease in noncurrent assets of \$369 million (\$765 million adjusted for currency) was primarily due to intangibles amortization, a decrease in long-term financing receivables as a result of declines from seasonally higher year-end balances; partially offset by an increase in prepaid pension assets and goodwill.

Long-term debt increased \$4,502 million (\$4,305 million adjusted for currency) primarily driven by new debt issuances; partially offset by reclassifications to short-term debt to reflect upcoming maturities.

Noncurrent liabilities (excluding debt) decreased \$789 million (\$917 million adjusted for currency) primarily driven by deferred taxes, deferred income, and retirement and postretirement benefit obligations.

Management Discussion – (continued)

Debt

Our funding requirements are continually monitored as we execute our strategies to manage the overall asset and liability profile. Additionally, we maintain sufficient flexibility to access global funding sources as needed.

(Dollars in millions)	At June 30, 2023	At December 31, 2022
Total debt	\$ 57,476	\$ 50,949
Financing segment debt*	\$ 10,551	\$ 12,872
Non-Financing debt	\$ 46,925	\$ 38,077

* Refer to Financing’s “Financial Position” on page 79 for additional details.

Total debt of \$57,476 million increased \$6,527 million (\$6,312 million adjusted for currency) from December 31, 2022, primarily driven by proceeds from issuances of \$9,482 million; partially offset by maturities of \$3,155 million.

Non-Financing debt of \$46,925 million increased \$8,848 million (\$8,656 million adjusted for currency) from December 31, 2022, primarily driven by our first quarter debt issuances to plan for debt maturity obligations in 2023 and 2024 as well as capital allocation priorities.

Financing segment debt of \$10,551 million decreased \$2,321 million (\$2,344 million adjusted for currency) from December 31, 2022, primarily due to lower funding requirements associated with financing receivables.

Financing provides financing solutions predominantly for IBM’s external client assets, and the debt used to fund Financing assets is primarily composed of intercompany loans. Total debt changes generally correspond with the level of client and commercial financing receivables, the level of cash and cash equivalents, the change in intercompany and external payables and the change in intercompany investment from IBM. The terms of the intercompany loans are set by the company to substantially match the term, currency and interest rate variability underlying the financing receivable. The Financing debt-to-equity ratio remained at 9.0 to 1 at June 30, 2023.

We measure Financing as a stand-alone entity, and accordingly, interest expense relating to debt supporting Financing’s external client and internal business is included in the “Financing Results of Operations” and in note 4, “Segments.”

Equity

Total equity increased \$250 million from December 31, 2022, primarily driven by an increase from net income of \$2,511 million and common stock of \$619 million; partially offset by dividends paid of \$3,007 million.

Cash Flow

Our cash flows from operating, investing and financing activities, as reflected in the Consolidated Statement of Cash Flows on page 7, are summarized in the table below. These amounts also include the cash flows associated with the Financing business.

(Dollars in millions)	2023	2022
For the six months ended June 30:		
Net cash provided by/(used in):		
Operating activities	\$ 6,412	\$ 4,569
Investing activities	(7,953)	(1,186)
Financing activities	2,978	(2,819)
Effect of exchange rate changes on cash, cash equivalents and restricted cash	(1)	(267)
Net change in cash, cash equivalents and restricted cash	\$ 1,436	\$ 297

Management Discussion – (continued)

Net cash provided by operating activities increased \$1,843 million as compared to the first six months of 2022. This was due to an increase in cash provided by financing receivables and improvements in sales cycle working capital; partially offset by an increase in performance-based compensation payments given our strong results in 2022.

Net cash used in investing activities increased \$6,767 million mainly driven by net purchases of marketable securities and other investments and a decrease in cash provided by divestitures; partially offset by a decrease in cash used in acquisitions.

Financing activities were a net source of cash of \$2,978 million in the first six months of 2023 compared to a net use of cash of \$2,819 million in the first six months of 2022. The year-to-year change of \$5,797 million is mainly due to an increase in net cash provided by debt transactions of \$5,736 million primarily driven by a higher level of net additions in the current year.

Results of Discontinued Operations

Income from discontinued operations, net of tax was \$2 million in the second quarter of 2023 compared to a loss of \$73 million in the prior-year period. For the first six months of 2023, loss from discontinued operations, net of tax was \$4 million compared to \$2 million in the prior-year period. The results for all periods reflect the net impact of changes in separation-related estimates and the settlement of assets and liabilities in accordance with the separation and distribution agreement. The prior-year results also reflect a gain on sale of a joint venture historically managed by Kyndryl, which was sold to Kyndryl in the first quarter of 2022 upon receiving regulatory approval.

Looking Forward

Clients and partners continue to view technology as a source of competitive advantage. Clients turn to us to speed up their transformation journeys, modernize applications and optimize their business workflows. At the same time, they continue to prioritize projects that focus on productivity and deliver quick time to value. To seize this opportunity, we are bringing new innovations to market, expanding strategic partnerships and making investments in targeted growth markets, while unlocking value through our productivity initiatives. We continue to build a stronger, more focused company that is closely aligned to the needs of our clients.

Hybrid Cloud and AI Strategy

We have made progress in our strategy around hybrid cloud and AI, the two key drivers of business innovation.

Hybrid cloud is the most widespread form of IT architecture. Red Hat OpenShift, our leading container platform based on open-source innovations, plays a crucial role in making this possible, along with IBM Software and Infrastructure. Our consultants use their technical and business knowledge to speed up clients’ digital transformation journeys and help drive adoption of our technology platforms, and our broad ecosystem of partners amplifies our reach and ability to meet client demand. **Artificial Intelligence (AI)** is a transformative technology that has the potential to unlock tremendous business value. Our focus is on enterprise AI, designed to address these opportunities and solve business problems. We are infusing AI into our software products and building products that address specific enterprise use cases such as digital labor, customer service and code generation. We recently announced watsonx, our enterprise-ready AI and data platform to help clients and partners capitalize on the AI opportunity. Watsonx delivers the value of foundation models to enterprises, enabling them to be more productive and will be the core technology platform for our AI capabilities. We have over 20,000 data and AI consultants and we recently launched our new Center of Excellence for Generative AI, staffed with over 1,000 consultants with specialized generative AI expertise. These investments we are making in products and skills will help us seize the AI opportunity.

To further advance our hybrid cloud and AI strategy, we continue to invest, both organically and inorganically, to deliver new innovation to our clients and to shape the technologies of the future. Watsonx is just one of many new technology innovations in 2023. Others include OpenShift AI, Hybrid Cloud Mesh, as well as sustainability tools to help clients advance their sustainability agendas. We continue to make advances in quantum computing that put us on a path toward building practical quantum computers that can solve hard problems in areas such as risk, finance and materials. To complement our innovations, we closed six acquisitions in the first half of 2023, and in July 2023, we announced the intent to acquire Apptio, a leader in financial and operational IT management and optimization software.

Management Discussion – (continued)

We are making progress on our productivity initiatives. Through our hybrid cloud and AI strategy, we are digitally transforming IBM as client zero, modernizing our IT infrastructure, optimizing the operating model, simplifying and automating our workflows and deploying AI across the enterprise. These productivity benefits free up spend for reinvestment and contribute to margin expansion.

Through first half 2023 our Consulting business continued to see sustained demand for larger transformation projects that deliver meaningful return on investment. At the same time, other projects considered to be more discretionary are being delayed, predominantly in the U.S. Our signings in the second quarter of 2023 were solid, and we addressed continued demand for technology-driven transformations as clients prioritize projects that drive cost savings and increase productivity.

We remain confident in our strategy and in the fundamentals of our business. Our balance sheet and liquidity position remain strong. At June 30, 2023 we had \$16.3 billion of cash and cash equivalents, restricted cash and marketable securities. We issued \$9.5 billion of debt in the first quarter of 2023 to prudently plan for our debt maturity obligations in 2023 and 2024 as well as capital allocation priorities. We continued to manage our debt levels while being acquisitive and without sacrificing investments in our business or our secure and modestly growing dividend policy.

“Today’s IBM” is a higher-growth, higher-value business with strong cash generation. We expect to continue our progress as a leading hybrid cloud and AI company with a focus on revenue growth and cash generation.

Retirement-Related Plans

Our pension plans are well funded. Contributions for all retirement-related plans are expected to be approximately \$2.0 billion in 2023, approximately flat compared to 2022, of which \$0.2 billion generally relates to legally required contributions to non-U.S. defined benefit and multi-employer plans. We expect 2023 pre-tax retirement-related plan cost to be approximately \$1.2 billion, a decrease of approximately \$6.5 billion compared to 2022. The decrease is primarily driven by a \$5.9 billion settlement charge in the third quarter of 2022 resulting from the transfer of a portion of the U.S. Qualified PPP to insurance companies. This estimate reflects current pension plan assumptions at December 31, 2022. Within total retirement-related plan cost, operating retirement-related plan cost is expected to be approximately \$1.2 billion, approximately flat versus 2022. Non-operating retirement-related plan cost is expected to be approximately \$0.1 billion, a decrease of approximately \$6.5 billion compared to 2022, primarily driven by the third-quarter 2022 settlement charge and lower recognized actuarial losses, partially offset by higher interest cost.

Currency Rate Fluctuations

Changes in the relative values of non-U.S. currencies to the USD affect our financial results and financial position. At June 30, 2023, currency changes resulted in assets and liabilities denominated in local currencies being translated into more dollars than at year-end 2022. We use financial hedging instruments to limit specific currency risks related to foreign currency-based transactions.

Movements in currency, and the fact that we do not hedge 100 percent of our currency exposures, will result in a currency impact to our revenues, profit and cash flows throughout 2023. We execute a hedging program which defers, versus eliminates, the volatility of currency impacts on our financial results. During periods of sustained movements in currency, the marketplace and competition adjust to the changing rates over time.

We translate revenue, cost and expense in our non-U.S. operations at current exchange rates in the reported period. References to “adjusted for currency” or “constant currency” reflect adjustments based upon a simple mathematical formula. However, this constant currency methodology that we utilize to disclose this information does not incorporate any operational actions that management could take to mitigate fluctuating currency rates. Based on the currency rate movements in the second quarter of 2023, revenue from continuing operations decreased 0.4 percent as reported but increased 0.4 percent at constant currency versus the second quarter of 2022. In the first six months of 2023, revenue from continuing operations was flat as reported and increased 2.3 percent at constant currency, compared to the same period in 2022. Currency translation and hedging impacted year-to-year pre-tax income growth and operating (non-GAAP) pre-tax income growth by approximately \$150 million in the second quarter of 2023, and approximately \$300 million in the first six months of 2023. From a segment perspective, currency translation and hedging impacted our Software pre-tax income margin year-to-year growth by over a point, and Infrastructure by about a point in the second quarter of 2023. In the first six months of 2023, currency translation and hedging impacted our Software pre-tax income margin year-to-year growth by

Management Discussion – (continued)

about a point, and Infrastructure by over a point. We view these amounts as a theoretical maximum impact to our as-reported financial results. Hedging and certain underlying foreign currency transaction gains and losses are allocated to our segment results. Considering the operational responses mentioned above, movements of exchange rates, and the nature and timing of hedging instruments, it is difficult to predict future currency impacts on any particular period.

For non-U.S. subsidiaries and branches that operate in U.S. dollars or whose economic environment is highly inflationary, translation adjustments are reflected in results of operations. Generally, we manage currency risk in these entities by linking prices and contracts to U.S. dollars.

Liquidity and Capital Resources

In our 2022 Annual Report, on pages 33 to 35, there is a discussion of our liquidity including two tables that present three years of data. The table presented on page 33 includes net cash from operating activities, cash and cash equivalents, restricted cash and short-term marketable securities, and the size of our global credit facilities for each of the past three years. For the six months ended, or at, as applicable, June 30, 2023, those amounts are \$6.4 billion of net cash from operating activities, \$16.3 billion of cash and cash equivalents, restricted cash and short-term marketable securities and \$10.0 billion in global credit facilities, respectively. While we have no current plans to draw on these credit facilities, they are available as back-up liquidity. On June 15, 2023, the company amended its existing \$2.5 billion Three-Year Credit Agreement and \$7.5 billion Five-Year Credit Agreement (the Credit Agreements) to extend the maturity dates to June 20, 2026 and June 22, 2028, respectively. Refer to note 12, “Borrowings,” for additional details on these credit facilities.

The major rating agencies' ratings on our debt securities at June 30, 2023 appear in the following table and remain unchanged from March 31, 2023.

	STANDARD AND POOR'S	MOODY'S INVESTORS SERVICE
IBM RATINGS:		
Senior long-term debt	A-	A3
Commercial paper	A-2	Prime-2

IBM has ample financial flexibility, supported by our strong liquidity position and cash flows, to operate at a single A credit rating. Debt levels have increased \$6.5 billion from December 31, 2022, primarily driven by debt issuances. In the first quarter of 2023, we issued \$9.5 billion of debt primarily to plan for our debt maturity obligations in 2023 and 2024 as well as capital allocation priorities.

We do not have “ratings trigger” provisions in our debt covenants or documentation, which would allow the holders to declare an event of default and seek to accelerate payments thereunder in the event of a change in credit rating. Our debt covenants are well within the required levels. Our contractual agreements governing derivative instruments contain standard market clauses which can trigger the termination of the agreement if our credit rating were to fall below investment grade. At June 30, 2023, the fair value of those instruments that were in a liability position was \$824 million, before any applicable netting, and this position is subject to fluctuations in fair value period to period based on the level of our outstanding instruments and market conditions. We have no other contractual arrangements that, in the event of a change in credit rating, would result in a material adverse effect on our financial position or liquidity.

Effective December 31, 2022, the use of LIBOR was substantially eliminated for purposes of any new financial contract executions. The UK’s Financial Conduct Authority (FCA) extended the phase out of LIBOR in the case of U.S. dollar settings for certain tenors until the end of June 2023. Any legacy USD LIBOR based financial contracts were addressed using the LIBOR rates published through the June 2023 extension period. The replacement of the LIBOR benchmark within the company’s risk management activities did not have a material impact in the consolidated financial results.

We prepare our Consolidated Statement of Cash Flows in accordance with applicable accounting standards for cash flow presentation on page 7 of this Form 10-Q and highlight causes and events underlying sources and uses of cash in that format on pages 73 and 74. For the purpose of running its business, IBM manages, monitors and analyzes cash flows in a different manner.

Management Discussion – (continued)

Management uses free cash flow as a measure to evaluate its operating results, plan shareholder return levels, strategic investments and assess its ability and need to incur and service debt. The entire free cash flow amount is not necessarily available for discretionary expenditures. We define free cash flow as net cash from operating activities less the change in Financing receivables and net capital expenditures, including the investment in software. A key objective of the Financing business is to generate strong returns on equity, and our Financing receivables are the basis for that growth. Accordingly, management considers Financing receivables as a profit-generating investment, not as working capital that should be minimized for efficiency. Therefore, management includes presentations of both free cash flow and net cash from operating activities that exclude the effect of Financing receivables.

The following is management’s view of cash flows for the first six months of 2023 and 2022 prepared in a manner consistent with the description above.

(Dollars in millions)

For the six months ended June 30:

	2023	2022*
Net cash from operating activities per GAAP	\$ 6,412	\$ 4,569
Less: change in Financing receivables	2,028	367
Net cash from operating activities, excluding Financing receivables	\$ 4,385	\$ 4,202
Capital expenditures, net	(944)	(871)
Free cash flow	\$ 3,441	\$ 3,331
Acquisitions	(356)	(958)
Divestitures	6	1,268
Dividends	(3,007)	(2,963)
Non-Financing debt	8,514	1,740
Other (includes Financing net receivables and Financing debt)	(1,109)	(2,197) **
Change in cash, cash equivalents, restricted cash and short-term marketable securities	\$ 7,489	\$ 221

* Includes immaterial cash flows from discontinued operations.

** Recast to conform to current-year presentation.

In the first six months of 2023, we generated \$3.4 billion in free cash flow, an increase of \$0.1 billion versus the prior-year period. The increase was driven by working capital efficiencies, cash from our profit performance and lower payments for structural actions; partially offset by higher performance-based compensation payments given our strong results in 2022 and higher cash taxes. In the first six months of 2023, we also continued to return value to shareholders with \$3.0 billion in dividends.

Events that could temporarily change the historical cash flow dynamics discussed previously and in our 2022 Annual Report include significant changes in operating results, material changes in geographic sources of cash, unexpected adverse impacts from litigation, future pension funding requirements, periods of severe downturn in the capital markets or the timing of tax payments. Whether any litigation has such an adverse impact will depend on a number of variables, which are more completely described in note 14, “Contingencies,” in this Form 10-Q. With respect to pension funding, we expect to make legally mandated pension plan contributions to certain non-U.S. defined benefit plans of approximately \$200 million in 2023. Contributions related to all retirement-related plans are expected to be approximately \$2.0 billion in 2023. Financial market performance could increase the legally mandated minimum contributions in certain non-U.S. countries that require more frequent remeasurement of the funded status. We are not quantifying any further impact from pension funding because it is not possible to predict future movements in the capital markets or changes in pension plan funding regulations. In 2023, we are not legally required to make any contributions to the U.S. defined benefit pension plans.

Our cash flows are sufficient to fund our current operations and obligations, including investing and financing activities such as dividends and debt service. When additional requirements arise, we have several liquidity options available. These options may include the ability to borrow additional funds at reasonable interest rates and utilizing our committed global credit facilities. With our share repurchase program suspended since the close of the Red Hat acquisition, our overall shareholder payout remains at a comfortable level and we remain fully committed to our secure and modestly growing dividend policy.

Management Discussion – (continued)

Financing

Financing is a reportable segment that is measured as a stand-alone entity. Financing facilitates IBM clients’ acquisition of IBM information technology systems, software and services by providing financing solutions in the areas where the company has the expertise, while generating solid returns on equity.

Results of Operations

(Dollars in millions)				Yr. to Yr. Percent Change
For the three months ended June 30:		2023	2022	
Revenue	\$	185	\$ 146	26.2 %
Pre-tax income	\$	64	\$ 102	(36.8) %

(Dollars in millions)				Yr. to Yr. Percent Change
For the six months ended June 30:		2023	2022	
Revenue	\$	380	\$ 300	26.8 %
Pre-tax income	\$	165	\$ 186	(11.3) %

For the three months ended June 30, 2023, financing revenue increased 26.2 percent as reported (27 percent adjusted for currency) compared to the prior year, driven by client financing revenue up \$36 million to \$181 million. For the six months ended June 30, 2023, financing revenue increased 26.8 percent as reported (29 percent adjusted for currency) compared to the prior year, driven by client financing up \$77 million to \$374 million. The increase in client financing revenue in both periods in 2023 was primarily driven by an increase in client financing asset yields.

Financing pre-tax income decreased 36.8 percent to \$64 million in the second quarter of 2023, compared to the prior year and the pre-tax margin of 34.9 percent decreased 34.8 points year to year. The decrease in pre-tax income for the second quarter was driven by an increase in expense, primarily due to year-to-year foreign currency impacts and higher unallocated reserve requirements in the current year. For the six months ended June 30, 2023, Financing pre-tax income decreased 11.3 percent to \$165 million compared to the prior year and the pre-tax margin of 43.3 percent decreased 18.6 points year to year, primarily driven by year-to-year foreign currency impacts.

Management Discussion – (continued)
Financial Position

(Dollars in millions)	At June 30, 2023	At December 31, 2022
Cash and cash equivalents	\$ 662	\$ 699
Client financing receivables:		
Net investment in sales-type and direct financing leases ⁽¹⁾	3,831	4,047
Client loans	6,722	8,329
Total client financing receivables	\$ 10,553	\$ 12,376
Commercial financing receivables:		
Held for investment	245	293
Held for sale	865	939
Other receivables	42	66
Total external receivables ⁽²⁾	\$ 11,705	\$ 13,674
Intercompany assets ⁽³⁾	453	988 ⁽⁴⁾
Other assets	346	395 ⁽⁴⁾
Total assets	<u>\$ 13,166</u>	<u>\$ 15,757</u>
Intercompany payables ⁽³⁾	\$ 427	\$ 637
Debt ⁽⁵⁾	10,551	12,872
Other liabilities	1,016	814
Total liabilities	\$ 11,994	\$ 14,323
Total equity	\$ 1,172	\$ 1,433
Total liabilities and equity	<u>\$ 13,166</u>	<u>\$ 15,757</u>

(1) Includes deferred initial direct costs which are expensed in IBM's consolidated financial results.

(2) The difference between the decrease in total external receivables (from \$13.7 billion in December 2022 to \$11.7 billion in June 2023) and the change in Financing segment's receivables disclosed in the free cash flow presentation on page 77 is primarily attributable to currency impacts.

(3) This entire amount is eliminated for purposes of IBM's consolidated financial results and therefore does not appear in the Consolidated Balance Sheet.

(4) Prior period amounts have been recast to conform to 2023 presentation.

(5) Financing segment debt is primarily composed of intercompany loans.

Total external receivables decreased \$1,970 million primarily due to collections of higher year-end balances. Intercompany assets decreased \$535 million primarily driven by intercompany financing receivables at December 31, 2022 that settled in the first half of 2023. These declines had corresponding reductions in debt funding.

We continue to apply our rigorous credit policies. Approximately 73 percent of the total external portfolio was with investment-grade clients with no direct exposure to consumers at both June 30, 2023 and December 31, 2022; a year-to-year increase of 2 points as compared to June 30, 2022. This investment grade percentage is based on the credit ratings of the companies in the portfolio and reflects certain mitigating actions taken to reduce the risk to IBM.

We have a long-standing practice of taking mitigation actions, in certain circumstances, to transfer credit risk to third parties. These actions may include credit insurance, financial guarantees, nonrecourse secured borrowings, transfers of receivables recorded as true sales in accordance with accounting guidance or sales of equipment under operating lease. Sale of receivables arrangements are also utilized in the normal course of business as part of our cash and liquidity management. For additional information relating to financing receivables refer to note 9, "Financing Receivables." Refer to pages 72 through 73 for additional information related to Financing segment receivables, allowance for credit losses and debt.

Management Discussion – (continued)

Return on Equity Calculation

(Dollars in millions)	For Three Months Ended June 30,		For Six Months Ended June 30,	
	2023	2022	2023	2022
Numerator:				
Financing after-tax income*	\$ 53	\$ 84	\$ 135	\$ 153
Annualized after-tax income (1)	\$ 211	\$ 336	\$ 270	\$ 305
Denominator:				
Average Financing equity (2)**	\$ 1,170	\$ 1,358	\$ 1,258	\$ 1,421
Financing return on equity (1)/(2)	18.1 %	24.8 %	21.5 %	21.5 %

* Calculated based upon an estimated tax rate principally based on Financing’s geographic mix of earnings as IBM’s provision for income taxes is determined on a consolidated basis.
** Average of the ending equity for Financing for the last two quarters and three quarters, for the three months ended June 30 and for the six months ended June 30, respectively.

Return on equity was 18.1 percent and 21.5 percent for the three and six months ended June 30, 2023, respectively, compared to 24.8 percent and 21.5 percent for the same periods in 2022. The changes in the three months ended June 30, 2023 were driven by a decrease in net income, partially offset by a lower average equity balance.

Residual Value

The estimated residual value represents the estimated fair value of the equipment under lease at the end of the lease. The company estimates the future fair value of leased equipment by using historical models, analyzing the current market for new and used equipment and obtaining forward-looking product information such as marketing plans and technology innovations.

The company optimizes the recovery of residual values by extending lease arrangements with, or selling leased equipment to existing clients and periodically reassesses the realizable value of its lease residual values.

The following table presents the recorded amount of unguaranteed residual value for sales-type and direct financing leases at June 30, 2023 and December 31, 2022. In addition, the table presents the run out of when the unguaranteed residual value assigned to equipment on leases at June 30, 2023 is expected to be returned to the company. The unguaranteed residual value for operating leases at June 30, 2023 and December 31, 2022 was not material.

Unguaranteed Residual Value

(Dollars in millions)	At December 31, 2022	At June 30, 2023	Estimated Run Out of June 30, 2023 Balance			
			2023	2024	2025	2026 and Beyond
Sales-type and direct financing leases	\$ 422	\$ 404	\$ 28	\$ 60	\$ 137	\$ 180

Management Discussion – (continued)

GAAP Reconciliation

The tables below provide a reconciliation of our income statement results as reported under GAAP to our operating earnings presentation which is a non-GAAP measure. Management’s calculation of operating (non-GAAP) earnings, as presented, may differ from similarly titled measures reported by other companies. Refer to the “Operating (non-GAAP) Earnings” section for management’s rationale for presenting operating earnings information.

(Dollars in millions except per share amounts)						
For the three months ended June 30, 2023:	GAAP	Acquisition-Related Adjustments	Retirement-Related Adjustments	U.S. Tax Reform Impacts	Kyndryl-Related Impacts	Operating (non-GAAP)
Gross profit	\$ 8,501	\$ 150	\$ —	\$ —	\$ —	\$ 8,650
Gross profit margin	54.9 %	1.0 pts.	— pts.	— pts.	— pts.	55.9 %
SG&A	\$ 4,900	\$ (245)	\$ —	\$ —	\$ —	\$ 4,655
Other (income) and expense	\$ (261)	\$ 0	\$ (1)	\$ —	\$ —	\$ (262)
Total expense and other (income)	\$ 6,501	\$ (246)	\$ (1)	\$ —	\$ —	\$ 6,254
Pre-tax income from continuing operations	\$ 2,000	\$ 395	\$ 1	\$ —	\$ —	\$ 2,396
Pre-tax margin from continuing operations	12.9 %	2.6 pts.	0.0 pts.	— pts.	— pts.	15.5 %
Provision for income taxes*	\$ 419	\$ 87	\$ (3)	\$ (110)	\$ —	\$ 393
Effective tax rate	21.0 %	0.2 pts.	(0.2) pts.	(4.6) pts.	— pts.	16.4 %
Income from continuing operations	\$ 1,581	\$ 308	\$ 5	\$ 110	\$ —	\$ 2,003
Income margin from continuing operations	10.2 %	2.0 pts.	0.0 pts.	0.7 pts.	— pts.	12.9 %
Diluted earnings per share from continuing operations	\$ 1.72	\$ 0.34	\$ 0.00	\$ 0.12	\$ —	\$ 2.18

(Dollars in millions except per share amounts)						
For the three months ended June 30, 2022:	GAAP	Acquisition-Related Adjustments	Retirement-Related Adjustments	U.S. Tax Reform Impacts	Kyndryl-Related Impacts	Operating (non-GAAP)
Gross profit	\$ 8,290	\$ 180	\$ —	\$ —	\$ —	\$ 8,470
Gross profit margin	53.4 %	1.2 pts.	— pts.	— pts.	— pts.	54.5 %
SG&A	\$ 4,855	\$ (279)	\$ —	\$ —	\$ 0	\$ 4,576
Other (income) and expense	\$ (81)	\$ (1)	\$ (192)	\$ —	\$ (145)	\$ (418)
Total expense and other (income)	\$ 6,568	\$ (280)	\$ (192)	\$ —	\$ (145)	\$ 5,952
Pre-tax income from continuing operations	\$ 1,722	\$ 460	\$ 192	\$ —	\$ 145	\$ 2,518
Pre-tax margin from continuing operations	11.1 %	3.0 pts.	1.2 pts.	— pts.	0.9 pts.	16.2 %
Provision for income taxes*	\$ 257	\$ 115	\$ 46	\$ (4)	\$ —	\$ 413
Effective tax rate	14.9 %	1.8 pts.	0.7 pts.	(0.2) pts.	(0.9) pts.	16.4 %
Income from continuing operations	\$ 1,465	\$ 345	\$ 146	\$ 4	\$ 145	\$ 2,105
Income margin from continuing operations	9.4 %	2.2 pts.	0.9 pts.	0.0 pts.	0.9 pts.	13.5 %
Diluted earnings per share from continuing operations	\$ 1.61	\$ 0.38	\$ 0.16	\$ 0.00	\$ 0.16	\$ 2.31

* The tax impact on operating (non-GAAP) pre-tax income from continuing operations is calculated under the same accounting principles applied to the GAAP pre-tax income which employs an annual effective tax rate method to the results.

Management Discussion – (continued)

(Dollars in millions except per share amounts)						
For the six months ended June 30, 2023:	GAAP	Acquisition-Related Adjustments	Retirement-Related Adjustments	U.S. Tax Reform Impacts	Kyndryl-Related Impacts	Operating (non-GAAP)
Gross profit	\$ 16,010	\$ 298	\$ —	\$ —	\$ —	\$ 16,308
Gross profit margin	53.9 %	1.0 pts.	— pts.	— pts.	— pts.	54.9 %
SG&A	\$ 9,754	\$ (491)	\$ —	\$ —	\$ —	\$ 9,263
Other (income) and expense	\$ (506)	\$ (2)	\$ 4	\$ —	\$ —	\$ (504)
Total expense and other (income)	\$ 12,952	\$ (493)	\$ 4	\$ —	\$ —	\$ 12,463
Pre-tax income from continuing operations	\$ 3,058	\$ 791	\$ (4)	\$ —	\$ —	\$ 3,845
Pre-tax margin from continuing operations	10.3 %	2.7 pts.	0.0 pts.	— pts.	— pts.	12.9 %
Provision for income taxes*	\$ 543	\$ 178	\$ (14)	\$ (115)	\$ —	\$ 593
Effective tax rate	17.8 %	1.0 pts.	(0.3) pts.	(3.0) pts.	— pts.	15.4 %
Income from continuing operations	\$ 2,515	\$ 613	\$ 10	\$ 115	\$ —	\$ 3,252
Income margin from continuing operations	8.5 %	2.1 pts.	0.0 pts.	0.4 pts.	— pts.	10.9 %
Diluted earnings per share from continuing operations	\$ 2.74	\$ 0.67	\$ 0.01	\$ 0.13	\$ —	\$ 3.54

(Dollars in millions except per share amounts)						
For the six months ended June 30, 2022:	GAAP	Acquisition-Related Adjustments	Retirement-Related Adjustments	U.S. Tax Reform Impacts	Kyndryl-Related Impacts	Operating (non-GAAP)
Gross profit	\$ 15,625	\$ 361	\$ —	\$ —	\$ —	\$ 15,986
Gross profit margin	52.6 %	1.2 pts.	— pts.	— pts.	— pts.	53.8 %
SG&A	\$ 9,452	\$ (565)	\$ —	\$ —	\$ 0	\$ 8,887
Other (income) and expense	\$ 166	\$ (1)	\$ (394)	\$ —	\$ (367)	\$ (596)
Total expense and other (income)	\$ 13,280	\$ (566)	\$ (394)	\$ —	\$ (367)	\$ 11,953
Pre-tax income from continuing operations	\$ 2,345	\$ 928	\$ 394	\$ —	\$ 367	\$ 4,033
Pre-tax margin from continuing operations	7.9 %	3.1 pts.	1.3 pts.	— pts.	1.2 pts.	13.6 %
Provision for income taxes*	\$ 218	\$ 224	\$ 104	\$ 112	\$ —	\$ 657
Effective tax rate	9.3 %	3.4 pts.	1.7 pts.	2.8 pts.	(0.8) pts.	16.3 %
Income from continuing operations	\$ 2,127	\$ 704	\$ 290	\$ (112)	\$ 367	\$ 3,376
Income margin from continuing operations	7.2 %	2.4 pts.	1.0 pts.	(0.4) pts.	1.2 pts.	11.4 %
Diluted earnings per share from continuing operations	\$ 2.34	\$ 0.77	\$ 0.32	\$ (0.12)	\$ 0.40	\$ 3.71

* The tax impact on operating (non-GAAP) pre-tax income from continuing operations is calculated under the same accounting principles applied to the GAAP pre-tax income which employs an annual effective tax rate method to the results.

Management Discussion – (continued)

Forward-Looking and Cautionary Statements

Except for the historical information and discussions contained herein, statements contained in this Form 10-Q may constitute forward-looking statements within the meaning of the Private Securities Litigation Reform Act of 1995. Forward-looking statements are based on the company’s current assumptions regarding future business and financial performance. These statements involve a number of risks, uncertainties and other factors that could cause actual results to differ materially, including, but not limited to, the following: a downturn in economic environment and client spending budgets; a failure of the company’s innovation initiatives; damage to the company’s reputation; risks from investing in growth opportunities; failure of the company’s intellectual property portfolio to prevent competitive offerings and the failure of the company to obtain necessary licenses; the company’s ability to successfully manage acquisitions, alliances and dispositions, including integration challenges, failure to achieve objectives, the assumption of liabilities, and higher debt levels; fluctuations in financial results; impact of local legal, economic, political, health and other conditions; the company’s failure to meet growth and productivity objectives; ineffective internal controls; the company’s use of accounting estimates; impairment of the company’s goodwill or amortizable intangible assets; the company’s ability to attract and retain key employees and its reliance on critical skills; impacts of relationships with critical suppliers; product quality issues; impacts of business with government clients; reliance on third party distribution channels and ecosystems; cybersecurity and data privacy considerations; adverse effects related to climate change and environmental matters, tax matters; legal proceedings and investigatory risks; the company’s pension plans; currency fluctuations and customer financing risks; impact of changes in market liquidity conditions and customer credit risk on receivables; potential failure of the separation of Kyndryl Holdings, Inc. to qualify for tax-free treatment; risk factors related to IBM securities; and other risks, uncertainties and factors discussed in the company’s Form 10-Qs, Form 10-K and in the company’s other filings with the U.S. Securities and Exchange Commission or in materials incorporated therein by reference. Any forward-looking statement in this Form 10-Q speaks only as of the date on which it is made. Except as required by law, the company assumes no obligation to update or revise any forward-looking statements.

Item 4. Controls and Procedures

The company’s management evaluated, with the participation of the Chief Executive Officer and Chief Financial Officer, the effectiveness of the company’s disclosure controls and procedures as of the end of the period covered by this report. Based on that evaluation, the Chief Executive Officer and Chief Financial Officer have concluded that the company’s disclosure controls and procedures were effective as of the end of the period covered by this report. There has been no change in the company’s internal control over financial reporting that occurred during the quarter covered by this report that has materially affected, or is reasonably likely to materially affect, the company’s internal control over financial reporting.

Part II — Other Information

Item 1. Legal Proceedings

Refer to note 14, “Contingencies,” in this Form 10-Q.

Item 2. Unregistered Sales of Equity Securities and Use of Proceeds and Issuer Repurchases of Equity Securities

The following table provides information relating to the company’s repurchase of common stock for the second quarter of 2023.

Period	Total Number of Shares Purchased	Average Price Paid per Share	Total Number of Shares Purchased as Part of Publicly Announced Program	Approximate Dollar Value of Shares that May Yet Be Purchased Under The Program*
April 1, 2023 - April 30, 2023	—	\$ —	—	\$ 2,007,611,768
May 1, 2023 - May 31, 2023	—	\$ —	—	\$ 2,007,611,768
June 1, 2023 - June 30, 2023	—	\$ —	—	\$ 2,007,611,768
Total	—	\$ —	—	

* On October 30, 2018, the Board of Directors authorized \$4.0 billion in funds for use in the company’s common stock repurchase program. The company stated that it would repurchase shares on the open market or in private transactions depending on market conditions. The common stock repurchase program does not have an expiration date. This table does not include shares tendered to satisfy the exercise price in connection with cashless exercises of employee stock options or shares tendered to satisfy tax withholding obligations in connection with employee equity awards. The company suspended its share repurchase program at the time of the Red Hat closing in 2019.

Item 6. Exhibits

Exhibit Number

10.1	Terms and Conditions of IBM LTPP Equity Awards, effective June 1, 2023.
10.2	Forms of LTPP equity award agreements for (i) stock options, restricted stock units, cash-settled restricted stock units, SARs, (ii) retention restricted stock units, cash-settled retention restricted stock units, and (iii) performance share units, as well as the Terms and Conditions of LTPP Equity Awards, effective July 1, 2023, in connection with the foregoing award agreements.
10.3	Annual Incentive Program, effective January 1, 2023.
31.1	Certification by principal executive officer pursuant to Rule 13A-14(a) or 15D-14(a) of the Securities Exchange Act of 1934, as adopted pursuant to Section 302 of the Sarbanes-Oxley Act of 2002.
31.2	Certification by principal financial officer pursuant to Rule 13A-14(a) or 15D-14(a) of the Securities Exchange Act of 1934, as adopted pursuant to Section 302 of the Sarbanes-Oxley Act of 2002.
32.1	Certification by principal executive officer pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002.
32.2	Certification by principal financial officer pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002.
101.INS	XBRL Instance Document – the instance document does not appear on the Interactive Data File because its XBRL tags are embedded within the Inline XBRL document
101.SCH	XBRL Taxonomy Extension Schema Document
101.CAL	XBRL Taxonomy Extension Calculation Linkbase Document
101.DEF	XBRL Taxonomy Extension Definition Linkbase Document
101.LAB	XBRL Taxonomy Extension Label Linkbase Document
101.PRE	XBRL Taxonomy Extension Presentation Linkbase Document
104	Cover Page Interactive Data File – the cover page interactive data file does not appear in the Interactive Data File because its XBRL tags are embedded within the Inline XBRL document

SIGNATURE

Pursuant to the requirements of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned thereunto duly authorized.

International Business Machines Corporation

(Registrant)

Date: July 25, 2023

By: /s/ Nicolás A. Fehring
Nicolás A. Fehring
Vice President and Controller